A map is a phase without a clock: what path-integrated attention learns, and what decides whether training finds it

Main paper, draft 2 (2026-10-21), from `docs/paper/expansion/SYNTHESIS_DRAFT2_PLAN.md`. *Appendix:* `docs/paper/MAIN_APPENDIX2.md`. *Word counts, the source of every number, deviations from the plan and open items:* `docs/paper/MAIN_DRAFT2_notes.md`. *Delete this line before submission.*

Abstract

Path-integrated rotary attention replaces RoPE's token index with a learned phase that sums a content-dependent step per token. We give an account of what such models learn and of what decides whether training finds a map. On 1024-move torus walks, where a map-free rule that retraces the last run of moves scores 0.843, one path-integrated layer scores 0.998 (8 of 8 seeds solved) against 0.731 for index RoPE. The rotation carries the prefix sum of the moves, so one layer compares positions at any gap; index attention must build that sum in its residual stream, and at matched depth one path layer is worth about three index layers on Dyck-2 and on a biased 1D walk, and more than four on the torus. We show that a map code is a phase without a clock, the phase an out-and-back adds, and bound how much a clock erodes the position match one attention pass can make as the moves since the last visit grow. Two choices decide whether training removes the clock. A signed step at per-head rank above the map's dimension has two routes that remove the clock and leave the map intact; at rank equal to the dimension no channel can be clock-free and a full map while a clock remains; a non-negative step has neither route from a small start. In line with this, rank 2 solves the torus on at most 2 of 8 seeds against 8 of 8 at rank 4, and non-negative steps on 0 of 8 against 8 of 8 for signed steps, although both constrained classes represent the map exactly and keep or re-find an installed solution. On a ring the sign gap holds at each step's best learning rate and vanishes when only the readout, not the increment, is non-negative. Told in English, the walk is learned by one path layer (0.979 vs 0.505), which cancels opposite directions.

1. Introduction

Rotary position encodings are integrators. Index RoPE (Su et al. 2024) adds a constant angle per token; content-dependent variants let each token set its own step (CARoPE, Veisi et al. 2025; CoPE, Golovneva et al. 2024; Selective RoPE, Movahedi et al. 2026; Mamba-3, Lahoti et al. 2026; MapFormer, Rambaud et al. 2026). When the step is driven by actions, the summed phase is a position on a map, and attention compares net displacements instead of token distances. MapFormer built this into a transformer and showed that it helps navigation, generalises to sparser and denser walks, extrapolates on Dyck-2 with one layer, and learns opposite latents for opposite actions. We take that as the starting point. What we add is a measurement of how much of the task needs a map, the exchange rate between path integration and index depth together with its mechanism, a theory of what decides whether training finds the map, and the same mechanism at work on navigation told in words.

The account turns on one decomposition. Split the phase a channel gains over an interval into a part that depends on net displacement, the *map*, and a part that grows with the number of moves whatever their direction, the *clock*. A map code is one whose used channels carry no clock: an out-and-back returns the phase to where it started. Path integration writes the prefix sum of the moves into the rotation, so a clock-free code lets one attention layer find the same place at any gap and across the wrap of the torus. A clock works against this: its phase error grows with the number of moves since the last visit, and Proposition 2 bounds the position match it leaves to one attention pass while that error stays below pi. Every model we see fail, index attention, a rank-2 content-to-angle map and a non-negative step, shows the matching measured signature: it keeps short-gap revisits and loses long-gap and wrap-only ones (Fig. 2).

Depth does not buy the map. Four index layers, or one block looped four times, learn local return rules that reach a few direction runs, and never find the global count that would give them the map (Sec. 6). So the map has to come from the step map, and whether training finds it depends on whether training can remove the clock. A signed step can remove it collectively: driving the latents of opposite moves to sum to zero cancels every channel's clock at once and leaves the map untouched. Above rank D, the map's dimension, each channel can also stop reading the clock on its own, at no cost to the map. A non-negative step has neither route from a small start, and at rank D no channel can be both clock-free and a full map while a clock remains (Proposition 4). The measurements follow this inventory on both axes (Fig. 1): the constrained classes represent the map exactly, keep or re-find an installed solution, and from scratch rarely find it, and the runs that fail from a small start end with a clock of the order of their map or, for a few rank-2 runs, with a map that has lost a dimension.

Contributions.

1. *Measuring map use.* Map-free floors (blank, n-gram, retrace) and revisit strata (short-gap, long-gap, wrap-only) are defined once and used throughout. A retrace rule scores 0.934 on MapFormer's 128-move walks and 0.843 at 1024 moves, where one path layer scores 0.998 against index RoPE's 0.731 (8/8 vs 0/8 SOLVED, p 0.0002) and only the path model holds long-gap and wrap-only revisits.
2. *Why one path layer suffices, and what depth buys index attention.* The prefix sum is in the rotation (proved). Index attention reaches the map only through a global count, which needs two layers in principle but a normaliser, fine precision and large width (cited construction). Trained index models instead learn local return rules whose reach grows with depth in direction runs (analysis). Registered exchange rates: about three index layers on Dyck-2 and on a biased ring, more than four on the torus.
3. *A theory of when the map is found.* The clock/map decomposition and the bound it puts on one-pass retrieval; a route inventory for each step class (proved); a conjecture for why rank D starves the collective route in two or more dimensions but not in one. Registered support on both axes: the solution exists, is kept or re-found, and is not found from scratch at rank 2 (at most 2/8, also at twice the budget) or with a non-negative step (0/8 against 8/8, also at each step's best of four learning rates on a ring). Causal tests of the routes: freezing the move latents removes the collective route (1/8 against 7/8), and a non-negative readout with a signed increment restores it (16/16).

4. *Readers and codes.* Loops, extra layers and a score rule change how a code is read and never change the code family (proved). Loops and depth raise rank-2 accuracy (+0.079 and +0.096, registered), while the codes they learn stay far from rank 4's (analysis). PoPE's polar score pins every kernel's peak at zero phase difference (proved), which helps a path phase and hurts an index phase; freeing its phase offset raises the index model by +0.122 and leaves the path model at ceiling (registered).
5. *Maps from words.* One path layer learns navigation told in English (registered). On the revisits retrace cannot reach it scores 0.956 (0.978 on fresh seeds) against 0.411 for two index layers, below the 0.506 constant (analysis). Its step table follows one rule: a word's offset from its place depends on its role, not its identity, and a per-move clock survives only in channels retrieval does not use.

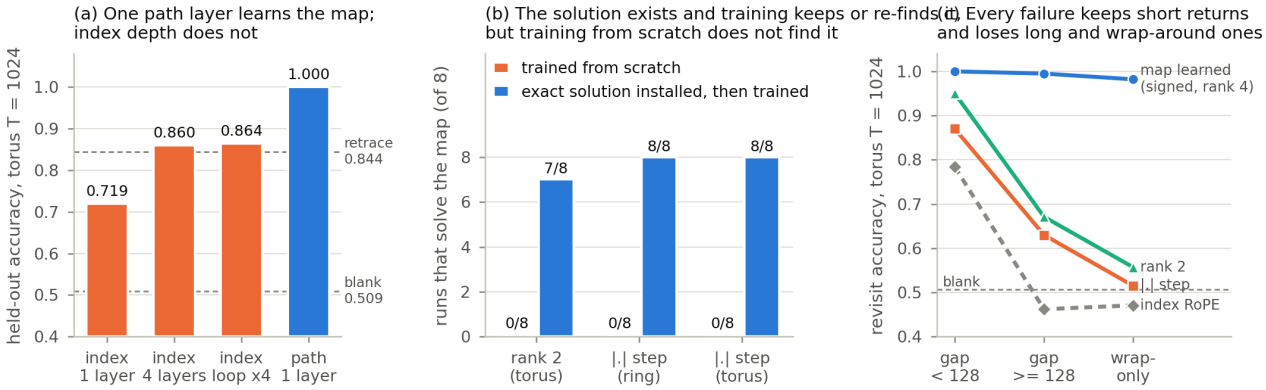


Figure 1 (`fig_summary`). The paper in three panels (torus, T = 1024 unless marked). (a) One path-integrated layer learns the map; index attention at one or four layers or looped four times does not (n = 4; dashed: this batch's retrace and always-blank floors, 0.844 and 0.509). (b) For both constrained classes a solution exists and training keeps it (rank 2: a rank-2 projection of a solved rank-4 model, 7/8 SOLVED after training; |.| on the ring: held 8/8) or re-finds it (|.| on the torus: perturbed early, 8/8 SOLVED at the end), while training from scratch solves 0/8. (c) Every failing one-layer model keeps short-gap revisits and loses long-gap and wrap-only ones; the model that learns the map holds all three (n = 8 per arm; dashed: always-blank within each stratum).

2. Background

2.1 Rotary position as an integrator. Write the phase of rotary channel c (a head and a two-dimensional block) at token t as theta_{c,t} = omega_c Phi_{c,t}, with Phi_{c,t} = sum_{u <= t} Delta_{c,u} the channel's summed step. The channel's contribution to the score of query t and key t' depends on position only through exp(-i omega_c (Phi_{c,t} - Phi_{c,t'})), the identity behind Mamba-3's Proposition 3 (App. E1). Howard et al. (2014) drive one bank of integrators by a rate alpha(t): a constant rate codes elapsed time (Case I), the speed codes distance travelled (Case II), and the signed velocity codes position, the same for every visit to a place, "like a bead on a wire" (Case III). The rotary families, as organised by GRAPE (Zhang et al. 2026), Selective RoPE and Mamba-3, sit in these three cases with an imaginary rate: index RoPE adds a constant angle per token; CoPE, CARoPE and non-negative content steps advance by a non-negative, content-dependent amount; MapFormer, Selective RoPE and Mamba-3's data-dependent RoPE add a signed

one, and alpha(t) is the selective-SSM step Delta_t. Velocity-modulated integrators have been trained with a real Laplace variable (Alipour et al. 2025), and RoPE has been derived as a velocity-controlled oscillator on given coordinates (GridPE, Li et al. 2024); the machine-learning models of the Howard and Tiganj labs keep the Laplace variable real (Mochizuki-Freeman et al. 2023; Kabir et al. 2025). In the rotary slot a signed rate is a bounded rotation, and the three cases differ not in what a channel can represent (Sec. 2.3) but in what the data and the optimiser make of it. TEM and TEM-t (Whittington et al. 2020, 2022) frame the Case III code as a "where" learned apart from the "what".

2.2 MapFormer. MapFormer's working-memory model (MapWM) rotates content queries and keys by theta_t = omega * cumsum(W_out W_in x_t), a low-rank content-to-angle map; observations learn near-zero steps and opposite actions opposite steps (their Fig. 3d). In their Table 2, one-layer MapWM with rank 2 solves 2D navigation at 128 steps, rank 1 does not, and RoPE, TAPE and CoPE with up to four layers fall short; they attribute CoPE's failure to the sign of its gate. We build on exactly this model. Our implementation reproduces their Table 2 2D result (0.989 for MapWM and 0.987 for MapEM, against 1.00 reported in v4 and 0.99 / 1.0 in v1; App. A4).

2.3 Representability. Real non-negative eigenvalues cannot track parity (Sarrof et al. 2024, Thm. 2; Grazzi et al. 2025, Thm. 1). Rotations escape this, and rotations by non-negative angles generate cyclic groups (Grazzi et al., Thm. 3; Mamba-3, Sec. 3.2). In the rotary slot the sign of a channel's step therefore does not limit what that channel can represent. Proposition 3 below extends this to our rank-limited step maps, so any difference between signed and non-negative steps in this paper is a difference in what training finds.

3. Setup and how map use is measured

Models. MapWM with one layer (up to four where stated), 2 heads, d_model 128, and step Delta = W_out W_in x through a rank-r latent. Unless stated, r = 4 and the latent is shared across heads; MapFormer's latent is per head, and per-head variants are named wherever rank is compared. *Per-head rank* is the rank of the map from token to the angles of one head. Three step parameterisations: *signed* (Delta = s = W_out W_in x), |.| (Delta = |s|, applied per channel after W_out) and *softplus* (Delta = softplus(s), so every step starts at ln 2). *Observation-step-free* variants give steps to the move tokens only. Each head has 32 two-dimensional rotary blocks (*channels*) with frequencies omega_i = 2 pi (1/Delta_max)^(i/31), Delta_max the grid size, as in MapFormer; omega is a free parameter whose sign is not constrained. The layer is otherwise a standard pre-norm transformer block (feed-forward width 4 d_model, dropout 0.1; about 204k parameters at one layer). The index baseline is RoPE with the canonical ladder base^(-2c/d), at 1-4 layers or as one block looped four times. We also pair PoPE (Gopalakrishnan et al. 2025), a rotary variant with a polar score, with its path-integrated version MapPoPE, and compare faithful CARoPE, CoPE and Selective RoPE (App. I).

Training and evaluation. AdamW (learning rate 1e-3, weight decay 0.05), 5% linear warm-up then cosine decay to 10% of the peak, 98 batches per epoch (16 walks per batch on the 1024-move torus, 128 on the ring and the 128-move torus), cross-entropy on revisited observations only (Dyck-2: learning rate 1e-4, next-token loss; App. A2). Torus

models are scored on 100 walks over one held-out map, so every torus number is transfer to a new map; ring models on 400 walks with freshly drawn maps.

Tasks. (i) *2D torus*: a 64 x 64 torus walked by runs of 1-10 repeated moves, interleaved with observations drawn from 16 objects. Walks have T = 1024 moves (2,048 tokens), 900 epochs; MapFormer's T = 128 is used for the floor analysis and the PoPE pair. (ii) *1D ring* of 32 cells, T = 128, forward-step probability p_plus (0.5 balanced; 1.0 never reverses), 300 epochs. (iii) *Dyck-2* at length 32 and nesting depth 12, scored by Hewitt et al.'s (2020) distance-averaged closing accuracy (chance 0.5). (iv) *Text world*: the torus walk rendered as English (58 words), T = 1024 words. (v) 2D and 3D tori of other sizes for rank against dimension. The loss and the accuracy are on *revisits*: observations at cells visited before, the only predictable targets.

Floors. For each task we measure the best map-free predictor: always-blank, the best n-gram over orders 0-5, and *retrace*: while the current run of moves undoes the previous run, copy the observation seen when the walk was last at the cell it is returning to. A score is evidence of a map when it clears these floors.

Strata. Revisits are split by what a map-free rule can reach. *Short-gap* revisits come fewer than 128 moves after the last visit to the cell; *long-gap* revisits come later; *wrap-only* revisits are to a cell seen before only from a different unwrapped position, a full lap of the torus away. Runs are at most 10 moves, so retrace never reaches a long-gap or wrap-only revisit; those two strata separate a map from a local rule. Figure 2 shows the strata for every model class in the paper, and we return to it throughout.

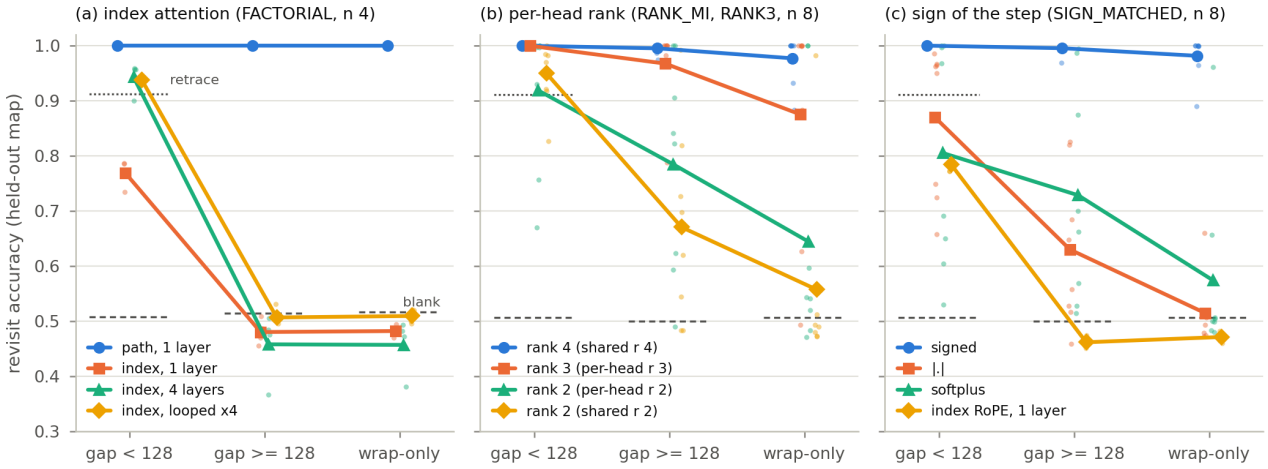


Figure 2 (`fig_strata_modes`). Revisit accuracy by stratum on the torus at T = 1024; lines: arm means, dots: runs. (a) Index attention (n = 4): depth and a loop help short-gap revisits only. (b) Per-head rank of the content-to-angle map (n = 8, same seeds and initial weights): rank 4 is near 1.000 in every stratum; rank 2 loses accuracy with the gap and most at wrap-only revisits. (c) Sign of the step (n = 8): signed near 1.000 everywhere; |.| and softplus fall from short gaps to wrap-only revisits; one-layer index RoPE for reference. Dashed: always-blank within each stratum; dotted: retrace within the short-gap stratum (0.91); floors on each batch's own walks.

Run classes and protocol. A run is *SOLVED* if its mean training loss over the final 5% of epochs is below 0.05; otherwise *STALLED* if its final-10% mean loss is within 5% of the preceding 10%, and *DESCENDING* otherwise. Every accuracy and SOLVED contrast

reported without the label "(analysis)" is a registered decision or a declared secondary of a pre-registered batch unless marked "not registered", with n = 8 seeds per arm unless stated, every model trained and tested at the same length and nesting depth, and exact permutation (accuracy) and Fisher (SOLVED) tests. Floors, strata floors and every number marked "(analysis)" were computed on stored streams and checkpoints, eval-only. App. A lists every batch with its seeds, its independent code check and its reproduction status; App. O lists every analysis script with its verification check.

4. Theory: maps and clocks in the phase

This section states what the results sections use; proofs are in App. E, and Table 1 gives the status of each statement. Notation: a channel c has angle a_c(x) = omega_c step_c(x) for token x; for a step map the latent of token x is z_x = W_in e_x and the channel's readout row is w_c, so step_c(x) = w_c . z_x before any |.| or softplus. The *usage* q_c of a channel is its mean query norm at move tokens times its mean key norm at observation tokens, normalised to sum to 1. A channel is *blind* to an inverse pair if both its angles are 0 mod 2 pi.

Proposition 1 (the prefix sum is in the phase; proved). The rotary score of channel c is Re(conj(q~*c) k~_c exp(-i omega_c (Phi*{c,t} - Phi_{c,s}))) with q~ the channel's query as a complex number. If every move token has a fixed step and every observation step is zero, theta_t = omega P_t, with P_t the unwrapped displacement after move t, and the score depends on position only through omega (P_t - P_s). One layer then compares two positions at any gap and any T, and wrapping costs nothing in a channel whose angle per move is a multiple of 2 pi / N.

Index attention sees position only through t - s, so "the moves in (s, t] sum to zero" must be computed in the residual stream. **Remark 1 (cited).** A global count gives it in two layers in principle: uniform causal attention averages the moves to P_t / n_t, an MLP rescales and takes the residue, and a second layer matches (Liu et al. 2023, Lemma 6; Yao et al. 2021). The construction needs a normaliser for the 1/n_t of causal averaging, precision 1/n_t (1/2048 per axis at the end of a torus walk), MLP width of order 4nT (64 x 2048 per axis here, against our 512), and a target whose sensitivity grows with the gap (Hahn and Rofin 2024, used for direction only). Rules that need no count, undoing the last run (retrace) or a fixed one-lap offset, are available to one layer. The measured exchange rates are therefore what training finds, not representational depth bounds.

Clock and map. For an inverse pair of moves (x, y) in channel c write the *clock* k_c = a_c(x) + a_c(y) and the *map* m_c = a_c(y). Over an interval with n_x moves x and n_y moves y, displacement d = n_y - n_x, the phase is m_c d + k_c n_x exactly. A *map code* has k_c = 0 mod 2 pi in every channel attention uses. To measure how far a trained code is from one we use a single ratio throughout the paper (Table 4, App. G3, H7): the *clock / map ratio* is the usage-weighted rms of wrap(k_c) over the usage-weighted rms of the map, with the map's size taken symmetrically over the pair, (wrap(a_c(x))^2 + wrap(a_c(y))^2) / 2, so that the readout does not depend on which move is called y; sums run over the inverse pairs and the channels of both heads (for one head, over its channels with its usage normalised to 1).

Proposition 2 (a bound on the position factor, and the tick rule; proved). (a) A zero-displacement revisit after n backward moves reads phase k_c n in channel c. If a usage share s of the channels has kappa <= |k_c| <= K, the same-cell position factor sum_c q_c cos(k_c n) is at most 1 - s (1 - cos(kappa n)) for n <= pi / K. The bound applies only while n |k| stays below pi: it says how fast a clock erodes the position match of a revisit, not that every clocked code fails at long gaps (by (b), a per-move tick of pi is invisible to revisits on the even torus). Index RoPE is the extreme case, in which every token advances every channel. (b) More generally, if the target is the key at the last visit to the same cell and returns come at lags with greatest common divisor g (g = 2 on the even torus), a channel can serve place-keyed retrieval only if its per-move *tick* is 0 mod 2 pi / g.

Proposition 3 (representability by wrapping; proved). If at most r tokens step, a |.| model of rank r reproduces any signed angle table exactly mod 2 pi; annihilating the observation embeddings in W_in leaves only the moves stepping, so every |.| model in this paper can represent the map. Softplus and exp can too: with the observations annihilated each steps softplus(0) = ln 2 or exp(0) = 1, and omega' = 2 pi / ln 2 or 2 pi makes that step a full turn.

Lemma 1 (relative-sign gauge; proved). Negating a channel's steps (through omega_c) together with the second coordinate of that channel's query and key projections leaves every attention score unchanged, since J R(phi) J = R(-phi) for J = diag(1, -1). A non-negative step therefore fixes only the relative signs of the tokens that step in a channel. For an inverse pair it removes cancellation through zero, angles (a, -a), while the wrapped code (a, 2 pi - a), the same function, remains.

Proposition 4 (route inventory; proved). (a) *Signed, collective route.* The clock vector is k = diag(omega) W u with u = z_x + z_y. The set {u = 0} cancels every channel at once and leaves the map free; gradient flow on a quadratic clock surrogate converges to it from every start; for almost every readout it is the only latent set that cancels every channel. (b) |.| *has neither route from a small start.* The same latent move ends at a(y) = a(x), a clock. At a generic readout no latent choice cancels more than 2r channels non-blindly. Per channel, an unwrapped |.| pair has |k_c| >= |m_c|, and its non-blind zero-clock points lie at |k_c| = 2 pi n, across |k_c| = pi from a small start. (c) *Rank.* Let c be the latent step common to all moves (with the mean observation step added when observations step) and U the map frame, the half-differences of opposite moves' latents. At per-head rank D, c lies in the plane U spans, c = U beta for a phantom displacement beta, so a channel is clock-free only by losing resolution along beta; above rank D the part of c outside the map plane can be cancelled channel by channel with the map untouched (*parking*). (d) *One dimension.* At D = 1 no readout change alters the drift-to-map ratio; only the collective route acts.

Proposition 5 (hiding; proved to first order) and Conjecture 1. At rank D >= 2 with isotropic resolution and a small drift, rotating a channel's resolution off the drift direction lowers the drift loss at zero first-order cost to the map (*hiding*). The collective route removes the drift at a rate proportional to the head's resolution along the drift direction, which hiding lowers. *Conjecture 1:* this feedback keeps the drift in place at rank D >= 2; in 1D there is nothing to hide along, which fits rank 1 learning the ring.

Remark 2 (exists, kept, not found). Let S be the solution set of a class and Bas the set of starting points from which training reaches S within the budget with probability at least one half. "A solution exists and an installed solution is kept, but training from scratch does not find it" says that S is non-empty, S lies in Bas, and the initialisation distribution puts little mass on Bas: the obstruction is where training starts, not what the class contains.

Proposition 6 (readers and codes; proved). In one layer, the query and key at a scored revisit depend on token identity only, so the model is a position-only reader of its own code. MapFormer computes theta once from the embeddings, so a loop or extra layers widen the reader and never change the code family (the set of codes the step map can produce). Any score of the form sum_c A_c cos(dphi_c + beta_c) splits target coherence into a code term C0 (the A-weighted cos dphi on the target) and a readout term C - C0; a score rule acts only on the second.

Proposition 7 (peak pinning; proved given delta = 0). PoPE's score sets beta_c = -delta_c with delta content-free and, in trained models, in its clamp's dead zone (delta = 0). Then every head's kernel, for every content pair, is a non-negative combination of cosines of dphi and peaks at dphi = 0. With a path phase dphi = 0 is the same place, the revisit target. With an index phase it is the current token, and with PoPE's frequencies (at most 1 rad per token) every content pair's kernel decreases over offsets 0-3, so no fixed far offset can be preferred in one layer.

Table 1. Status of the theory (proofs in App. E).

statement	status	checked against
Prop. 1, prefix sum in the phase	proved	same-cell attention of path models at long gaps (Sec. 6)
Remark 1, two-layer count	cited (Liu et al. Lemma 6; Yao et al.)	probes: no count in torus index models (Sec. 6)
Prop. 2, position-factor bound and tick rule	proved	bound: consistent with the strata of every failing model (Fig. 2), not a test of them; tick rule: text-world channels (Sec. 8)
Prop. 3 and Lemma 1, representability and gauge	proved	constructions exact to 1.4e-6 rad (App. D)
Prop. 4, route inventory	proved	end-state clocks (Sec. 7.1); frozen latents; hybrid readout (Sec. 7.3)
Prop. 5, hiding at first order	proved under small drift, isotropy, near-isometry	drift on the least-resolved axis (Sec. 7.2)
Conjecture 1, hiding starves the collective route	conjecture	consistent with rank 1 learning the ring; not tested
Remark 2, basins	definitional	existence and continuation batches (Secs. 7.2-7.3)
Prop. 6, readers and codes	proved	one-pass reader fits (Sec. 7.4)
Prop. 7, peak pinning	proved given delta = 0	delta-freedom batch (Sec. 7.4)
Prop. 8, CARoPE cannot cancel	proved	trained CARoPE models (Sec. 7.3)

5. One path layer solves the long-walk map

Short walks mostly do not need a map (analysis). On MapFormer's T = 128 torus walks retrace scores 0.934 (blank 0.507, n-gram 0.598), and no revisit is wrap-only. The walk moves in runs, and a run that reverses the previous one returns, step by step, to cells seen moments earlier; over 128 moves nearly every revisit is of this local kind. At T = 1024 retrace falls to 0.843 (blank 0.506, n-gram 0.575) and 7.9% of revisits are wrap-only (Table 2). At T = 128 the path model still beats index attention (path 0.971 at rank 2 and 1.000 at rank 4, index 0.805; position effect +0.243 at rank 2), but index RoPE falls below retrace there and the rank-2 path model clears it by only 0.04.

Table 2. Torus floors (analysis, on the same evaluation walks as the models) and accuracies at MapFormer's walk length and at ours (held-out map; n = 8 per model).

T (moves)	wrap-only revisits	blank	n-gram	retrace	index RoPE, 1 layer	path, rank 2	path, rank 4
128	0.000	0.507	0.598	0.934	0.805	0.971	1.000
1024	0.079	0.506	0.575	0.843	0.731	0.894	0.998

One path layer solves the long-walk map; index attention does not. On the torus at T = 1024 one path layer scores 0.998 +/- 0.004 (8/8 SOLVED) against 0.731 +/- 0.008 for index RoPE (0/8), a difference of +0.267 (permutation and Fisher p 0.0002). Index RoPE sits below retrace on every run (0.72-0.74; analysis): it does not learn even the map-free rule. Two further registered batches give the same contrast: 1.000 against 0.722 (n = 8) and 1.000 against 0.719 (n = 4; Fig. 3c). As Proposition 1 says, the path model's score depends on P_t - P_s directly, and it uses that at every gap.

Where the difference lives. By stratum (short gap / long gap / wrap-only), the path model scores 1.000 / 0.995 / 0.982 and one-layer index RoPE 0.784 / 0.462 / 0.471, against stratum floors of 0.506 / 0.500 / 0.507 and retrace 0.911 on short gaps (analysis) (Fig. 2c). Index attention gains its margin over the constant on short gaps, where retrace does better still; long-gap and wrap-only revisits require the map, and only the path model has it.

6. What index depth buys

Depth is the obvious way to give index attention more computation, so we ask how many index layers one path layer is worth, with both trained and tested at the same length and nesting depth.

Dyck-2: about three index layers. A bracket string is a walk on a stack: an opening bracket steps forward, its closing partner steps back, and MapFormer's models encode the two as opposite directions. At length 32 and nesting depth 12 one path layer scores 0.996 (1.000 at 2-4 layers), and index attention 0.648 / 0.889 / 0.964 / 0.982 at 1-4 layers, against a best stack-free floor of 0.594 (Fig. 3a). The position main effect, averaged over two encodings (MapWM vs RoPE and MapPoPE vs PoPE; App. F), is +0.353 / +0.130 / +0.045 / +0.024 at 1-4 layers, positive on 8/8 seeds at each depth (sign-flip p 0.0078). With three times the budget four index layers reach the ceiling (0.997-0.998), and training on a mixture of depths 4-12 keeps an effect of +0.110 at depth 12 (8/8). The

index arms are not held back by their frequency ladder: base 32 and base 10000 agree to 0.001 at four layers.

A biased 1D walk: the same rate. The ring dials how often a step is undone: at p_plus 0.5 the walk is balanced, at 0.9 one segment in ten runs backwards, and at 1.0 the walk never reverses and position is a count. On fresh seeds (n = 6) one path layer scores 1.000 at every bias, and index attention with 1 / 2 / 3 layers 0.710 / 0.949 / 0.998 at p_plus 0.5, 0.677 / 0.962 / 0.998 at 0.75 and 0.824 / 0.987 / 0.999 at 0.9 (Fig. 3b; declared secondary). Wherever a step can be undone, three index layers come within 0.002 of the path layer; at p_plus 1.0 a single index layer reaches 1.000.

The 2D torus: more than four. At T = 1024 (n = 4) one path layer scores 1.000 (4/4 SOLVED) against index attention at one layer 0.719, at four real layers 0.860 and as one block looped four times 0.864 (each 0/4; complete separations, p 0.029) (Fig. 3c). Depth and the loop each help (+0.140 and +0.145), and the loop matches four real layers with a quarter of the parameters (203,925 against 798,741). Four layers or a loop lift short-gap revisits from 0.769 to 0.943 / 0.938 and leave long-gap (0.458 / 0.507) and wrap-only (0.457 / 0.509) revisits at the constant floor (0.515 / 0.517; Fig. 2a).

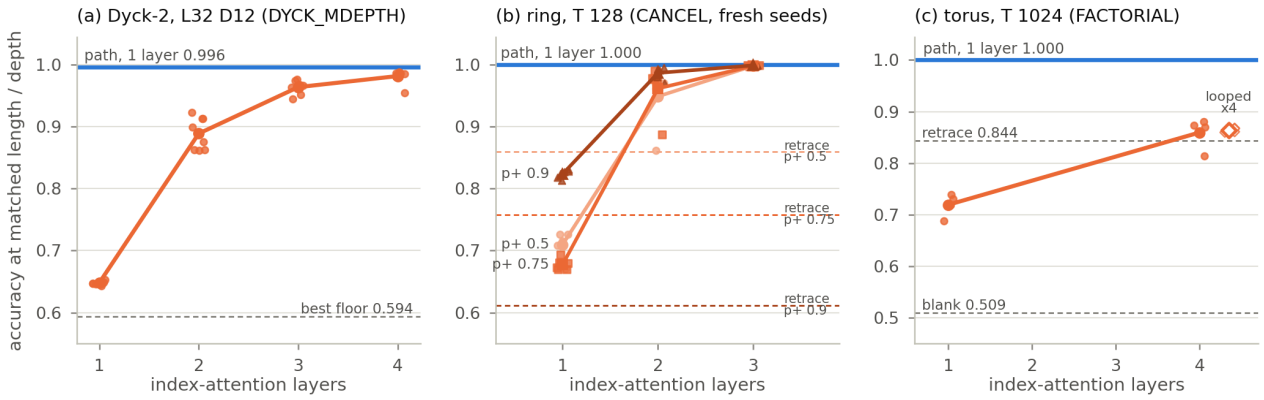


Figure 3 (`fig_exchange_rate`). Accuracy of index-RoPE attention against its number of layers (orange; dots: runs, line: means) beside one layer of path integration (blue line), each trained and tested at the same length and depth. (a) Dyck-2 at length 32 and depth 12 (Hewitt closing accuracy; n = 8): index attention needs about three layers to approach one path layer (0.964 at 3, 0.982 at 4, against 0.996). (b) Ring of 32 cells at T = 128 for three move biases (n = 6 fresh seeds; declared secondary): three index layers reach one path layer (0.998-0.999 against 1.000). (c) Torus at T = 1024 (n = 4): four index layers (0.860) or one layer looped four times (0.864) stay far below one path layer (1.000), just above this batch's retrace floor (0.844; the 0.843 of Table 2 is the floor on the sign batch's walks). Dashed lines: map-free floors (analysis; Dyck: the best stack-free floor; ring: retrace per bias, on another stream of the same task; torus: always-blank and retrace on this batch's own walks).

Why the rate differs by task. A revisit asks for an earlier observation at the same cell, that is, for a key whose moves since then sum to zero. Remark 1 offers index attention a global count or rules that need none. The stored models show that training finds only the second kind, with a reach that differs by task (analysis; Table 3, App. F). One index layer learns retrace and the one-lap offset: on the ring it scores 0.68-0.80 on retraces, 0.99-1.00 on one-lap returns once they are common, and the floor on everything else at

p_plus 0.5 and 0.75 (0.487 against 0.491-0.493; at 0.9 an unexplained 0.592 against 0.502), so at p_plus 1.0, where every revisit is a one-lap return, it matches the path layer. Three index layers on the ring are flat in the number of direction runs a revisit spans (0.997 / 0.993 / 0.972 at 3-4 / 5-8 / 9-16 runs, wrap-only revisits included) and carry a displacement code over the last ~64 moves (linear-probe R^2 0.72 for the displacement over the last 64 moves, read at moves 96-127); displacement older than 64 moves is not decodable (R^2 0.01), so no global count is detectable on the ring either, and on 128-move walks a 64-move window covers most revisits. On the torus no index model carries a count (R^2 at most 0.21). Four layers or the loop solve retraces (0.996 / 1.000) and decay with the number of runs spanned, at a fixed gap as well as overall: at gaps of 32-127 moves four layers score 0.874 at 2 runs and 0.493 at 17 or more (seed means). Their last layer puts 0.55-0.58 of its attention on the remembered cell for retraces and 0.00-0.01 at long gaps, where the path layer puts 0.81. Nearly all long-gap revisits span 17 or more runs (98.5%; median 37), beyond that reach. Short-gap accuracy decomposes over these classes: the retrace floor is (0.680 x 1 + 0.148 x 0.5) / 0.828 = 0.911, and four layers give (0.680 x 0.996 + 0.148 x 0.699) / 0.828 = 0.943, so depth adds the exact retrace rule plus partial returns over 3-8 runs.

Table 3. Torus at T = 1024: accuracy by revisit class (analysis; FACTORIAL models, n = 4, eval mode; "runs" = maximal runs of one move between the two visits; share = fraction of all revisits; class contrasts at n = 4 carry no test).

revisit class	share	index 1 layer	index 4 layers	index loop x4	path 1 layer
short gap, retrace	0.680	0.833	0.996	1.000	1.000
short gap, not retrace: 3-4 runs	0.031	0.477	0.926	0.977	1.000
... 5-8 runs	0.056	0.476	0.735	0.645	1.000
... 9-16 runs	0.052	0.474	0.561	0.499	1.000
... 17+ runs	0.009	0.472	0.493	0.502	1.000
long gap (>= 128 moves)	0.090	0.480	0.458	0.507	1.000
wrap-only	0.083	0.482	0.457	0.509	1.000

So depth buys index attention a local reach measured in direction runs, and the torus rate is "more than four" because its long-range revisits lie beyond that reach and the global count is not found. The same four passes do something else on a rank-2 path code: they lift long-gap revisits from 0.671 to 0.927-0.990 and wrap-only revisits from 0.557 to 0.747-0.887 (Sec. 7.4). Extra passes add accuracy where the code carries position information that one pass cannot read, and none where it carries none.

7. What decides whether training finds the map

Two choices of the content-to-angle map decide whether the torus map is learned: its per-head rank and the sign parameterisation of its step. For each, the constrained class fails from scratch, a solution in the class exists and is constructed explicitly, and that solution, installed and trained, is kept or re-found. Section 4 says what to look for: whether training removes the clock, and by which route.

7.1 The clock and the signature

Training from a small start ends in one of two places on either axis (Table 4; analysis; the clock / map ratio of Sec. 4). The models that succeed end with clock-free heads: the signed and rank-3/4 arms at arm means 0.001-0.051, with no head above 0.14, and every solved rank-2 run with at least one clean head (0.03-0.07; two of the four keep a drifting second head, 0.54 and 0.68). The models that fail end with a clock of the order of their map: the non-negative arms at 1.25-1.57 (arm means; for an unwrapped |.| channel a ratio of at least 1 holds by construction, so the informative part is that the channels did not wrap) and the unsolved rank-2 runs at 0.54-0.69 (medians 0.58-0.79). The exceptions are informative too: 4 of the 20 unsolved rank-2 runs end with clock-free heads whose two map axes have collapsed onto one direction (a map that has lost a dimension; App. G3), and one more sits just above the SOLVED threshold (final loss 0.051). The measured strata show the signature that Proposition 2's bound leads one to expect: shared rank 2 scores 0.950 / 0.671 / 0.557 on short-gap / long-gap / wrap-only revisits and |.| 0.870 / 0.630 / 0.515, against 1.000 / 0.995 / 0.982 for the signed rank-4 model (Fig. 2b-c). The bound alone does not predict failure: two signed torus runs started with angle sums past pi solve with clock / map 1.39 and 1.78, with 0.43 and 0.99 of their usage on channels whose per-pair clock is near pi, where the bound says nothing (App. H7). Weighted by usage, signed steps cancel opposite moves mod 2 pi in 70-93% of channels and |.| steps in 2-10% (analysis; App. H6).

Table 4. Clock at the end of training (analysis; clock / map ratio of Sec. 4, one value per run, arm means unless marked; SOLVED of 8). For unwrapped |.| channels the ratio is at least 1 by construction; the informative rows are the frozen-latent signed arm and the rank arms.

arm	SOLVED	clock / map
ring, signed (three batches)	8/8, 7/8, 8/8	0.015-0.018
ring, |.| from small starts (three batches)	0/8, 0/8, 1/8	1.47-1.57
ring, signed with move latents frozen, |.| start	1/8	1.18 (the solved run 0.06)
ring, |.| from a start past pi	8/8	0.32 (wrapped; 0.82 of usage on cancelling channels)
torus, signed / |.|, observation-step-free	8/8 / 1/8	0.001 / 1.25
torus, rank 2, unsolved runs (three arms)	0/8, 2/8, 2/8	0.54-0.69 (medians 0.58-0.79); 4 of 20 runs collapsed, clock-free
torus, rank 2, solved runs (4): six clean heads / two drifting heads (per head)	4 runs	0.03-0.07 / 0.54, 0.68
torus, rank 3 and rank 4, solved runs	6/8, 8/8, 8/8	0.043-0.051

7.2 Per-head rank of the content-to-angle map

Per-head rank decides it. On the torus at T = 1024 we compare six bottlenecks, all built from the same initial weights at each seed (Table 5, Fig. 4a). Every arm whose heads each read a rank-2 map solves 0-2 of 8; rank 3 solves 6/8 and rank 4 8/8, whether the latent is shared or per head. At matched sharing (per-head rank 4 against a block-

diagonal rank-4 latent in which each head reads rank 2) the split is 8/8 against 2/8 (Fisher and permutation p 0.0070). Per-head rank 3 improves on per-head rank 2 by +0.102 in accuracy (permutation p 0.027, Holm 0.054). Total latent size does not track success: the per-head rank-2 arm and the shared rank-4 arm both have four latent dimensions and land on opposite sides. Nor does the initial angle scale: two rank-2 arms start at the same scale as rank 4 and fail. Most rank-2 runs flatten at losses of 0.2-0.5, while the rank-3 and rank-4 medians cross the SOLVED threshold at epochs 478-532 (App. G). With MapFormer's Table 2, the picture is: rank 1 cannot represent a 2D map; rank 2 can, and learns 128-step walks; at 1024 moves reliable learning needs per-head rank above the map's dimension.

Table 5. SOLVED (of 8) and held-out accuracy at T = 1024 (mean over 8 seeds) by per-head rank; torus, 900 epochs, every arm built from the same initial weights at each seed.

per-head rank	arm	SOLVED	accuracy
2	one rank-2 latent shared by the heads	0/8	0.894
2	per-head rank-2 latents	2/8 (1/6 on seeds 2-7)	0.885
2	block-diagonal rank 4 (each head reads rank 2)	2/8	0.948
3	per-head rank 3	6/8	0.987
4	shared rank 4	8/8	0.998
4	per-head rank 4	8/8	0.999

Twice the budget does not rescue rank 2. Trained for 1800 epochs from scratch, rank 2 solves 0/8 (0.908) against rank 4's 7/8 (Fisher p 0.0014; accuracy +0.086, permutation p 0.0003). PoPE's polar score does not rescue it either: 3/12 SOLVED under either score at T = 1024, with rank 4 at 8/8 under both (Sec. 7.4 gives the reason).

The rank-2 solution exists and is held. The rank-2 projection of each solved rank-4 model scores 0.9955 frozen (at least 0.981 on every seed). Trained from there it solves 7/8, as many as the rank-4 control trained from its own solution (7/8), against 1/8 for rank 2 trained further from its own stalled weights (accuracy p 0.0009) (Fig. 5a). A rank-2 map that solves the torus exists, is kept by training, and is not reached from scratch.

Rank against the map's dimension. Rank equal to the map's dimension D is hard in 2D and 3D and not in 1D (Fig. 4b). On a 2D torus of 32 cells per side rank 2 solves 1/8 against rank 3's 8/8 (Fisher p 0.0014). On a 3D torus of 10 cells per side rank 3 solves 1/8 (rank 4, 4/8), and rank 4 solves an 18-per-side 3D torus 8/8 (1.000). On the 1D ring rank 1 solves 7/8 against rank 2's 8/8, and with the drift forbidden by construction rank 1 solves 8/8 in a median of 24 epochs instead of 58.

Why the extra rank helps. Proposition 4(c) gives a channel above rank D a map-free way to stop reading the clock. The stored checkpoints are consistent with it but do not discriminate it from plain removal of the clock through the latent (analysis; App. G4). Here the clock is c = a + o, the mean move latent a plus the mean observation latent o, the drift a walk accumulates per move and observation. In solved heads of rank above D, a and o are each 0.02-0.04 of a map step and point in opposite directions (cosine -0.79 to -0.999), so they cancel; at T = 1024 both lie mostly in the map plane. What is left of c (0.001-0.025 of a map step) lies almost entirely outside the plane: off-plane share 0.96-1.00 in nine of ten sets of heads, against 0.40-0.71 for a random direction (median),

read at 0.09-0.27 of the map's gain. Parking and removal through the latent would both leave such a residual. Drifting rank-D heads drift along their least-resolved axis (50 of 55 runs, counting each network once, against 25.1 expected for a random direction), the geometry Proposition 5 leads to. A fresh rank-4 step map trained on the stalled rank-2 networks removes the drift in 16 of 16 heads: its latent clock shrinks 14-85x in absolute size, and the residual (0.002-0.005 of a map step) lies off the plane (share 0.91-1.00). At rank 1 on the ring, Proposition 4(d) leaves only the collective route, and Conjecture 1 has nothing to starve: rank 1 learns.

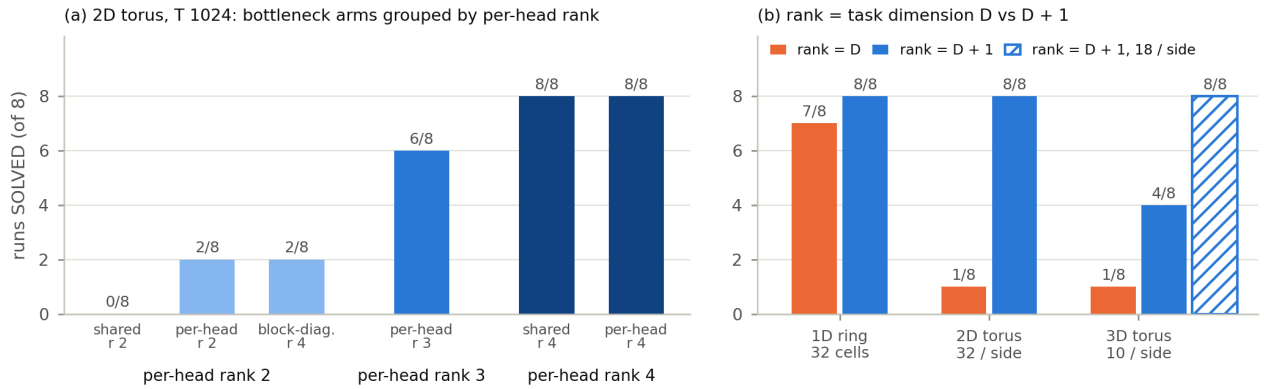


Figure 4 (`fig_rank_solved`). Runs (of 8) that solve the map, by the per-head rank of the content-to-angle map. (a) Torus, T = 1024, six bottlenecks built from the same initial weights: every arm whose heads each read a rank-2 map solves 0-2/8, rank 3 solves 6/8 and rank 4 8/8, whether the latent is shared or per head. (b) Rank equal to the task's dimension D against D + 1 (separate batches and recipes): on the 1D ring rank 1 solves 7/8 (rank 2, 8/8); on a 2D torus rank 2 solves 1/8 (rank 3, 8/8); on a small 3D torus (10 per side) rank 3 solves 1/8 and rank 4 4/8, and rank 4 solves 8/8 on an 18-per-side torus with few wrap-only revisits (hatched). Torus budget 900 epochs; at 1800 epochs rank 2 still solves 0/8.

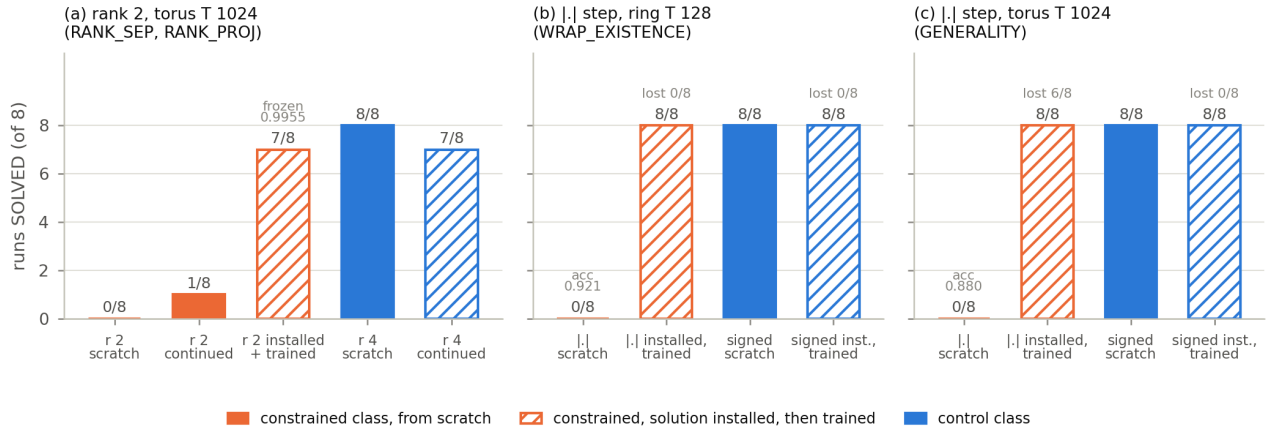


Figure 5 (`fig_existence`). A solution exists in each constrained class and training keeps or re-finds it, but training from scratch does not find it. Bars: runs SOLVED of 8 (hatched: the solution was installed and then trained). (a) Rank 2 on the torus: a rank-2 projection of a solved rank-4 model scores 0.9955 frozen and is held when trained (7/8), while rank 2 from scratch solves 0/8, and 1/8 when the stalled runs are trained further. (b) The |.| step on the ring: an installed exact non-negative solution is held under training (8/8

SOLVED, lost on 0/8), while |.| from scratch solves 0/8 against signed 8/8. (c) The |.| step on the torus: the installed solution is perturbed (loss above 0.05 on 6/8) and re-found (8/8 SOLVED); from scratch 0/8 against 8/8. "Lost": the maximum per-epoch training loss after the start exceeds 0.3, or exceeds 10x the signed control's maximum at that seed and 0.05.

7.3 The sign of the phase step

Non-negative steps rarely learn the map from scratch. On the torus at T = 1024 the |.| step scores 0.821 against 0.998 for the signed step, -0.177 (permutation p 0.0002), and solves 0/8 against 8/8; softplus scores 0.781 and solves 1/8 (-0.218, p 0.0003). On the ring |.| solves 2/8 and 1/8 against 8/8 at p_plus 0.5 and 0.75 (Fisher p 0.007 and 0.0014). Observation-step-free models show the same: 0/8 against 8/8 on the ring and on the torus (0.880 against 1.000). The training curves separate (Fig. 6): the signed median crosses the SOLVED threshold at epoch 480 / 57 / 309 / 183 in four batches and keeps descending, while most |.| runs flatten at a loss of 0.3-0.5.

The cost holds at each step's best learning rate, without weight decay and at 2 layers d_model 256 (Table 6). With each step at its own best of four learning rates on the ring (the |.| optimum, 3e-3, is bracketed by the grid), |.| solves 2/8 against signed 8/8 (selection-aware p 0.0070 on SOLVED and 0.0057 on accuracy). Without weight decay the ring gives 0/8 against 8/8 (-0.127); at a third of the learning rate, 0/8 against 6/8. A 2-layer, d_model 256 torus model with eight times the parameters gives 2/8 against 8/8 (0.967 against 0.999, p 0.0005). Across the learning-rate grid, higher rates hurt the signed step first (8, 8, 4 and 1 of 8 solved at 1e-3, 3e-3, 1e-2 and 3e-2) and do not help |.| past 3e-3 (1, 2, 1 and 0 of 8). Two further non-negative increments behave alike at the two reference rates: exp, the LRU parameterisation, and softplus, both of which start every token at the same positive step, solve 1/8 to 3/8 against signed 8/8 (App. H).

Table 6. The sign cost by task, model and recipe (observation-step-free models unless marked "full"; n = 8 per arm).

task and model	recipe	|.| accuracy, SOLVED	signed accuracy, SOLVED	test
torus, 1 layer, full	default (lr 1e-3, weight decay 0.05)	0.821, 0/8	0.998, 8/8	perm p 0.0002; Fisher 0.0002
torus, 1 layer	default	0.880, 0/8	1.000, 8/8	perm p 0.0002; Fisher 0.0002
torus, 2 layers, d_model 256	default	0.967, 2/8	0.999, 8/8	perm p 0.0005; Fisher 0.0070
ring, full, p_plus 0.5	default	0.973, 2/8	1.000, 8/8	Fisher 0.0070
ring	default	0.911, 1/8	1.000, 8/8	perm p 0.0002; Fisher 0.0014
ring	weight decay 0	0.873, 0/8	1.000, 8/8	perm p 0.0002; Fisher 0.0002
ring	lr / 3 (3.3e-4)	0.841, 0/8	0.950, 6/8	Fisher 0.0070
ring		0.963, 2/8	1.000, 8/8	

task and model	recipe	|.| accuracy, SOLVED	signed accuracy, SOLVED	test
	each step at its best of four rates (|.| 3e-3, signed 1e-3)			selection-aware p 0.0070 (SOLVED), 0.0057 (accuracy)

The non-negative solution exists, and training keeps or re-finds it. Proposition 3 constructs it: each solved signed observation-step-free model is converted exactly into |.| form, with the move latents re-solved to orthogonal directions, each channel's frequency set to max(|omega|, 1) and each step to the signed angle reduced mod 2 pi. Angles are unchanged mod 2 pi (error at most 1.4e-6 rad), frozen accuracy equals the source's on 8/8 seeds, and the weights stay within about 5x of the signed model's, so the test is posed at the scale training uses (App. D). On the ring the constructed |.| model, trained for 300 more epochs, stays solved on 8/8 seeds and never leaves the solved region, keeping its cancelling code in the channels attention uses (98%, usage-weighted). On the torus, trained at learning rate 1e-4, it is perturbed early (peak loss 0.11-0.21), re-enters the solved region for good between epochs 119 and 204, and ends solved on 8/8 seeds at accuracy 1.000. From scratch the same class solves 0/8 on both tasks (Fig. 5b-c).

Why the non-negative step does not find it. Proposition 4(b) says the |.| class lacks both routes from a small start, and the stored models show the consequences (analysis; App. H). At initialisation a latent move produces exactly the change of angle sums the signed step needs (share 1.000) but only 0.34 of what |.| needs (0.56 usage-weighted); the rest must come from per-channel readout changes. With the content of a solved model held fixed, the loss along the straight path from the start to the |.| solution rises above its running minimum by 0.6-1.9 nats on every seed, against 0.04-0.48 for the signed path (complete separation). Rotating one move latent toward the negative of the other, with nothing else changed, takes the signed model from 0.30 to 0.72 accuracy and leaves the |.| model at 0.30, since under |.| that move makes a clock. The cone of Proposition 4(b) is where trained |.| ring models end: 0.25-0.36 of their sub-pi channels lie within 10% of its boundary, where one move's step is zero, against 0.056 at initialisation. In full training the content co-adapts along the collective route; with the content frozen, the step map alone rarely takes it from the same-direction start (exploratory: these CPU runs are not deterministic, and single runs moved by up to 0.43 in accuracy on re-runs; App. H7).

The cost is carried by the increment, not the readout. A hybrid whose readout matrix is confined to the non-negative orthant, a_c(x) = omega_c |w_c| . z_x, but whose latent and hence increment are signed, keeps the collective route. On the ring it solves 8/8 at both reference rates, at accuracy 1.000 on every seed, exactly as signed, against |.| 0/8 and 2/8 (Fisher p 0.0002 and 0.007), and ends where Proposition 4(a) says: antipodal move latents (median |z_x + z_y| / (|z_x| + |z_y|) 0.043 and 0.002; cosine -0.996 and -1.000). A |.| control with the move latents tied antipodal by construction, whose two steps are then equal and whose code is a pure clock, solves 0/16. Non-negativity in the readout costs nothing; a non-negative increment, which removes both the collective route and the near-side cancelling point, is what costs.

How signed steps reach the map. Started from the |.| start, with both moves rotating the same way, signed steps still solve 7/8 on the ring (|.| from the same start: 0/8; Fisher

p 0.0014). They end with 98-99% of channels opposite-signed (analysis): they cancel opposite moves through zero, which Lemma 1 shows the non-negative step excludes for an inverse pair (Fig. 7). The escape is collective. Freezing the two move latents, with W_out and omega still free, blocks it: 1/8 against 7/8 (Fisher p 0.010), and the frozen-latent model ends where the non-negative step ends (clock / map 1.18, Table 4). With the latents free, one move's latent moves toward the negative of the other's and every channel cancels together (82% of channels end opposite-signed, against 15% with the latents frozen). MapFormer described this antipodal end state (their Figs. 3d and 9b); this is the causal test of how it is reached. A |.| step has only the wrapped branch, and on the ring it learns the map when its opposite-move angles start with sums past pi (8/8, cancelling by wrapping; Fig. 7d).

Published content phases also miss the torus map. Faithful CARoPE cannot represent cancellation. *Proposition 8 (proved):* CARoPE advances every channel by an amount in (0, 1] per token, so the shortest return's phase lies in (0, 4], never 0 mod 2 pi, and every channel is either a clock or blind over an excursion (App. E5). It scores 0.667 on the torus (0/8), below index RoPE's 0.722 (p 0.040). Faithful CoPE can represent ring cancellation (we construct the solution; App. I) yet solves 0/8 on the ring (0.887 against signed 1.000, p 0.0002) while beating index RoPE by +0.157, and it fails a 256-cell line that never wraps (1/16 against signed 8/8), so its obstacle is not what it can represent. Faithful Selective RoPE, a signed content-dependent phase in its authors' best language-model configuration, builds a phase it relies on (it beats index RoPE by +0.236, and zeroing its angle drops it from 0.961 to 0.507) but does not learn the torus map within 900 epochs: 0/8 SOLVED against MapWM's 8/8 in the same batch (0.961 against 1.000, p 0.0002), with every run still descending. Its readouts show why the map is harder for it: every token steps the phase, and an observation steps 0.86-0.96 as far as a move (MapWM: 0.027), so what is seen moves where the model thinks it is. Its state-tracking configuration carries a positive bias that keeps every learned step non-negative, and it fails as the non-negative steps do (0.769, 0/8).

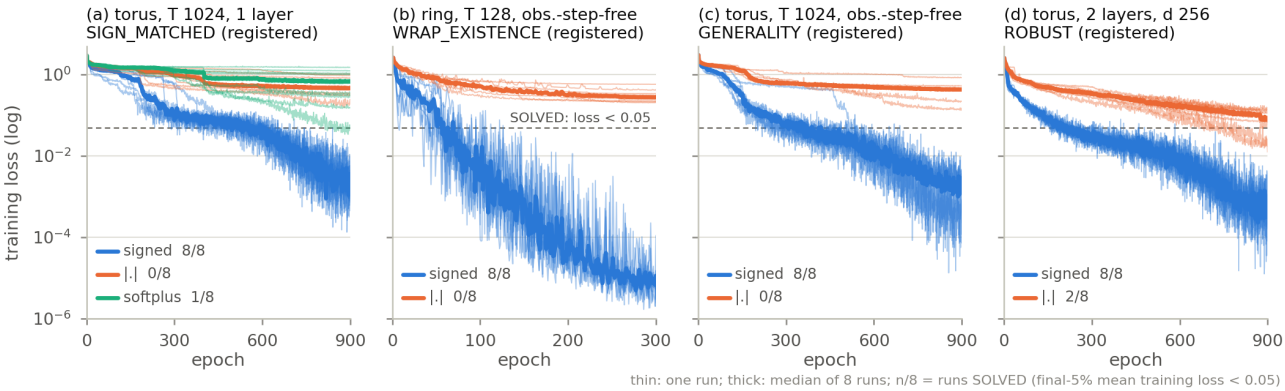


Figure 6 (`fig_loss_curves`). Training loss per epoch for the signed step (blue) and the non-negative steps |.| (orange) and softplus (green), every run of four batches (thin lines; thick: median of 8 runs; log scale; training loss with dropout on). Dashed line: the SOLVED threshold (final-5% mean training loss below 0.05); n/8 = runs SOLVED. (a) Torus, T = 1024, one layer: signed 8/8, |.| 0/8, softplus 1/8. (b) Ring of 32 cells, observation-step-free: 8/8 against 0/8. (c) Torus, observation-step-free: 8/8 against 0/8. (d) As (c) at two layers and d_model 256: 8/8 against 2/8. The signed median crosses the

threshold at epoch 480 / 57 / 309 / 183 of 900 / 300 / 900 / 900 and keeps descending; the |.| median never does.

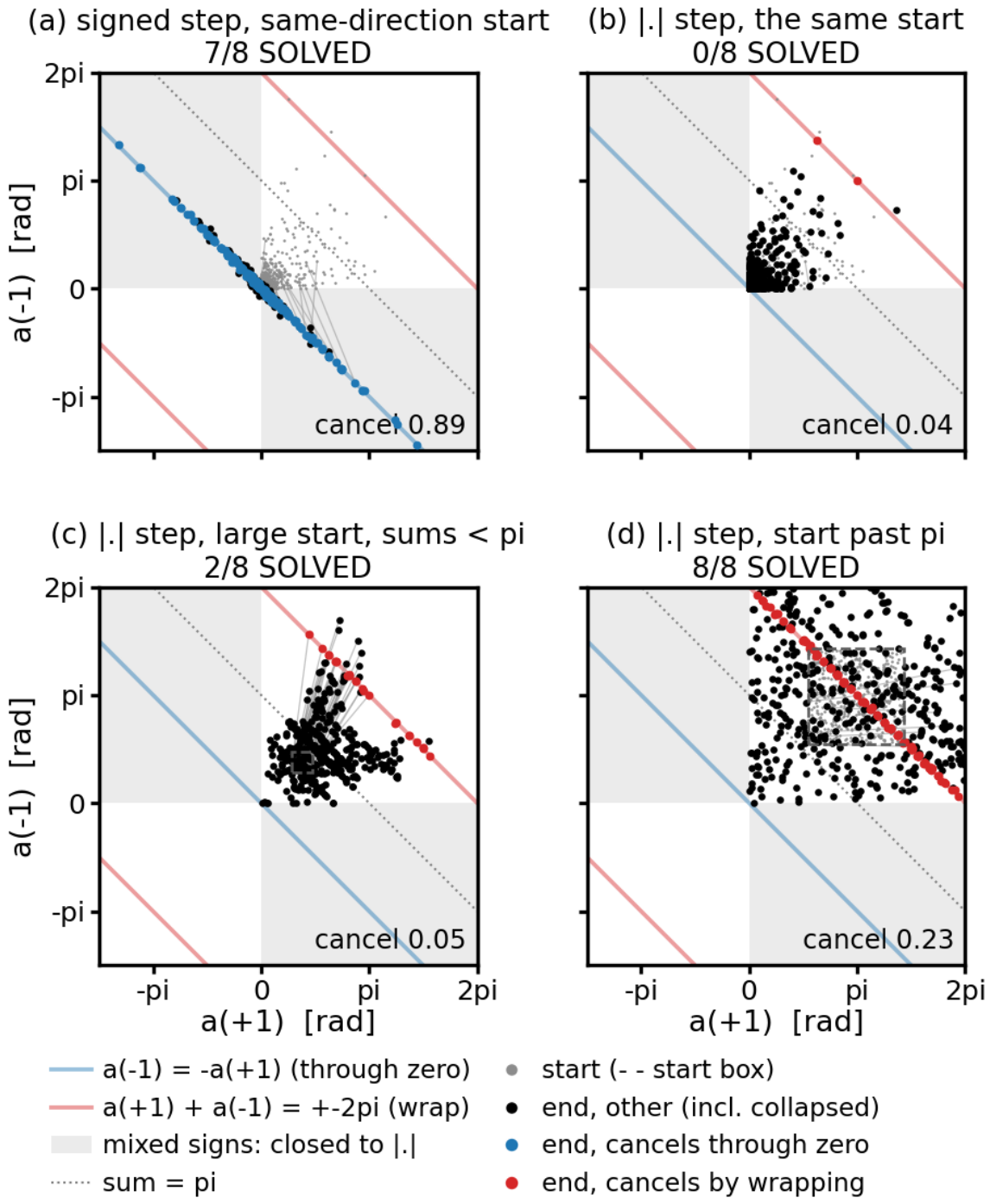


Figure 7 (`fig_mechanism`; ring, 8 runs per panel; analysis of a registered batch). Move angles (a(+1), a(-1)) of every rotary channel at the start (grey) and after training (dots). A channel cancels opposite moves when a(+1) + a(-1) = 0 mod 2 pi: on the blue line (through zero, reached only with opposite signs) or the red line (by wrapping). Grey shading marks the two mixed-sign quadrants, the only region the |.| step cannot reach (omega's sign is free, so both angles may be negative); dotted: a(+1) + a(-1) = pi; dashed boxes: start ranges. (a) Signed step from the same-sign start: 7/8 runs solve; one move's angle crosses zero, and 89% of channels cancel at the end (6% at the start). (b) The |.| step from the same start: 0/8 solve, 4% cancelling. (c) The |.| step from a large start

whose sums stay below pi: 2/8 solve, 5% cancelling, all by wrapping. (d) The |.| step from a large start whose sums exceed pi: 8/8 solve; cancelling rises from 7% to 23% of channels, all by wrapping (App. H).

7.4 Readers cannot repair a code

More computation reads a rank-2 code; it does not repair it. Looping the single block four times (identical parameters and initialisation) or using four real layers raises rank-2 accuracy from 0.894 to 0.973 and 0.990 (permutation p 0.0034 and 0.0012; SOLVED 2/8 and 5/8 against 0/8; rank 4 0.998, 8/8). By Proposition 6 neither change alters the code family; both change how a code is read. Fitting a one-pass, position-only reader to each model's own phase recovers the one-layer models' accuracy (rank 2 0.866 against 0.894; rank 4 1.000 against 0.998) but falls 0.32-0.44 short of the looped and deep models on long-gap and wrap-only revisits (analysis; App. M). Their codes are not measurably more readable by one pass than plain rank 2's (0.893 and 0.897 against 0.866, below minimum detectable effects of 0.07-0.08) and stay far from rank 4's (p 0.0002). Run with one pass, the looped weights score 0.381 (mean; 7 of 8 seeds below the 0.51 floor); the right cell is found only at passes 2-4. Only 1 of the 16 runs finds a two-dimensional code (rank 4: 8/8); half end with x and y moving the phase along one direction. Downstream computation compensates for the rank-2 deficit and leaves it in place.

Where the angle comes from and how content uses it are separate choices. Any rotary score can be written sum_c A_c cos(dphi_c + beta_c): the phase difference dphi says where, and content sets an amplitude A and, in RoPE, a phase beta. Path integration changes dphi; PoPE's polar score removes beta, and Proposition 7 then pins every kernel's peak at zero phase difference. The two predictions have opposite signs, and so do the results. On the short-walk torus PoPE's score raises the path model (MapPoPE 0.999 against MapWM 0.971) and lowers the index model (PoPE 0.679 against RoPE 0.805), an interaction of +0.154 (8/8). Probes give the mechanism (analysis; App. L). For the path model the pinned peak aligns the kernel with the target (target coherence 0.995 against 0.70 for MapWM). For the index model no content pair's kernel prefers offset 3 over offset 1 (0% of pairs, against 53-100% for RoPE, whose kernel is a comb at the retrace offsets), and 0.110 of the 0.126 accuracy gap lies on reversals whose target sits three or more steps back. Freeing PoPE's phase offset tests this directly (registered): the index model rises by +0.122 (8/8 seeds, sign-flip p 0.0078), most on multi-step reversals (+0.180), its kernel now prefers offset 3 over offset 1 in 50-100% of content pairs, with peaks at offsets 3, 4, 7 and 8, and the path model stays at ceiling (-0.018, p 0.31) while its target alignment falls from 0.987 to 0.62. PoPE's pinned offset is the lever: it helps a path phase and hurts an index phase.

A score rule repairs the readout, not the code. On the long-walk torus at rank 2 the polar score repairs short-gap revisits mostly through the readout term (target coherence 0.617 to 0.960, p 0.0002, with the code term moving +0.06) and leaves the code term low on long-gap revisits (0.80 / 0.91 under the two scores, against 0.99 at rank 4) and wrap-only revisits (0.40 / 0.48, against 0.97 at rank 4 under either score) (analysis; each stratum read on the head that carries its same-cell attention). The rank-2 phase does not place long-gap or wrapped revisits at zero phase, and no score rule can make it: under either score rank 2 solves 3/12 (registered).

8. Maps from words

The text world asks whether the path model can find its actions when nobody marks them. Each move of the torus walk is written as a sentence: a direction word chosen from three synonyms, verbs that do not move the walker, optional adverbs and fillers, and, after some moves, an aside that names an object without visiting it ("she thought about a cat ."). Nothing in the input says which words move the walker (vocabulary and step tables: App. J). A walk of 1024 words is about 132 moves, so the retrace rule of Sec. 5 scores 0.934 here (analysis), beside the constant (0.505) and reversal-copy (0.597) floors.

One path layer learns navigation told in English. On fresh seeds (n = 6) one path layer scores 0.979 +/- 0.048 (6/6 SOLVED) against 0.505 for one-layer index RoPE (0/6), +0.474 (permutation and Fisher p 0.0022); two index layers reach 0.793 (0/6). With the attention-dropout evaluation scale (App. C9) the path layer scores 0.993 over all eight seeds. Both index models stay below retrace on every seed (the best two-layer seeds reach 0.927 and 0.926 against 0.934). The decisive revisits are the 13.4% that retrace cannot reach, places reached by a route other than undoing the last run (Table 7; analysis, all 200 evaluation walks): there the path layer scores 0.956 over all eight seeds (re-scored; 0.978 on the fresh seeds, 0.934 in eval mode), while two index layers score 0.411 and one index layer 0.454, both below the constant 'nothing' (0.506). Of these revisits, the 6.6% whose cell holds an object are the ones retrace gets wrong (on blank cells its default 'nothing' is right by chance); there the path layer scores 0.925 (0.967 on the fresh seeds), two index layers 0.129, the best constant 0.088 ('nothing' excluded by construction), and one index layer 0.004, since it predicts 'nothing'. Two index layers learn part of retrace and nothing that needs the map.

Table 7. Text world: accuracy by whether the retrace rule can reach a revisit (analysis; registered batch, all 200 evaluation walks of the registered stream, 6,381 revisits; mean over 8 seeds, fresh seeds 2-7 in brackets; path models re-scored with the attention-dropout scale unless marked).

model (n = 8)	all revisits	retrace applies (n 5,525)	retrace cannot reach (n 856)	... cell holds an object (n 423)
path, 1 layer	0.993	0.999	0.956 (0.978)	0.925 (0.967)
path, 1 layer, eval mode	0.969	0.975	0.934 (0.958)	0.884 (0.927)
index RoPE, 2 layers	0.772	0.828	0.411 (seeds 0.34-0.55)	0.129
index RoPE, 1 layer	0.505	0.513	0.454	0.004
best constant	0.505 ('nothing')	--	0.506 ('nothing')	0.088 ('cat')

What the step table holds. On every seed, opposite direction words cancel once the component common to the four directions is removed (0.002-0.012), synonyms collapse to one step (cosine 1.000), and zeroing the direction steps drops accuracy to the constant floor (0.39-0.51) (declared secondaries; Fig. 8). The rest of the vocabulary is organised by how often a word occurs per move (analysis, 32 models): adverbs and fillers, whose count per move varies, get steps of 0.004-0.03 of a move; words that occur once per move

(verb, seeing words, object, full stop) may carry a common tick; aside sentences get sizeable word steps that cancel within the sentence, leaving the aside's noun 0.04-0.16 of a move off the place. Words whose count per move varies get no net step; the steps that remain belong to words that occur once per move or cancel within their sentence.

The offset is why learned steps beat being told. A model whose steps are restricted to the twelve direction words stops at 0.970-0.974 on every seed, and its errors name objects told in an aside at that place: with no other word stepping, the aside's noun sits at the place's phase. Moving each aside back onto its place's phase at evaluation, with the phase after it unchanged, drops the learned-step models at ceiling from 0.989-1.000 to 0.952-0.975, with 71-100% of their errors now of the same kind; offsetting the oracle's aside words by half a move lifts it from 0.962-0.981 (on the analysis stream) to 0.999-1.000 on every seed (analysis). "Only actions move" is the wrong separation in language: some words that are not actions need steps, so that what is mentioned is not filed where it is not.

Half the models also count moves, in channels they do not use. On four of the eight registered seeds the direction words' common component points along the verbs' step (cosine +0.999 to +1.000), and 31-38 of 64 channels drift between visits to the same cell (declared secondary; drift from the first visit). Across 24 learned-step models (analysis), 10 carry such a per-move tick, which drifts in 26-38 channels between consecutive visits. It counts moves: an aside sentence adds at most 0.06 of a tick, a filler word at most 0.007. In Howard et al.'s terms these channels integrate distance travelled (Case II) beside the others' position (Case III). Proposition 2(b) says such channels cannot serve retrieval by place, and they do not: the tick alone predicts which channels drift (agreement 0.98-1.00; informative on the 10 clock models, where the predicted and observed counts of drifting channels match exactly), 79-99% of retrieval usage sits in channels whose tick is zero modulo pi (on the even torus a return always takes an even number of moves), and removing the 26-38 clock channels from the score costs at most 0.009 on each of the seven clock models at ceiling, while removing as many map channels collapses accuracy (to 0.03-0.65, means over random draws). The three clock models below ceiling still lean on their clock channels (removal costs 0.09-0.26). The clock seeds' extra dependence on non-direction words follows the direction words' common component in the used channels (Spearman +0.905 over the 8 registered seeds, exact permutation p 0.0046), which those words cancel within each move.

What must and must not move. The same rule covers the what-to-where leak on a task with new objects in every episode. A step that depends on an object's identity spreads objects at one place over identity-dependent offsets; removing the objects' steps, or normalising the step input, lets every run converge (16/16 SOLVED against 0/8 for MapWM; 8/8 against 0/8 on fresh seeds; App. K). A token's offset from its place should depend on its role, not on its identity.

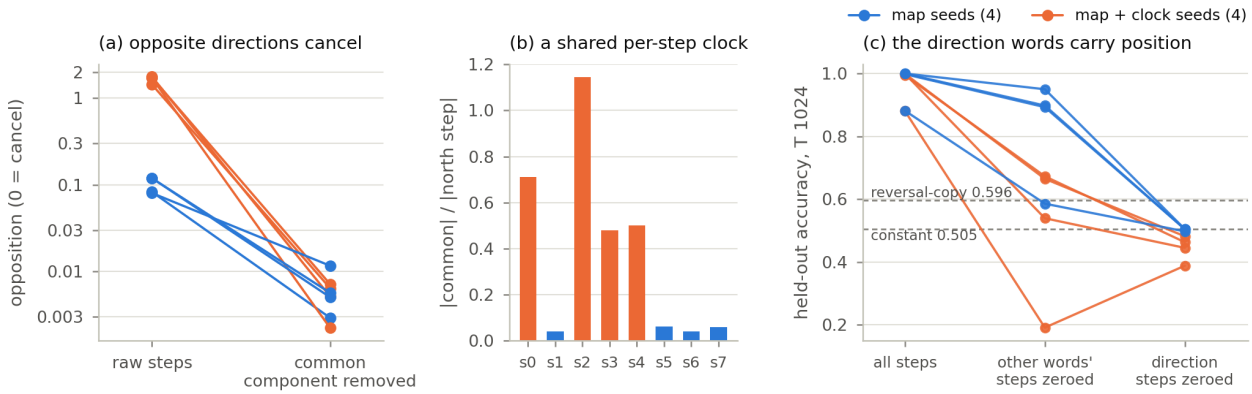


Figure 8 (`fig_textworld`). What a one-layer path model learns from a torus walk told in English (n = 8 seeds). (a) Opposition of the learned direction-word steps, |A + B| / |A| averaged over north/south and west/east (0 = opposites cancel): after removing the component common to all four directions, opposites cancel to 0.002-0.012 on every seed. (b) That common component is small on four seeds (blue, 4-6% of a direction step) and large on the other four (orange, 0.48-1.14), where it points along the verbs' step: a per-move clock beside the map (31-38 of 64 phase channels drift between visits to the same cell, counted from the first visit). (c) Held-out accuracy with every step, with the steps of all non-direction words zeroed, and with the direction words' steps zeroed: removing the direction steps drops every seed to the constant floor (0.505, dashed) or below. Eval mode without the dropout re-scale (App. C9).

9. Discussion and related work

What the two failure axes share. For per-head rank 2 and for non-negative steps, a solution in the class exists (the rank-2 projection scores 0.9955; the |.| construction is exact), training keeps or re-finds it (7/8 at rank 2; held 8/8 on the ring and re-found 8/8 on the torus for |.|), and training from scratch does not reach it (0-2/8; 0/8). In Remark 2's terms the initialisation sits outside the basin of a non-empty, stable solution set. The route inventory accounts for this, proved for the routes and conjectured for the rank-D dynamics, and the two axes fail differently inside it. A non-negative increment has no collective route and its per-channel targets lie past pi; restoring a signed increment under a non-negative readout restores learning (16/16). At rank D the collective route exists, but no channel can be both clock-free and a full map while a clock remains, and in two or more dimensions hiding offers a free local move that, by Conjecture 1, starves the collective route; one more latent dimension adds a local move, parking, that costs the map nothing. Both failures end in the same measured place: a clock of the order of the map (or, in a few rank-2 runs, a map that has lost a dimension), and strata that keep short-gap revisits and lose the rest. Giving the rest of the network four times the computation or the parameters reads that code better and does not find the rank-2 solution: the deficit is local to the content-to-angle map. Shakerinava et al. (2026) report "representable but not learned" for non-Abelian rotations in diagonal SSMs; we find it in single-layer Abelian tasks, produced by one rank or parameterisation choice, and give its routes.

Not every search failure is of this kind. Single-origin MapEM represents k-back recency exactly (installed and frozen, 1.000) and finds the rewind per query token, and

four times the budget closes most of its deficit (0.578 to 0.928; not registered), whereas rank 2 stays at 0/8 with twice the budget (App. N).

Design guidance. Keep the phase increment signed; a non-negative readout is harmless. Give each head rank above the map's dimension (D + 1). Keep observations from stepping the phase: a step that depends on what is seen moves where the model thinks it is. Measure map use where retracing no longer suffices, against measured map-free floors and by stratum: the long-gap and wrap-only revisits separate a map from a local rule, and an aggregate accuracy near 0.9 can be either. To tell whether a trained model's phase is a map or a clock, read the used-channel clock ratio, not the growth of the phase with the token index.

What and where. In the long-walk regime, the model that learns the map separates where from what, and the models that fail do not (analysis; App. K). For the signed rank-4 model at T = 1024, one shared position kernel per head times a per-pair content gain plus offset reproduces its accuracy exactly (0.998); removing token identity within type from the score costs 0.032 (0.966; 8/8 seeds lower, sign-flip p 0.008, uncorrected; Holm over the 24 contrasts 0.08); and the content-averaged score carries 92.5% of the score's within-row sum of squares (SS(P) / SS(S)). Token type (move or observation) is essential: without it the model falls to 0.113. The non-negative and index models lose 0.25-0.31 when token identity within type is removed, and a shared kernel times a gain does not restore them; within the non-negative arms, accuracy tracks the share that survives the removal (Spearman 0.84 over 16 runs, p 5e-5). Separation is a property of the solution training finds, not of the architecture. MapFormer showed the separation in the learned embeddings (their Fig. 9); these checks show that the models rely on it where a map is needed, in the sense TEM proposed.

The integrator view. A learned, signed, content-dependent rate can realise all three of Howard et al.'s cases, and which one a channel realises is set by what the readout keys on. Split each step into the frequency-weighted mean, a clock, and a content part, and write rho_c for the clock part over the content part at the training length (Remark 3, App. E13). For targets keyed to a place reached at variable lags the used channels must be clock-free (Proposition 2(b)), and the models put them there: torus 0.064-0.079, text world 0.06-0.23 (23 of 24 learned-step models; the unconverged one 2.5; analysis). For targets keyed to a count back, the used channels are clocks: k-back recency, the same one-layer signed class, gives 21.8-22.1. One text-world model can do both at once, with distance travelled in idle channels and position in used ones. In the rotary slot the cases coincide in what they can represent (Proposition 3), so the sign result is about learnability: a non-negative step could hold a Case III code and does not find one.

Text. On ordinary text at small scale (n = 3 pilots; the ratios are analysis), the channels attention uses are clock-dominated (rho 9.9-12.3 on TinyStories, Eldan and Li 2023; 4.5-6.6 on code; against 0.06-0.23 in the text world), MapWM matches index RoPE on TinyStories (1.5053 against 1.5050 nats per token), and at matched length path integration costs on code (+0.0056 bits per byte, 3/3 seeds). The phase grows linearly with the token index there (R^2 0.994-0.999), as it does in text-world models that carry a perfect map, where the growth is carried by channels attention does not use. The used-channel ratio separates the two; the growth exponent does not.

Egocentric actions. The cumulative phase assumes allocentric moves. With turn-and-forward actions the step depends on the heading, and recoding the actions allocentrically, what head-direction cells supply, restores path integration's advantage (position effect +0.050 to +0.488, n = 8, in a short-walk recipe; not registered; App. A5).

Related work. *Optimisation, not expressivity.* Shakerinava et al. (2026) find that input-dependent rotations in unit-modulus SSMs can represent non-Abelian state tracking but do not learn it from scratch, while near-solution starts recover it. Grazzi et al. (2025) and Mamba-3 own the representability of rotations with non-negative angles; ours is the learnability cost in the rotary slot and its route account. Orientation-fixing parameterisations have a history: scoRNN fixes the +/-1 diagonal of its Cayley transform (Helfrich et al. 2018); a reparameterised weight w = |u| cannot change sign under gradient flow (Amid and Warmuth 2020, Prop. 2; Woodworth et al. 2020); the LRU initialises its phases near zero (Orvieto et al. 2023). With fixed frequencies the parameter symmetries of rotary attention are per-plane complex scalars (Wang and Wang 2025; Tran et al. 2026); a learnable step adds the reflection of Lemma 1. *Depth and recurrence.* Depth can substitute for recurrence through shortcuts (Liu et al. 2023; Yao et al. 2021), though shallow transformers learn sensitive functions unreliably (Hahn and Rofin 2024), while Selective RoPE and PaTH (Yang et al. 2025) solve parity or Flip-Flop with one layer. We give the rate at matched depth, the mechanism trained models use, and where it breaks. *Integrators.* Howard et al. (2014) for the three rate cases; GridPE for RoPE as a velocity-controlled oscillator; TEM and TEM-t for the "where" apart from the "what".

10. Limitations

- The models are small (1-4 layers, 2 heads, d_model 128-256) and the tasks synthetic; natural text appears only as a three-seed pilot. Actions are allocentric translations.
- "Not found" is relative to the stated budgets (300-1800 epochs). Learning rates were swept on the ring only. On the torus, in the 2-layer d_model 256 model, the sign gap is a quarter of the one-layer gap in accuracy and most |.| runs were still descending, so at scale part of the cost may be speed. Selective RoPE's runs were all still descending at 900 epochs. The exp and softplus increments start at a constant positive step and were run at two fixed rates.
- Rank results use two heads; our latent is shared across heads where MapFormer's is per head (per-head arms are reported). The torus exchange rate is a lower bound measured at n = 4.
- The theory of Sec. 4 was developed after the registered batches. Its proved statements concern the step parameterisation and quadratic surrogates; the hiding feedback at rank D is a conjecture, and the per-channel dynamics of |.| are not derived. The checks marked (analysis) are post hoc, on one batch per arm, without fresh-seed replication; their thresholds and statistics chosen after the data were seen are listed in App. O, and one CPU analysis (the step-map basin) is not deterministic and is reported as exploratory.
- The text world is scripted (58 words, one movement clause per move, context-free direction words), and its walks are short enough that retrace scores 0.934; the map evidence there is the revisits retrace cannot reach and the mechanism.
- No recurrent or SSM baseline.

References

Titles and author lists checked against arXiv, OpenReview, Crossref, ACL Anthology or PMLR metadata on 2026-10-15/16; entries marked * were not re-checked (venue, or the whole entry where stated in the notes file).

- Alipour, A., James, T., Brown, J. W. and Tiganj, Z. Self-supervised learning of scale-invariant neural representations of space and time. J. Comput. Neurosci. 53:131, 2025*.
- Amid, E. and Warmuth, M. K. Reparameterizing mirror descent as gradient descent. NeurIPS 2020*; arXiv:2002.10487.
- Eldan, R. and Li, Y. TinyStories: how small can language models be and still speak coherent English? arXiv:2305.07759, 2023*.
- Golovneva, O., Wang, T., Weston, J. and Sukhbaatar, S. Contextual position encoding: learning to count what's important (CoPE). arXiv:2405.18719, 2024.
- Gopalakrishnan, A., Csordas, R., Schmidhuber, J. and Mozer, M. C. Decoupling the "what" and "where" with polar coordinate positional embedding (PoPE). arXiv: 2509.10534, 2025*.
- Grazzi, R., Siems, J., Zela, A., Franke, J. K. H., Hutter, F. and Pontil, M. Unlocking state-tracking in linear RNNs through negative eigenvalues. ICLR 2025*; arXiv: 2411.12537.
- Hahn, M. and Rofin, M. Why are sensitive functions hard for transformers? ACL 2024 (long papers); arXiv:2402.09963.
- Helfrich, K., Willmott, D. and Ye, Q. Orthogonal recurrent neural networks with scaled Cayley transform. ICML 2018*; arXiv:1707.09520.
- Hewitt, J., Hahn, M., Ganguli, S., Liang, P. and Manning, C. D. RNNs can generate bounded hierarchical languages with optimal memory. EMNLP 2020*; arXiv: 2010.07515.
- Howard, M. W., MacDonald, C. J., Tiganj, Z., Shankar, K. H., Du, Q., Hasselmo, M. E. and Eichenbaum, H. A unified mathematical framework for coding time, space, and sequences in the hippocampal region. J. Neurosci. 34(13):4692-4707, 2014.
- Kabir, M. R., Mochizuki-Freeman, J. and Tiganj, Z. Deep reinforcement learning with time-scale invariant memory. AAAI 2025*; arXiv:2412.15292.
- Lahoti, A., Li, K. Y., Chen, B., Wang, C., Bick, A., Kolter, J. Z., Dao, T. and Gu, A. Mamba-3: improved sequence modeling using state space principles. ICLR 2026; arXiv:2603.15569.
- Li, B., Wu, Y., Huang, N. and Zhang, W. GridPE: a grid cell-inspired unified position embedding for arbitrary-dimensional spaces. arXiv:2406.07049, 2024*.
- Liu, B., Ash, J. T., Goel, S., Krishnamurthy, A. and Zhang, C. Transformers learn shortcuts to automata. ICLR 2023; arXiv:2210.10749.
- Mochizuki-Freeman, J., Maini, S. S. and Tiganj, Z. Characterizing neural activity in cognitively inspired RL agents during an evidence accumulation task. IJCNN 2023*.
- Movahedi, S., Carstensen, T., Afzal, A., Hutter, F., Orvieto, A. and Cevher, V. Selective rotary position embedding. ICLR 2026; arXiv:2511.17388.
- Orvieto, A., Smith, S. L., Gu, A., Fernando, A., Gulcehre, C., Pascanu, R. and De, S. Resurrecting recurrent neural networks for long sequences (LRU). ICML 2023*; arXiv:2303.06349.

- Rambaud, V., Mascarenhas, S. and Lakretz, Y. MapFormer: self-supervised learning of cognitive maps with input-dependent positional embeddings. arXiv:2511.19279 (v4, 10 May 2026).
- Sarrof, Y., Veitsman, Y. and Hahn, M. The expressive capacity of state space models: a formal language perspective. NeurIPS 2024; arXiv:2405.17394.
- Shakerinava, M., Khavari, B., Ravanbakhsh, S. and Chandar, S. The expressive limits of diagonal SSMs for state-tracking. ICLR 2026; arXiv:2603.01959.
- Su, J., Lu, Y., Pan, S., Murtadha, A., Wen, B. and Liu, Y. RoFormer: enhanced transformer with rotary position embedding. Neurocomputing 568:127063, 2024*; arXiv:2104.09864.
- Tran, V.-H., Bui, V. K., Trinh, V.-H., Lai Ngoc, T. and Nguyen, T. M. Functional equivalence in attention: a comprehensive study with applications to linear mode connectivity. ICML 2026 (PMLR 306); arXiv:2606.17830.
- Veisi, A., Fartoot, D. and Amirzadeh, H. Context-aware rotary position embedding (CARoPE). arXiv:2507.23083, 2025.
- Wang, H. and Wang, K. Complete characterization of gauge symmetries in transformer architectures. Symmetry and Geometry in Neural Representations (NeurReps), PMLR 282:622-663, 2025.
- Whittington, J. C. R., Muller, T. H., Mark, S., Chen, G., Barry, C., Burgess, N. and Behrens, T. E. J. The Tolman-Eichenbaum Machine: unifying space and relational memory through generalization in the hippocampal formation (TEM). Cell 183(5): 1249-1263, 2020.
- Whittington, J. C. R., Warren, J. and Behrens, T. E. J. Relating transformers to models and neural representations of the hippocampal formation (TEM-t). ICLR 2022; arXiv:2112.04035.
- Woodworth, B., Gunasekar, S., Lee, J. D., Moroshko, E., Savarese, P., Golan, I., Soudry, D. and Srebro, N. Kernel and rich regimes in overparametrized models. COLT 2020*; arXiv:2002.09277.
- Yang, S., Shen, Y., Wen, K., Tan, S., Mishra, M., Ren, L., Panda, R. and Kim, Y. PaTH attention: position encoding via accumulating Householder transformations. NeurIPS 2025; arXiv:2505.16381.
- Yao, S., Peng, B., Papadimitriou, C. and Narasimhan, K. Self-attention networks can process bounded hierarchical languages. ACL 2021*; arXiv:2105.11115.
- Zhang, Y., Chen, Z., Liu, Y., Qin, Z., Yuan, H., Xu, K., Yuan, Y., Gu, Q. and Yao, A. C.-C. Group representational position encoding (GRAPE). ICLR 2026; arXiv: 2512.07805.

Appendix to "A map is a phase without a clock: what path-integrated attention learns, and what decides whether training finds it"

Draft 2 (2026-10-21), for `docs/paper/MAIN_DRAFT2.md`. Every number is copied from the file named beside it; a CORRECTED or AUDIT block at the top of a results file supersedes its body. "Registered" = a pre-registered decision or declared secondary; "(analysis)" = computed on stored streams or checkpoints, eval-only, never changing a verdict; its script is listed in App. O. App. E states and proves every theory statement the main text uses; statements not used are in `docs/paper/APPENDIX_theory_v3.md`. Delete this paragraph before submission.

Contents. A: models, recipes, batches. B: floors. C: per-seed tables, SOLVED-threshold sensitivity, evaluation scale. D: constructions. E: theory. F: exchange-rate details. G: rank details. H: sign details. I: CARoPE, CoPE and Selective RoPE. J: text world. K: what and where; the leak. L: PoPE's score. M: loops and depth on a rank-2 code. N: EM's recency deficit, a contrast case. O: analysis scripts.

A. Models, recipes, batches

A1. Architecture

component	setting
layers	1 (Sec. 6: 1-4 real layers and one block looped x4; ROBUST torus: 2 layers, d_model 256)
heads, width	2 heads, d_model 128, d_head 64; feed-forward 4 x d_model with GELU; LayerNorm before attention and before the feed-forward; dropout 0.1 on attention weights, attention output and feed-forward output
rotary channels	32 two-dimensional blocks per head, 64 channels; every query and key coordinate is rotated
score	channel c contributes q_c^T R(theta_{t',c} - theta_{t,c}) k_c (content Q and K rotated by the path angle; not additive)
step map	Delta = W_out W_in x, W_in: 128 -> r, W_out: r -> 64, no bias; r = 4 shared across heads unless stated (MapFormer's latent is per head); per-head variants: one W_in / W_out block per head; block-diagonal rank 4: the shared rank-4 latent with the off-diagonal blocks of W_out held at zero (each head reads rank 2)
frequencies	omega_c = 2 pi (1 / Delta_max)^(i / 31) for block i = 0..31, the same ladder in each head; Delta_max = grid size (torus 64, ring 32, line 256); omega is a free parameter, its sign unconstrained
vocabulary	torus: 4 moves + 17 observation tokens (16 types + blank); ring and line: 2 moves + 17
parameters	about 204k for the one-layer torus MapWM; 798,741 for four index layers; 203,925 for the looped block (`FACTORIAL_RESULTS.md`); 1.59M for the ROBUST torus model

Step parameterisations (`model.py`, `model_sign.py`, `model_wrap_exist.py`): signed (`Signed_r4` = `Vanilla_r4`), |.| (`Abs_r4`; the absolute value is applied after W_out on each of the 64 channel steps, adding no parameters), softplus (`Pos_r4`; softplus(0) = 0.693, so every token starts at about the same step), observation-step-free (`SignedAO_r4`, `AbsAO_r4` on the ring,

`SignedA04_r4`, `AbsA04_r4` on the torus: every non-move token's step is 0 by construction). Index RoPE (`RoPE`): canonical ladder base^(-2c / d_head), base 10000.

A2. Recipes

	torus (Secs. 5-8)	**ring and line (Secs. 6, 7)**	**Dyck-2 (Sec. 6)**
optimiser	AdamW, weight decay 0.05	AdamW, weight decay 0.05	AdamW, weight decay 0.01
peak learning rate	1e-3	1e-3 (ROBUST and LRSWEEP vary it)	1e-4
schedule	5% linear warm-up, cosine to 10% of the peak	same	same
batch	16 walks of T = 1024 moves (2,048 tokens); T = 128 batches of 128	128 walks of T = 128	128 sequences
budget	98 batches per epoch, 900 epochs (1800 in Sec. 7.2; 300 at T = 128)	98 batches per epoch, 300 epochs	560,000 sequences (1x) or 1,680,000 (3x)
loss	cross-entropy on revisited observations	same	next-token cross-entropy on every position
training map	one per run	redrawn for every walk	--
evaluation	100 walks on one held-out map, eval mode	400 walks, maps redrawn	L32 D12 test set

Run classes (`stats_core.classify_run`, per-epoch training loss): SOLVED if the mean of the final 5% of epochs is below 0.05; otherwise STALLED if the final-10% mean is within 5% of the preceding 10% mean, else DESCENDING. Continuations of constructed models (App. D): ring 300 epochs at lr 1e-3 (schedule re-warmed, fresh optimiser state, fresh walk stream); torus 300 epochs at lr 1e-4, fixed before launch (about the rate at which the source solution was reached).

A3. Batches

"Code check" = an independent, results-blind code review, whose findings were amended into the pre-registration before any result was read (before launch, except TEXTWORLD and LEAK, checked while the batch trained). "Reproduced" = the analysis was re-run from stored checkpoints by an independent reproduction audit (`docs/paper/review_draft5/REPRO_AUDIT.md`, `REPRO_AUDIT_nonsign.md`, 2026-10-16) and matched (spot re-evaluations within 7e-4); batches that landed after that audit read "no". "Attention": standard, or `--fast-attn` (SDPA; agrees to ~1e-6 in logits, not bit-identical in training). Pilots ran on seeds outside the batch except where noted.

batch	**main text**	**task**	**arms**	**seeds**	**epochs**	**attention**	**code check**	**reproduced**
PAPER2X2	5, 7.4	torus T = 128	RoPE, PoPE, MapWM r2 / r4, MapPoPE r2 / r4	0-7	300	standard	no	yes
SIGN_MATCHED		torus T = 1024		0-7	900	standard	no (authors'	yes

batch	main text	task	arms	seeds	epochs	attention	code check	reproduced
	5, 7.3, 9		signed, |.|, softplus, RoPE				pre-launch gate)	
GENERALITY	5, 7.3	torus; ring p_plus 0.5	obs.-step-free signed / |.| and constructions; signed, CARoPE, CoPE, RoPE	700-707	900 (ring 300); +300	fast-attn (torus)	yes	yes
FACTORIAL	6	torus T = 1024	path 1 layer; index 1, 4 layers, looped x4	500-503 (n = 4)	900	fast-attn	yes	yes
DYCK_MDEPTH	6	Dyck-2 L32 D12	MapWM, MapPoPE (r2), RoPE, PoPE at 1-4 layers; 3x; mixture	0-7	1x / 3x	standard	no	yes
CANCEL	6	ring p_plus 0.5-1.0	path 1 layer; index 1-3 layers	0-7; main text 2-7 (0-1 were the pilot)	300	standard	no	yes
RANK_MI	7.2	torus	shared r2, per-head r2, shared r4	0-7 (per-head r2 s0-1 were a pilot)	900	standard	no	yes
RANK_SEP	7.2	torus	block-diagonal r4, per-head r4	0-7	900	standard	no	yes
RANK3	7.2	torus	per-head r3	0-7	900	standard	no	yes
LOOP_RANK_E1800 (part 1)	7.2	torus	shared r2, shared r4	0-7	1800	standard	no	yes
RANK_PROJ	7.2	torus	projected r2 (frozen, trained), r4 control, r2 continued	0-7	+900	standard	no	yes
RANK_ND	7.2	2D (32 per side), 3D (10 per side) tori	per-head rank D, D + 1	0-7	900	standard	yes	yes
RANK_WRAP	7.2	2D and 3D tori of two sizes	per-head rank D + 1	0-7	900	standard	yes	yes

batch	main text	task	arms	seeds	epochs	attention	code check	reproduced
RANK_TESTS	7.2	ring; torus escape test	rank 1, 2 (+ variants); lifted / fresh rank 4	20-27; 0-7	300; 600	standard	yes	yes
SCORE_RANK	7.2, 7.4	torus T = 1024	MapWM / PoPE score at r2 and r4	r2 10-21, r4 10-17	900	standard	yes	no
LOOP_RANK	6, 7.4	torus T = 1024	r2 1 layer, r2 looped x4, r2 4 layers, r4	0-7 (A and C reused; A s0 reproduced bitwise)	900	standard	no	yes
RING_DIAL	7.3	ring p_plus 0.5 / 0.75 / 0.9	signed, |.|	30-37	300	standard	yes	yes
WRAP_EXISTENCE	7.3	ring p_plus 0.5	obs.-step-free signed / |.|, constructions	600-607	300 + 300	standard	yes	yes
LRSWEEP	7.3	ring	obs.-step-free signed / |.| at 1e-3 / 3e-3 / 1e-2 / 3e-2	1500-1507	300	standard	yes	no (windowed run classes cross-checked)
ROBUST	7.3	ring (four recipes); torus 2 layers d 256	obs.-step-free signed / |.|	1400-1407	300; 900	fast-attn (torus)	yes	yes
ACCOUNTS	7.3	ring; line	T1 continuations; T2 frozen latents; T4 CoPE on a line	600-607; 1000-1007; 1100-1115	300	standard	yes	yes
MECHANISM	7.3	ring p_plus 0.5	eight start / step arms	800-807	300	standard	yes	yes
SIGNROUTE	7.3	ring, moves +1/-1	obs.-step-free signed, |.|, hybrid, abs_tied, exp, softplus at lr 1e-3 and 3e-3	1800-1807	300	standard	yes	no
SROPE_TORUS	7.3	torus T = 1024	Selective RoPE (best LM config), without SiLU, state-tracking	1700-1707	900	fast-attn	yes	no

batch	main text	task	arms	seeds	epochs	attention	code check	reproduced
			config, generator swap; MapWM; RoPE					
DELTA_FREE	7.4	torus T = 128, r2	PoPE / MapPoPE with delta pinned, free, frozen-random (path)	1600-1607	300	standard	yes	no
TEXTWORLD	8	torus in words	path 1 layer; index 1, 2 layers	0-7; main text 2-7 (0-1 were the pilot)	900	standard	yes	yes
TW_NORMSTEP	8	torus in words	MapWM, NormStep, DirOnly (direction words only step)	10-17	900	standard	yes	yes
LEAK	8	new-object torus	MapWM, ActOnly, NormStep	0-7	900	standard	yes	yes
GAIN_PHASE (K)	--	new-object torus	step x score 2x2	8-15	900	standard	yes	no
RING_BUDGET (H)	--	ring p_plus 0.5	|.| at 300 and 900 epochs	40-47	300 / 900	standard	yes	yes
TORUS_START (H)	--	torus	start arms	1200-1207	900	fast-attn	yes	yes
GAUGE (H)	--	ring, moves +1/-1 or +1/+2	obs.-step-free signed / |.|, RoPE	1300-1307	300	standard	yes	yes
WRAP_COPE (I)	--	ring	CoPE table starts	900-907	300	standard	yes	yes
MAPPOPE_PAIR (L)	--	torus T = 128	MapWM r2, MapPoPE r2 (32 / 64 angles)	10-25	300	standard	yes	no
RECENCY_EM, WARM, NOLEAK, SPREAD, SEARCH (N)	9	k-back recency	MapEM variants and installs	0-7	300 (SPREAD 1200)	standard	no	no
KNOB_SWEEP (A5)	9	torus T = 128, 16-	MapWM, RoPE under action variants	8 seeds	16	standard	not registered	no

batch	main text	task	arms	seeds	epochs	attention	code check	reproduced
		epoch recipe						

ACCOUNTS T1's weight-decay-0.05 arms are bit-identical retrains of WRAP_EXISTENCE's constructed arms (a second readout of the same runs, not a replication). The shared rank-2 arm of every rank contrast is one stored set of eight runs. SIGN_MATCHED's signed arm is the source of the T = 1024 what/where analysis (App. K).

A4. Reproduction of MapFormer v4 Table 2 (2D)

With the paper's recipe (16 epochs x 98 batches, linear decay, lr 3e-4; `python3 -m mapformer.main --device cuda --epochs 16 --n-batches 98`), our MapWM scores 0.989 +/- 0.010 and our MapEM (single p_0) 0.987 +/- 0.012 on fresh maps at T = 128 (`PAPER_TASK_ACCURACY.md`), against 1.00 / 1.00 in v4 and 0.99 / 1.0 in v1 (MapWM-r2, MapEM-os, 2D IID).

A5. Egocentric actions (Sec. 9; `KNOB_SWEEP_n8.md`, not registered)

Torus at T = 128 with MapFormer's 16-epoch recipe, n = 8, held-out map. With allocentric moves (baseline) path integration's position effect over index RoPE is +0.438 (MapWM 0.967, RoPE 0.529). With egocentric turn-and-forward actions ("rotate") it is +0.050 (0.558 vs 0.508): a cumulative phase cannot integrate a heading-dependent step. Recoding the same walks into allocentric moves ("allocentric") restores it to +0.488 (0.996 vs 0.508). The index arm sits at the 0.508 floor in both action conditions, so these are boundary statements about the path model, not exchange rates.

B. Floors

A floor is the accuracy of the best map-free predictor on the same kind of evaluation walk. Rows marked (analysis) were computed after the batches.

task	predictor	value	stream	source
torus T = 128 (analysis)	blank / n-gram (orders 0-5) / retrace	0.507 / 0.598 / 0.934 (retrace per seed 0.928-0.946); wrap-only share 0.000	PAPER2X2 eval walks, seeds 0-7	`docs/audits/2026-10-16/paper2x2_T128_floors_out.txt`
torus T = 1024 (analysis)	blank / n-gram / retrace	0.506 / 0.575 / 0.843 (retrace per seed 0.837-0.849); wrap-only share 0.079	SIGN_MATCHED eval walks, seeds 0-7 (shared by RANK_MI, RANK_SEP, RANK3)	`docs/audits/2026-10-16/sign_floors_out.txt`
torus T = 1024 (analysis)	same	0.508 / 0.578 / 0.843; wrap-only 0.081	GENERALITY, seeds 700-707	`generality_floors_out.txt` (same directory)

task	predictor	value	stream	source
torus T = 1024, 2 layers d 256 (analysis)	same	0.509 / 0.578 / 0.844; every run of both arms above retrace (lowest |.| 0.910)	ROBUST, seeds 1400-1407	`robust_floors_out.txt`
torus T = 1024 (analysis)	blank / retrace	0.509 / 0.844	FACTORIAL, seeds 500-503	`docs/paper/figs/fig_factorial_strata_out.txt`
ring p_plus 0.5 / 0.75 / 0.9 (analysis)	retrace	0.860 / 0.758 / 0.612 (blank 0.497-0.499; 0.864 at p_plus 0.5 on the registered eval stream)	2000 walks per bias, another stream of the same task	`docs/audits/2026-10-16/ring_floors_out.txt`
ring p_plus 0.5 / 0.75 / 0.9 / 1.0	better of best constant and n-gram (orders 1-5)	0.554 / 0.544 / 0.520 / 0.498	1000 held-out walks	`CANCEL_PREREG.md`, `CANCEL_RESULTS.md`
line (256 cells)	blank	0.503	ACCOUNTS T4	`ACCOUNTS_RESULTS.md`
ring, moves +1/+2	blank (retrace reduces to it)	~0.50	GAUGE	`GAUGE_RESULTS.md`
torus in words	best constant / reversal-copy	0.505 / 0.597	TEXTWORLD eval set	`TEXTWORLD_RESULTS.md`
Dyck-2 L32 D12	best stack-free A2f floor (n-grams of orders 1-6)	0.594	gate G4	`DYCK_MDEPTH_RESULTS.md`

Per-stratum floors (analysis). Torus, seeds 0-7 (`docs/paper/figs/torus_strata_floors_s0-7.json`): best constant 0.506 / 0.500 / 0.507 (short gap / long gap / wrap-only), retrace 0.911 on short gaps and equal to the constant on the others (walk runs are at most 10 moves, so a retrace never reaches a gap of 128 moves or a wrap-only revisit). FACTORIAL's stream: 0.508 / 0.515 / 0.517, retrace 0.912. Ring (`ring_strata_floors_out.txt`): retrace 0.9998 at gaps 2-7, 0.943 at 8-31, 0.570 at 32-127, 0.493 wrap-only; constant 0.49-0.50 in every stratum.

Against its own seed's retrace floor (analysis, torus T = 1024): signed falls below on 0/8 runs, |.| on 3/8 (0.637, 0.690, 0.717), softplus on 4/8, index RoPE on 8/8.

Further floors (analysis unless marked).

task	predictor	value	stream	source
torus T = 1024				`SCORE_RANK_RESULTS.md`

task	predictor	value	stream	source
	blank / n-gram / retrace	0.507 / 0.576 / 0.843	SCORE_RANK (registered in its results file)	
torus T = 1024	blank / n-gram / retrace	0.509 / 0.578 / 0.845	SROPE_TORUS, seeds 1700-1707	`SROPE_TORUS_RESULTS.md`; `docs/audits/2026-10-20/srope_torus_floors_out.txt`
torus T = 128	retrace	0.938	DELTA_FREE, seeds 1600-1607	`DELTA_FREE_RESULTS.md`; `docs/audits/2026-10-20/delta_free_floors_out.txt`
torus in words	always-'nothing' / one-step reversal-copy / retrace	0.5054 / 0.5965 / 0.9337	TEXTWORLD registered eval stream	`docs/paper/expansion/scripts/language/tw_retrace_floor_out.txt`
torus in words	same	0.5120 / 0.6024 / 0.9365	TW_NORMSTEP eval stream	same
torus in words, revisits retrace cannot reach	best constant	0.506 ('nothing'; 856 of 6,381 revisits, 13.4%)	all 200 walks of the TEXTWORLD stream	`docs/paper/expansion/scripts/language/tw_strata_200walks_out.txt`
torus in words, ... whose cell holds an object (the revisits retrace gets wrong)	best constant ('nothing' excluded by construction)	0.088 ('cat'; 423 of 6,381, 6.6%)	same	same
ring, moves +1/-1, observation-step-free	n-gram / retrace	0.554 / 0.860	SIGNROUTE reference	`SIGNROUTE_RESULTS.md`

C. Per-seed tables

Accuracy at the training length on held-out maps, eval mode. SOLV / STAL / DESC = SOLVED / STALLED / DESCENDING; a number in brackets is the final-5% training loss.

C1. Torus sign batch (Secs. 5, 7.3; `SIGN_MATCHED_ANALYSIS.txt`)

seed	signed	|.|	softplus	index RoPE
0	1.000 SOLV (0.003)	0.931 DESC (0.201)	0.590 STAL (1.250)	0.730 STAL (0.983)
1	0.997 SOLV (0.003)	0.917 DESC (0.314)	0.527 STAL (1.510)	0.729 STAL (0.939)
2	1.000 SOLV (0.005)	0.868 STAL (0.492)	0.630 STAL (1.084)	0.720 STAL (0.931)
3	1.000 SOLV (0.008)	0.637 STAL (1.043)	0.972 DESC (0.158)	0.739 STAL (0.910)
4	1.000 SOLV (0.004)	0.690 DESC (0.962)	0.674 STAL (1.009)	0.737 STAL (0.928)
5	0.990 SOLV (0.003)	0.888 STAL (0.446)	0.935 DESC (0.295)	0.744 STAL (0.885)
6	1.000 SOLV (0.004)	0.922 STAL (0.349)	0.922 DESC (0.345)	0.722 STAL (0.956)

seed	signed	|.|	softplus	index RoPE
7	1.000 SOLV (0.004)	0.717 STAL (0.834)	0.995 SOLV (0.049)	0.728 STAL (0.976)
mean	0.998	0.821	0.781	0.731

Registered: |.| - signed -0.177 (permutation p 0.0002; 0/8 vs 8/8, Fisher p 0.0002); softplus - signed -0.218 (p 0.0003; 1/8 vs 8/8, p 0.0014); signed - RoPE +0.267 (p 0.0002; 8/8 vs 0/8, p 0.0002). r(final loss, accuracy) -0.992 over 32 runs.

C2. Torus, index depth (Sec. 6; `FACTORIAL_{L1,L4,LOOP}.json`)

seed	path, 1 layer	index, 1 layer	index, 4 layers	index, looped x4
500	1.000	0.719	0.814	0.868
501	1.000	0.731	0.870	0.861
502	1.000	0.688	0.874	0.860
503	1.000	0.739	0.881	0.868

Every path run SOLVED, every index run STALLED. Registered C0-C3: path beats index at one layer (+0.281), four layers (+0.141) and loop x4 (+0.136); depth and loop each help index (+0.140, +0.145); every firing a complete separation (p 0.029).

C3. Ring exchange rate, fresh seeds 2-7 (Sec. 6; `docs/audits/2026-09-27/fresh_seeds.txt`)

p_plus	path 1 layer	index 1 layer	gap to path at 1 / 2 / 3 index layers
0.5	1.000	0.710 +/- 0.020	+0.290 / +0.051 / +0.002
0.75	1.000	0.677 +/- 0.009	+0.323 / +0.038 / +0.002
0.9	1.000	0.824 +/- 0.007	+0.176 / +0.013 / +0.001
1.0	1.000	1.000	0 / 0 / 0

All eight seeds (`CANCEL_RESULTS.md`): index 1 / 2 / 3 layers 0.717 / 0.954 / 0.998 (p_plus 0.5), 0.676 / 0.969 / 0.998 (0.75), 0.825 / 0.988 / 0.999 (0.9), 1.000 (1.0); path 1.000 in every cell (32/32 SOLVED); registered exchange rate k = 3, 3, 3, 1 index layers.

C4. Rank (Sec. 7.2; `RANK_MI_ANALYSIS.txt`, `RANK_SEP_ANALYSIS.txt`, `RANK3_ANALYSIS.txt`, `RANK_SEP.json`, `RANK3.json`)

seed	shared r2	per-head r2	block-diag r4	per-head r3	shared r4	per-head r4
0	0.849 STAL (0.587)	0.656 STAL (1.049)	0.879 STAL (0.482)	0.942 DESC (0.243)	1.000 SOLV (0.004)	0.996 SOLV (0.004)
1	0.998 DESC (0.051)	1.000 SOLV (0.007)	0.936 DESC (0.302)	0.990 SOLV (0.003)	1.000 SOLV (0.003)	1.000 SOLV (0.010)
2	0.917 DESC (0.308)	1.000 SOLV (0.003)	1.000 SOLV (0.010)	1.000 SOLV (0.003)	1.000 SOLV (0.003)	1.000 SOLV (0.004)
3	0.917 DESC (0.321)	0.723 STAL (0.910)	0.945 DESC (0.234)	0.967 DESC (0.134)	1.000 SOLV (0.003)	1.000 SOLV (0.004)
4	0.947 DESC (0.177)	0.959 DESC (0.194)	0.964 DESC (0.202)	1.000 SOLV (0.003)	1.000 SOLV (0.002)	0.998 SOLV (0.002)

seed	shared r2	per-head r2	block-diag r4	per-head r3	shared r4	per-head r4
5	0.907 STAL (0.404)	0.949 DESC (0.211)	0.917 DESC (0.310)	1.000 SOLV (0.006)	0.994 SOLV (0.004)	1.000 SOLV (0.008)
6	0.773 STAL (0.675)	0.943 DESC (0.248)	1.000 SOLV (0.010)	1.000 SOLV (0.003)	1.000 SOLV (0.004)	1.000 SOLV (0.005)
7	0.842 STAL (0.605)	0.851 STAL (0.539)	0.947 DESC (0.166)	1.000 SOLV (0.005)	0.987 SOLV (0.010)	1.000 SOLV (0.003)
mean	0.894, 0/8	0.885, 2/8	0.948, 2/8	0.987, 6/8	0.998, 8/8	0.999, 8/8

Per-head r2 seeds 0-1 were a pilot run before registration (1/6 SOLVED on seeds 2-7). Registered: per-head rank 4 vs block-diagonal 8/8 vs 2/8 (Fisher and permutation p 0.0070); shared r4 vs block-diagonal the same (it also changes per-head rank); sharing (per-head r4 vs shared r4, +0.001) and W_out scale (block-diagonal vs per-head r2, +0.063) not detected; per-head r3 vs r2 +0.102 (permutation p 0.027; SOLVED 6/8 vs 2/8, Fisher p 0.13); r4 vs r3 +0.012, not detected. Shared r4 vs shared r2: 8/8 vs 0/8 (Fisher p 0.0002), +0.104 (p 0.0005).

Twice the budget (`LOOP_RANK_E1800_P1_ANALYSIS.txt`, 1800 epochs from scratch): shared r2 0/8 (final-5% losses 0.489, 0.383, 0.390, 0.340, 0.141, 0.231, 0.305, 0.571; 3 STALLED, 5 DESCENDING), accuracy 0.908; shared r4 7/8 (s5 STALLED at 0.180), 0.994.

C5. Sign, ring and torus (Sec. 7.3)

Ring dial (`RING_DIAL_VERDICTS.json`; full models, S = SOLVED):

seed	0.5 signed	0.5 |.|	0.75 signed	0.75 |.|	0.9 signed	0.9 |.|
30	1.000 S	1.000 S	1.000 S	0.994 S	1.000 S	1.000 S
31	1.000 S	0.977	1.000 S	0.958	1.000 S	0.999 S
32	1.000 S	0.987	1.000 S	0.957	1.000 S	0.885
33	1.000 S	0.964	1.000 S	0.954	1.000 S	1.000 S
34	1.000 S	1.000 S	1.000 S	0.964	1.000 S	1.000 S
35	1.000 S	0.944	1.000 S	0.967	1.000 S	0.825
36	1.000 S	0.977	1.000 S	0.993	1.000 S	0.960
37	1.000 S	0.931	1.000 S	0.958	1.000 S	0.964

Registered: p_plus 0.5 -0.028 (permutation p 0.007), 2/8 vs 8/8 (Fisher p 0.0070); 0.75 -0.032 (p 0.0002), 1/8 vs 8/8 (p 0.0014); at 0.9 the contrast did not reach the registered threshold (-0.046, below the 0.05 materiality; 4/8 vs 8/8, p 0.077).

Observation-step-free ring from scratch and constructed (`WRAP_EXISTENCE_ANALYSIS.txt`):

seed	signed	|.|	constructed |.|: frozen -> trained (max continuation loss)	constructed signed: frozen -> trained
600	1.000 S	0.905 STAL	1.000 -> 1.000 S (0.041)	1.000 -> 1.000 S (0.019)
601	1.000 S	0.949 STAL	1.000 -> 1.000 S (0.030)	1.000 -> 1.000 S (0.042)
602			1.000 -> 1.000 S (0.154)	1.000 -> 1.000 S (0.072)

seed	signed	|.|	constructed |.|: frozen -> trained (max continuation loss)	constructed signed: frozen -> trained
	1.000 S	0.853 STAL		
603	1.000 S	0.951 STAL	1.000 -> 1.000 S (0.046)	1.000 -> 1.000 S (0.017)
604	1.000 S	0.907 STAL	1.000 -> 1.000 S (0.024)	1.000 -> 1.000 S (0.039)
605	1.000 S	0.930 STAL	1.000 -> 1.000 S (0.032)	1.000 -> 1.000 S (0.028)
606	1.000 S	0.950 STAL	1.000 -> 1.000 S (0.031)	1.000 -> 1.000 S (0.112)
607	1.000 S	0.922 STAL	1.000 -> 1.000 S (0.036)	1.000 -> 1.000 S (0.015)

Registered: from scratch 0.921 vs 1.000, -0.079 (permutation p 0.0002), 0/8 vs 8/8 (Fisher p 0.0002); constructed |.| HELD (8/8 SOLVED, 0/8 LOST).

Observation-step-free torus (`GENERALITY_ANALYSIS.txt`):

seed	signed	|.|	constructed signed (lr 1e-4)	constructed |.| (lr 1e-4): peak loss, epoch of peak / of re-entry
700	1.000 S	0.874 STAL	1.000 S	1.000 S; 0.168, 29 / 153
701	1.000 S	0.707 STAL	1.000 S	1.000 S; 0.110, 40 / 121
702	1.000 S	0.946 DESC	1.000 S	1.000 S; 0.173, 19 / 204
703	1.000 S	0.879 STAL	1.000 S	1.000 S; 0.143, 52 / 149
704	1.000 S	0.892 STAL	1.000 S	1.000 S; 0.131, 13 / 119
705	1.000 S	0.907 STAL	1.000 S	1.000 S; 0.155, 13 / 148
706	1.000 S	0.877 STAL	1.000 S	1.000 S; 0.208, 36 / 164
707	1.000 S	0.960 DESC	1.000 S	1.000 S; 0.117, 23 / 139

Registered: from scratch 0.880 vs 1.000, -0.120 (p 0.0002), 0/8 vs 8/8; the constructed |.| model leaves the solved region on 6/8 seeds (peak loss 0.11-0.21; signed control 0.004-0.017) and ends SOLVED on 8/8 at 1.000: by the reading fixed before launch, perturbed and re-found.

Recipe and scale (`ROBUST_ANALYSIS.txt`, seeds 1400-1407; accuracy at the training length, class):

```
ring  seed | def:SO | def:AO | wd0:SO | wd0:AO | lrx3:SO | lrx3:AO | lrd3:SO | lrd3:AO
1400 | 1.0000 SOLV | 0.8230 STAL | 1.0000 SOLV | 0.9078 DESC | 1.0000 SOLV | 1.0000 DESC |
1.0000 SOLV | 0.8893 STAL
```

```
1401 | 1.0000 SOLV | 0.9155 STAL | 1.0000 SOLV | 0.6239 STAL | 1.0000 SOLV | 0.9596 STAL |
0.6194 STAL | 0.6221 STAL
1402 | 1.0000 SOLV | 0.9237 STAL | 1.0000 SOLV | 0.8878 STAL | 1.0000 SOLV | 0.9702 STAL |
1.0000 SOLV | 0.8319 STAL
1403 | 1.0000 SOLV | 0.9136 STAL | 1.0000 SOLV | 0.8800 STAL | 1.0000 SOLV | 0.9995 SOLV |
1.0000 SOLV | 0.9167 STAL
1404 | 1.0000 SOLV | 0.9118 STAL | 1.0000 SOLV | 0.9119 STAL | 1.0000 SOLV | 1.0000 SOLV |
1.0000 SOLV | 0.8313 STAL
1405 | 1.0000 SOLV | 0.8791 STAL | 1.0000 SOLV | 0.8858 STAL | 1.0000 SOLV | 0.9650 STAL |
1.0000 SOLV | 0.8526 STAL
1406 | 1.0000 SOLV | 0.9211 DESC | 1.0000 SOLV | 0.9110 STAL | 1.0000 SOLV | 1.0000 SOLV |
0.9794 STAL | 0.9062 STAL
1407 | 1.0000 SOLV | 0.9997 SOLV | 1.0000 SOLV | 0.9735 DESC | 1.0000 SOLV | 0.9998 SOLV |
1.0000 SOLV | 0.8801 STAL
torus 1400 SO 1.0000 SOLV | AO 0.9941 DESC     1404 SO 1.0000 SOLV | AO 0.9935 SOLV
      1401 SO 1.0000 SOLV | AO 0.9981 SOLV     1405 SO 1.0000 SOLV | AO 0.9706 DESC
      1402 SO 0.9941 SOLV | AO 0.9705 DESC     1406 SO 1.0000 SOLV | AO 0.9098 DESC
      1403 SO 1.0000 SOLV | AO 0.9454 DESC     1407 SO 1.0000 SOLV | AO 0.9544 DESC
```

Learning-rate sweep (`LRSWEEP_RESULTS.md`, seeds 1500-1507): per-seed table in H2.

C6. Text world (Sec. 8; `TEXTWORLD_ANALYSIS.txt`)

seed	path, 1 layer	index, 1 layer	index, 2 layers
0 (pilot)	0.881	0.504	0.695
1 (pilot)	0.998	0.504	0.723
2	0.999	0.503	0.927
3	0.995	0.506	0.722
4	0.999	0.505	0.926
5	1.000	0.506	0.706
6	1.000	0.506	0.760
7	0.882	0.504	0.720

Fresh seeds 2-7 (`docs/audits/2026-09-27/fresh_seeds.txt`): path 0.979 +/- 0.048 (6/6 SOLVED), index 1 layer 0.505 (0/6), index 2 layers 0.793 (0/6); path - index one layer +0.474, permutation and Fisher p 0.0022. All eight seeds: path 0.969 +/- 0.054 (7/8; seed 0 still descending), +0.464 (p 0.0002; Fisher p 0.0014); path - index two layers +0.197 (p 0.0009).

C7. Dyck-2

Per-seed values are in `DYCK_MDEPTH_RESULTS.json` (cells `T12|{RoPE,PoPE,MapWM,MapPoPE}|{1-4}L|L32D12`); every one of the eight seeds has a positive position effect at every depth (sign-flip p 0.0078 per depth).

C8. SOLVED-threshold sensitivity

Sign batches (`docs/audits/2026-10-15/reanalyses/solved_thresholds_out.md`; predictions and decision rules written before the re-analysis; two-sided Fisher p in brackets):

batch	contrast	loss < 0.02	loss < 0.05 (registered)	loss < 0.1	accuracy >= 0.99
SIGN_MATCHED torus	|.| vs signed	0/8 vs 8/8 (0.0002)	0/8 vs 8/8 (0.0002)	0/8 vs 8/8 (0.0002)	0/8 vs 7/8 (0.0014)
SIGN_MATCHED torus	softplus vs signed	0/8 vs 8/8 (0.0002)	1/8 vs 8/8 (0.0014)	1/8 vs 8/8 (0.0014)	1/8 vs 7/8 (0.0101)
RING_DIAL p 0.5	|.| vs signed	2/8 vs 8/8 (0.0070)	2/8 vs 8/8 (0.0070)	4/8 vs 8/8 (0.0769)	2/8 vs 8/8 (0.0070)
RING_DIAL p 0.75	|.| vs signed	1/8 vs 8/8 (0.0014)	1/8 vs 8/8 (0.0014)	2/8 vs 8/8 (0.0070)	2/8 vs 8/8 (0.0070)
WRAP_EXISTENCE ring	|.| vs signed	0/8 vs 8/8 (0.0002)	0/8 vs 8/8 (0.0002)	0/8 vs 8/8 (0.0002)	0/8 vs 8/8 (0.0002)
MECHANISM ring	|.| vs signed, same-direction start	0/8 vs 7/8 (0.0014)	0/8 vs 7/8 (0.0014)	0/8 vs 7/8 (0.0014)	0/8 vs 7/8 (0.0014)
ACCOUNTS T2 ring	frozen vs free move latents	1/8 vs 7/8 (0.0101)	1/8 vs 7/8 (0.0101)	1/8 vs 7/8 (0.0101)	1/8 vs 7/8 (0.0101)
GENERALITY torus	|.| vs signed	0/8 vs 8/8 (0.0002)	0/8 vs 8/8 (0.0002)	0/8 vs 8/8 (0.0002)	0/8 vs 8/8 (0.0002)
GENERALITY ring	CoPE vs signed	0/8 vs 8/8 (0.0002)	0/8 vs 8/8 (0.0002)	1/8 vs 8/8 (0.0014)	0/8 vs 8/8 (0.0002)

ROBUST (declared secondary, loss < 0.1): ring default 1/8 vs 8/8 (p 0.0014), weight decay 0 1/8 vs 8/8 (0.0014), lr / 3 0/8 vs 7/8 (0.0014), lr x3 4/8 vs 8/8 (0.077), torus 2 layers d 256 5/8 vs 8/8 (0.20); on that torus cell the accuracy contrast (0.967 vs 0.999, p 0.0005) carries the result. LRSWEEP (loss < 0.1, selection redone): 2/8 vs 8/8, same label.

Rank batches (analysis; read from the per-seed final-5% losses in C4): at loss < 0.02 the counts are 0 / 2 / 2 / 6 / 8 / 8 of 8 (shared r2, per-head r2, block-diagonal r4, per-head r3, shared r4, per-head r4), identical to the registered 0.05; at loss < 0.1 they are 1 / 2 / 2 / 6 / 8 / 8 (shared r2 seed 1 at 0.051 enters).

C9. Attention-dropout evaluation scale

Models trained with attention dropout expect the 1/(1 - p) attention scale; evaluated with it, below-ceiling one-layer runs score higher. The registered text-world evaluation re-run with the scale (`docs/audits/2026-10-04/DROPOUT_RESCORE.md`, all 24 runs) gives path 0.969 -> 0.993, index one layer 0.505 -> 0.506, index two layers 0.772 -> 0.772; path - index one layer +0.464 -> +0.488 (p 0.0002). The re-score was a declared secondary of RING_DIAL, RING_BUDGET, WRAP_EXISTENCE, GENERALITY, MECHANISM and WRAP_COPE and changed no label there; it was applied to SIGN_MATCHED by the same audit with no verdict changed. ACCOUNTS, TORUS_START, GAUGE and ROBUST were not re-scored (the correction does not apply at two layers). Main-text accuracies are eval mode without the scale except where marked re-scored (Sec. 8, Table 7, App. J).

C10. Sign route (Sec. 7.3; `SIGNROUTE_RESULTS.md`, registered; ring, observation-step-free, seeds 1800-1807)

Accuracy at T = 128 and run class (S SOLVED, T STALLED, D DESCENDING; none diverged):

```
lr     arm       1800   1801   1802   1803   1804   1805   1806   1807
1e-3   signed    1.000S 1.000S 1.000S 1.000S 1.000S 1.000S 1.000S 1.000S
1e-3   abs       0.912T 0.874T 0.902T 0.926T 0.918T 0.968T 0.939T 0.902T
1e-3   hybrid    1.000S 1.000S 1.000S 1.000S 1.000S 1.000S 1.000S 1.000S
1e-3   abs_tied  0.748T 0.748T 0.672T 0.723T 0.749T 0.751T 0.748T 0.770T
1e-3   exp       0.904T 0.973T 0.929T 0.957T 0.852T 0.958T 0.936D 0.972S
1e-3   softplus  0.841T 0.794T 1.000S 0.984T 0.734T 0.861T 0.909T 0.906D
3e-3   signed    1.000S 1.000S 1.000S 1.000S 1.000S 1.000S 1.000S 1.000S
3e-3   abs       0.921T 0.944T 0.935T 0.968T 0.949T 1.000S 0.945T 1.000S
3e-3   hybrid    1.000S 1.000S 1.000S 1.000S 1.000S 1.000S 1.000S 1.000S
3e-3   abs_tied  0.743T 0.748T 0.652T 0.778T 0.775T 0.748T 0.774T 0.769T
3e-3   exp       0.897T 0.959D 1.000S 1.000D 0.902T 0.967D 0.960D 0.977D
3e-3   softplus  0.954T 0.959D 0.999S 1.000S 0.964T 0.975D 1.000S 0.880T
```

D. Constructions

D1. The rank-2 projection (Sec. 7.2; `RANK_PROJ_RESULTS.md`)

Each solved shared rank-4 model (seeds 0-7, a stored batch) is projected onto the top two directions of its action latent over the vocabulary (top-2 spectral energy 0.99952-0.99994), every other weight kept, and installed in the shared rank-2 architecture. Frozen, the projections score 0.9955 at T = 1024 against their sources' 0.9966 (at least 0.981 on every seed; per seed 0.9993, 1.0000, 0.9988, 1.0000, 0.9850, 0.9806, 1.0000, 1.0000). Trained 900 epochs from a learning-rate restart at 1e-3, 7/8 are SOLVED (one DESCENDING), as for the rank-4 control trained from its own solution (7/8); rank 2 continued from its own 900-epoch weights solves 1/8. Registered verdict: STABLE.

D2. Per channel (Lemma P1.0 in v3)

For a signed step Delta and omega != 0, Delta' = Delta mod (2 pi / |omega|) lies in [0, 2 pi / |omega|) and omega Delta' = omega Delta (mod 2 pi) for either sign of omega. The model reads each channel's angle only through cos and sin, so the model with Delta' is the same function.

D3. The balanced |.| construction (Proposition 3(a), E2; the registered existence test)

Given a solved signed observation-step-free model with m stepping move tokens (m = 2 ring, 4 torus; r = 4):

1. Re-solve W_in so that the move latents are c e_1, ..., c e_m (c = the source's mean move-latent norm). W_in acts only on the move embeddings, so this is a free gauge.
2. Per channel set omega' = max(|omega|, 1) and the steps to remainder(omega Delta, 2 pi) / omega' (angles in [0, 2 pi) for either sign of omega).

3. Set W_out[:, j] = step_j / c for the m move columns and 0 elsewhere; copy every other weight.

The signed control is built by the same function with the unwrapped signed angle (steps omega Delta / omega'), so the two differ only in the wrap. On the torus m = 4 = r, exactly at Proposition 3(a)'s limit.

Exactness. Ring: frozen accuracy = source on 8/8; maximum angle error 1.2e-6 rad (|.|), 6.1e-7 (signed); maximum logit difference 5.1e-4 at T = 128. Torus: frozen = source on 8/8; maximum angle error 1.4e-6 rad (|.|), 5.8e-7 (signed). *Weight scale.* Ring: maximum |W_out| 0.31-0.38 in the signed sources, 1.07-1.70 built; |W_in| 0.15-0.21 -> 0.19-0.33. Torus: maximum |W_out| 0.48-0.58 -> 1.23-1.55, maximum |omega| 0.68-0.96 -> 1.00. Within about 5x of the trained scale.

Why the scale matters. A construction that keeps the source's W_in and solves W_out at minimum norm is also exact, but in a solved signed model the two move latents are nearly antiparallel (cosine -0.987 to -0.998), so W_out must be huge (up to 16,320 against 0.31-0.38). Adam's per-entry steps then lose that solution in the warm-up (8/8 LOST) before re-finding it (7/8 SOLVED). An existence test is informative only near a trained scale, which the balanced construction provides.

D4. The full |.| arms

To show that the task, not only a given signed table, is representable by a full |.| model (observations stepping), W_in is chosen to project onto two directions orthogonal to all 17 observation embeddings and W_out is solved per channel (`docs/audits/2026-10-12/ring_dial_audit.md` M1). In the real modules: |.| steps >= 0, observation-token |Delta| <= 1.7e-6, cos / sin equal to the signed code's to 6.3e-4 over a 512-step walk, circular mismatch <= 4.6e-4 on 2,000 same-cell revisit pairs.

D5. Softplus (Proposition 3(f), E2)

Softplus cannot reach 0, but observation steps need only be 0 mod 2 pi. With W_in annihilating every observation embedding, each observation steps softplus(0) = ln 2 in every channel, and omega'_c = 2 pi / ln 2 makes that exactly one turn; the move columns are set to softplus^{-1}((remainder(Theta, 2 pi) + 2 pi) / omega'). Built in the real softplus model from a solved ring model (`docs/paper/review_draft5/theory_check_extra.py` (2), CPU): minimum step 0.693 > 0, angle error mod 2 pi 8.4e-15 rad over all 19 tokens, logits equal to the source's to 2.8e-11 (logit scale 21.6). On the torus m = 4 = r, so the same construction applies.

D6. Continuations and the LOST rule

A constructed model is trained further on a fresh walk stream and compared with its signed control at the same seed. *LOST* (registered): the run's maximum per-epoch training loss during the continuation exceeds 0.3, or exceeds 10x its control's maximum at that seed and 0.05. *HELD* = SOLVED on at least n - 1 seeds, mean accuracy at least control - 0.01, and LOST on at most one seed. Ring (lr 1e-3): HELD, 8/8 SOLVED, 0/8 LOST (maximum per-epoch loss 0.024-0.154 against the control's 0.015-0.112). Torus (lr

1e-4): LOST on 6/8 and SOLVED on 8/8 at the end, read by the rule fixed before launch as perturbed and re-found; at lr 1e-3 (descriptive) 6/8 SOLVED (0.988).

D7. What the held model keeps (ACCOUNTS T1, registered)

Weighted by usage (mean query norm at move tokens times mean key norm at observation tokens, per channel), the held ring |.| model keeps its cancelling code from 0.99 at construction to 0.98 after training; unweighted (every channel equal) the share falls from 0.87 to 0.18, because channels attention does not read leave the code. Weight decay 0 vs 0.05 changes nothing detected (-0.003).

E. Theory

Labels used here: **proved** (a proof is given), **cited** (a published construction applied to our setting), **conjecture** (a derivation whose assumptions or dynamics are not checked). Main-text numbering: Proposition 1 = E6.1; Proposition 2 = E7.1 + E7.2; Proposition 3 = E2 (a), (f), (g); Lemma 1 = E3; Proposition 4 = E8.1-E8.6; Proposition 5 and Conjecture 1 = E9.2-E9.3; Remark 2 = E10; Proposition 6 = E11; Proposition 7 = E12; Proposition 8 = E5; Remark 3 = E13; Remark 1 = E6.2. The check scripts named "(check)" are CPU-only and eval-only; those in `docs/paper/appendix_checks/` were refereed with the v3 theory appendix, those in `docs/paper/expansion/scripts/` are listed in App. O.

Notation. A rotary channel is c = (h, b), head h and block b, acting on query and key coordinates (2b, 2b + 1) of head h; R(phi) is the 2D rotation. The phase is theta_{t,c} = omega_c sum_{u <= t} S_{c, x_u}, with S = W_out W_in E the C x V step table (rank at most r) and Theta = diag(omega) S the angle table; omega is free in sign. For token x the latent is z_x = W_in e_x and channel c's readout row is w_c, so the pre-nonlinearity step is w_c . z_x and the angle a_c(x) = omega_c w_c . z_x (signed), omega_c |w_c . z_x| (|.|), omega_c softplus(w_c . z_x) or omega_c exp(w_c . z_x). For an inverse pair of tokens (x, y), such as opposite moves, write the clock k_c = a_c(x) + a_c(y), the map m_c = a_c(y) and delta_c = (a_c(y) - a_c(x)) / 2. Channel c *cancels through zero* if k_c = 0 with a_c(x) ! = 0, *cancels by wrapping* if k_c is a non-zero multiple of 2 pi, and is *blind* if both angles are 0 mod 2 pi. *Usage* q_c of a channel is its mean query norm at move tokens times its mean key norm at observation tokens (exact per token at layer 0).

E1. Score identity (proved; the identity is Mamba-3's Proposition 3)

With alpha piecewise constant, alpha = Delta_u on (u - 1, u], the homogeneous part of Howard et al.'s integrator, dF/dt = -s alpha(t) F, has propagator F(t) = exp(-s (Phi_t - Phi_{t'})) F(t'). With s = i omega, channel c of the rotary score is q_c^T R(theta_s - theta_t) k_c = Re(conj(q~*c) k~_c exp(-i omega_c (Phi_t - Phi_s))), with q~ = q*{2b} + i q_{2b+1}.

Proof. On each interval the ODE is linear with constant coefficient -s Delta_u; flows over tokens compose by adding exponents. R(theta) acts on q~ as multiplication by e^{i theta}, so (R(theta_t) q)^T (R(theta_s) k) = Re(conj(q~) k~ e^{-i (theta_t - theta_s)}). [] With input, the zero-order-hold solution is F(t) = e^{-s Phi_t} F(0) + sum_{u <= t} e^{-s (Phi_t - Phi_u)} ((1 - e^{-s Delta_u}) / s) f_u; a zero-rate token writes nothing there, whereas in attention a zero-step token still writes its key and value at full strength. *Check*

(`a0_propagator.py`; `review_draft5/theory_check_extra.py` (1)): 3e-14 (propagator), 4e-15 (score), 1.1e-14 (input term).

E2. Representability by wrapping (proved)

(a) At most r stepping tokens. Let S be a signed table with at most m <= r non-zero columns (stepping tokens x_1, ..., x_m), whose embeddings are linearly independent modulo the span of the other embeddings (generic when V <= d_model, here 19 or 21 <= 128; automatic in observation-step-free models). Then a |.| model with the same embeddings and content weights has an angle table equal to Theta mod 2 pi, entrywise.

Proof. Pick W_in' with W_in' e_{x_j} = u_j (the j-th unit vector of R^r) and W_in' e_y = 0 for every other token. Set omega' = max(|omega|, 1) per channel and W_out'[:, j] = remainder(Theta_{., x_j}, 2 pi) / omega' >= 0, other columns zero. Then W_out' W_in' E is entrywise non-negative, so |.| leaves it unchanged, and it reproduces each stepping column mod 2 pi. [] *Full arms:* choosing W_in' to annihilate every observation embedding leaves only the moves stepping, so every |.| model in the paper can represent any code in which only the moves step. *Check* (`p1_wrap.py`): angle error <= 7e-14, logits <= 8e-14.

(f) Softplus. Under the same assumptions, send the stepping embeddings to u_1, ..., u_m and every other embedding to 0, set omega'*c = 2 pi / ln 2 for every c, and W_out'[c, j] = softplus^{-1}((remainder(Theta*{c,x_j}, 2 pi) + 2 pi) / omega'_c). Every non-stepping token then steps softplus(0) = ln 2, an angle of exactly 2 pi, and every stepping token's angle is Theta mod 2 pi. *Proof.* softplus is a bijection from R onto (0, inf); the targets lie in [ln 2, 2 ln 2); omega' ln 2 = 2 pi. [] *Check* (`theory_check_extra.py` (2)): built in the real softplus model, logits equal to the source's to 2.8e-11.

(g) Exp. The same construction with omega'*c = 2 pi and W_out'[c, j] = ln((remainder(Theta*{c,x_j}, 2 pi) + 2 pi) / 2 pi) makes every non-stepping token step exp(0) = 1, an angle of exactly 2 pi, and every stepping token's angle Theta mod 2 pi. *Proof.* exp is a bijection from R onto (0, inf); the targets (remainder + 2 pi) / 2 pi lie in [1, 2). [] Not built in a trained model.

What representability does not cover. For a rank-1 |.| model and two stepping tokens with steps of strictly opposite sign, exact representation mod 2 pi fails for almost every signed table (v3 P1(c), proved); rank-1 non-negative codes approximate any table mod 2 pi at a large angle scale (P1(d), Kronecker). Every claim in the paper uses (a), (f) or the full-arm construction, which are exact.

E3. The relative-sign gauge (Lemma 1; proved)

Take a MapWM model with any number of layers and a set F of channels. Form a second model by (a) negating every token's step in each channel of F (by negating omega_c; in the signed class equivalently row c of W_out, which is not available in the |.| class since |.| is applied after W_out) and (b) negating output coordinate h d_head + 2b + 1 of the query and key projections (weight row and bias entry, every layer) for each c = (h, b) in F. The two models have identical attention scores at every layer, hence identical logits.

Proof. With J = diag(1, -1), J R(phi) J = R(-phi) and J is orthogonal. (a) replaces theta_{t,c} by -theta_{t,c}; (b) replaces q_c, k_c by J q_c, J k_c, exactly including biases. The channel term becomes (J q_c)^T R(-(theta_s - theta_t)) (J k_c) = q_c^T R(theta_s - theta_t) k_c.

Every layer uses the same cos / sin, so the scores agree layer by layer. [] *Check* (`t1_gauge.py`): 35 channels flipped and reflected give bitwise-identical logits (control without the reflection: 6.6).

Corollaries. (i) The |.| class realises a signed angle table Theta exactly as real numbers, with the same latent, iff every row of Theta is sign-coherent. (ii) It therefore excludes no map *function* for which a code with sign-coherent rows exists, and that code needs no wrapping: one stepping token (parity, Z_m counting) or two moves that step the same way (+1 and +2: a(+2) = 2 a(+1)). (iii) Up to the gauge, the excluded tables are exactly those with a row containing entries of strictly opposite sign. For an inverse pair every through-zero cancelling channel is such a row; cancellation by wrapping, (a, 2 pi - a), is allowed whenever E2 applies. The constraint removes the through-zero branch of the inverse-pair problem, not cancellation.

E4. One channel's geometry (proved)

Take one channel and an inverse pair (x, y) with angles (a_x, a_y), everything else fixed; the loss is a function on the angle torus. Write sigma = a_x + a_y.

E4.1 (cut circle). (a) The closed quadrant [0, inf)^2 maps onto the torus, so the signed and |.| classes attain the same loss values in this channel. (b) The non-blind cancelling set in the quadrant consists of the segments sigma = 2 pi k, k >= 1; the through-zero line meets the quadrant only at the blind origin. (c) From a start in the open quadrant with sigma < 2 pi, the infimum distance (in angle coordinates) to a non-blind cancelling point is min(sigma, 2 pi - sigma) / sqrt 2 in the signed class and (2 pi - sigma) / sqrt 2 in the |.| class; they coincide iff sigma >= pi. (d) Any continuous |.| path from |sigma| < pi to a non-blind cancelling point passes through |sigma| = pi.

Proof. (a), (b) from the quotient map. (c) The distance to sigma = 2 pi k is |sigma - 2 pi k| / sqrt 2; the |.| class has only k >= 1. (d) |sigma| is continuous along a |.| path and a non-blind cancelling point has |sigma| >= 2 pi. [] *Check* (`p3_p5_geometry.py`): a brute-force grid matches both formulas to 2e-4.

E4.2 (sticky wall). Let s be the pre-|.| step of one token in one channel and a = omega |s|, with L(s) = l(|s|) and l continuously differentiable near 0. (a) For s != 0, dL/ds = omega sign(s) dL/da. (b) If l'(0) > 0, s = 0 is a strict local minimum of L along s with every other parameter fixed. (c) Gradient descent on s with step eta, l' in [mu, M] on [0, rho], mu > 0, eta M <= rho and |s_0| <= rho, gives |s_{k+1}| <= max(|s_k| - eta mu, eta M). *Proof.* Chain rule; first-order expansion; case split. [] In parameter space s = w_c . z_x is shared across tokens and channels, so (b) is not a local minimum of the loss. In trained |.| models the wall is the cone boundary of E8.2 (App. H7).

E5. CARoPE cannot cancel (Proposition 8; proved)

CARoPE (faithful, `model_mono_baselines.MapFormerWM_CARoPEF`) advances channel i of head h by f_h(x)^i per token, with f in (0, 1), so every per-token advance lies in (0, 1]; channel 0 advances by exactly 1, an index clock. (a) The phase of any interval of k >= 1 tokens lies in (0, k]. The shortest return spans 3 tokens between the move query and the observation key, and 4 between two observation tokens, so its phase lies in (0, 4], a subset of (0, 2 pi): no channel, at any parameter value, reads every return at phase 0. (b) Fix eps in (0, pi -

2). If a channel reads each return of an excursion of J moves out and J back within eps of 0 mod 2 pi, then the move-driven phase of any interval with at most 2J moves lies in [0, 2 eps]: every channel is either a clock or blind over that range.

Proof. (a) Sum k numbers in (0, 1]. (b) The depth-j return phase P_j satisfies P_1 in (0, 3] and P_{j+1} - P_j in (0, 4]; the arc (eps, 2 pi - eps) is longer than 4 when eps < pi - 2, so a sequence starting in (0, eps] with increments at most 4 cannot jump it; induction gives P_j in (0, eps], hence J (c_+ + c_-) <= eps. [] *Check* (`p6_carope.py`): eight trained ring CARoPE models; 10,331 random draws satisfy the bound.

E6. Path integration supplies the prefix sum; index attention needs a count

E6.1 (Proposition 1; proved). Tokens interleave moves and observations, [a_1, o_1, a_2, o_2, ...]; o_t is predicted at the token a_t. Let P_t be the unwrapped displacement after move t. The map is redrawn per evaluation, so the only source of o_t is an earlier observation o_s with P_s = P_t (mod N per axis). In MapWM, with observation steps zero and a fixed step vector per move, theta_t = omega P_t and score(t, s) = q^T R(omega (P_t - P_s)) k: the predicate is computed outside attention, exactly (a float32 cumsum), for every gap and every T. In a channel whose frequency times the move step is a multiple of 2 pi / N, the score is N-periodic in P, and wrap-around costs nothing. [] In an index-RoPE layer score(t, s) = q(h_t)^T R((t - s) Omega) k(h_s): position enters only through t - s, so the predicate "the moves in (s, t] sum to zero" must be written into the residual stream h.

E6.2 (Remark 1; cited). *Route A, a global count.* Layer 1: one head with constant query and key attends uniformly to the prefix; its value carries the move vector on move tokens, so its output is P_t / n_t. A feature proportional to 1/n_t is available in the same layer (for example a head with a fixed self-attention bonus). The MLP rescales to P_t and forms periodic features cos(2 pi k P_t / N), sin(2 pi k P_t / N); layer 2 matches on them. This is Liu et al.'s (2023) Lemma 6 followed by a match, and Yao et al.'s (2021) depth-2 Dyck construction (as summarised in Liu et al., App. A.5). In RASP terms the task has depth 2 in 1D and 2D alike. Its costs: a normaliser (Lemma 6 uses a padding token and an absolute channel log(2T - t); our torus and ring sequences have neither, Dyck has a BOS token); precision 1/n_t (1/256 at the end of a ring walk, 1/2048 per axis at the end of a torus walk); MLP width 4nT and weights up to 4nT + 2 in Lemma 6's mod gate (32 x 256 for the ring, 64 x 2048 per axis for the torus; our MLP has width 512); and sensitivity growing with the gap (Hahn and Rofin 2024, Cor. 5 and Thm. 6, Boolean inputs; used for direction only). *Route B, local return rules.* Retrace (while the current run reverses the previous one, the cell is the one seen 2j moves, 4j tokens, earlier: a bounded-window rule) and the one-lap clock (when the walk never reverses, the cell N moves back) need no count. A return that spans k direction runs is a function of those k runs; selecting its offset needs a summary of k runs, and stacking layers extends the reach. The reach is measured (App. F), not derived.

E7. Clock, map, the position-factor bound and the tick rule (proved)

E7.0 (phase decomposition). Over an interval with n_x moves x and n_y moves y, displacement d = n_y - n_x, the phase of channel c is a_c(y) n_y + a_c(x) n_x = m_c d + k_c n_x, exactly.

E7.1 (Proposition 2(a), a bound on the position factor). Let a revisit at displacement 0 have n backward moves; its phase in channel c is k_c n. For usage weights q_c (sum 1), the same-cell position factor F(n) = sum_c q_c cos(k_c n) equals 1 at n = 0; if a usage share s of the channels has kappa <= |k_c| <= K (unwrapped), then F(n) <= 1 - s (1 - cos(kappa n)) for n <= pi / K.

Proof. phi = m d + k n with d = 0. For n <= pi / K, kappa n <= |k_c| n <= pi, and cos is decreasing on [0, pi], so cos(k_c n) <= cos(kappa n) on the share s and <= 1 elsewhere. [] The bound covers n <= pi / K only. It says how fast a clock erodes the position factor of a revisit; it is not a theorem that a clocked code loses long-gap revisits, and near |k_c| n = pi it says nothing: two signed torus models started past pi solve with 0.43 and 0.99 of their usage on channels whose per-pair clock is near pi (App. H7). Index RoPE is the extreme case: every token, observations included, advances every channel, so the whole code is clock. Wrap-only revisits also need the lattice condition N m_c = 0 mod 2 pi.

E7.1' (the measured clock / map ratio). Every clock / map number in the paper (Table 4, G3, H7) is rho_cm = sqrt(sum_{pairs} sum_c q_c wrap(k_c)^2 / sum_{pairs} sum_c q_c (wrap(a_c(x))^2 + wrap(a_c(y))^2) / 2), with the usage q normalised over the channels of both heads (one value per run), or over one head's channels (per head). The symmetric denominator makes it invariant to which move of a pair is called y (rule 8); for unwrapped |.| channels |k_c| >= |m_c| (E8.1), so rho_cm >= 1 there by construction. Generator: `scripts/sign/clock_table.py`.

E7.2 (Proposition 2(b), tick rule). Split each move's contribution in channel c into the map part and a tick tau_c, the phase every move adds whatever its direction. If the target at t is the key at the last visit to the same cell, matched pairs come at lags L drawn from a distribution, and with the map part cancelling over the closed loop the matched phase difference is L tau_c. A channel scores matched pairs consistently only if L tau_c is constant mod 2 pi over the lags that occur, i.e. tau_c = 0 mod 2 pi / g, g the greatest common divisor of the return lags (g = 2 on a bipartite lattice such as the even torus). Otherwise its matched score decorrelates once |wrap(tau_c)| sd(L) exceeds about 1 rad. [] A pi tick is therefore invisible to revisits on the even torus: it encodes the parity of the cell.

E8. Route inventory (Proposition 4; proved)

E8.1 (the |.| clock floor). In the |.| class, for an inverse pair in channel c (one omega_c of either sign): (a) a_c(x) and a_c(y) have the same sign, so |k_c| = |a_c(x)| + |a_c(y)| >= max(|m_c|, 2 |delta_c|), with |k_c| = |m_c| iff a_c(x) = 0 and |k_c| = 2 |delta_c| iff one of the two angles is 0. (b) As real numbers k_c = 0 iff the channel is blind, so the non-blind zero-clock points are exactly the wrap sheets |k_c| in 2 pi Z_{>=1}. (c) On a zero-displacement revisit with n = g/2 backward moves the phase is k_c g / 2 >= |m_c| g / 2; if |k_c| < 2 pi it is within eps of 0 mod 2 pi only for g <= 2 eps / |k_c| <= 2 eps / |m_c| (or g near a multiple of 4 pi / |k_c|).

Proof. (a) same sign of omega_c times non-negative numbers; triangle equality cases. (b) from (a). (c) from E7.0. [] An unwrapped |.| channel is at least as much clock as map, and the better it resolves neighbouring cells, the shorter the backtracks it can read. E4.1 is the geometric form of (b). *Check* (`sign/sign_clock.py`): zero violations of the inequality in 4,059 sub-pi |.| channels.

E8.2 (the collective route of the signed class). Signed class, inverse pair, readout W in R^{C x r} of full column rank, omega_c != 0. (a) The clock vector is k = diag(omega) W u with u = z_x + z_y. The latent pairs at which every channel has zero clock form the r-dimensional linear subspace {u = 0}; the map part (z_y - z_x)/2 is free on it, so it contains non-blind points. (b) Along z_y(tau) = (1 - tau) z_y - tau z_x (W, omega fixed), u(tau) = (1 - tau) u(0), and every channel's clock is (1 - tau) k_c(0). (c) For the quadratic surrogate R(z) = (1/2) sum_c q_c k_c^2, gradient flow on (z_x, z_y) gives du/dt = -2 H u with H = W^T diag(omega) diag(q) diag(omega) W, positive definite when the used channels' rows span R^r, and the map part is constant: u(t) = exp(-2 H t) u(0) -> 0 from every start, and {u = 0} is the only stationary set. (d) For almost every (W, omega), {u = 0} is the only latent set at which every channel cancels mod 2 pi when C > r.

Proof. (a)-(c) linear algebra: k_c = omega_c (w_c . z_x + w_c . z_y) and dR/dz_x = dR/dz_y = H u. (d) diag(omega) W u is in 2 pi Z^C iff (2 pi n_c / omega_c)_c lies in col(W); for fixed n != 0 the map omega -> (2 pi n_c / omega_c) is a diffeomorphism of (R \ 0)^C onto itself, so the preimage of the r-dimensional subspace col(W) is null when r < C; take the countable union over n. [] The surrogate guarantee concerns the clock term alone; with the content frozen, gradient descent on the step map rarely takes this route from the same-direction start (App. H7).

E8.3 (no collective route for |.| on an inverse pair). |.| class, C channels, rank r. (a) The latent route of E8.2(b) ends, under |.|, at a(y) = a(x) in every channel, since |w . (-z_x)| = |w . z_x|: a clock. (b) For almost every (W, omega), at that fixed (W, omega) no latent pair (z_x, z_y) makes more than r channels of each sign-coherence class (pre-|.| steps of equal or of opposite sign), hence more than 2r channels, cancel non-blindly. (c) Hence every |.| map code with more than 2r cancelling channels has its readout tuned channel by channel: in step units the channel-c condition is |w_c . z_x| + |w_c . z_y| = 2 pi n_c / |omega_c|, an offset that differs across the frequency ladder (omega_max / omega_min = 32 on the ring, 64 on the torus).

Proof. (a) evenness of |.|. (b) Fix a set S of 2r + 1 channels, a sign pattern per channel and non-zero integers n_c (a non-blind |.| channel cancels only by wrapping, E8.1(b)). On the sign region k_c = omega_c (e_c(x) w_c . z_x + e_c(y) w_c . z_y) is linear in zeta = (z_x, z_y) in R^{2r}, k = diag(omega) N zeta, and the conditions k_c = 2 pi n_c for c in S hold for some zeta iff (2 pi n_c / omega_c)_{c in S} lies in col(N), a subspace of dimension <= 2r < 2r + 1 that does not depend on omega. As in E8.2(d) the set of omega_S for which it holds is null; Fubini and a countable union finish. Within one coherence class the sum depends on zeta only through z_x + z_y or z_x - z_y, an r-vector, giving the per-class bound. [] *Check* (`sign/prop_checks.py` (1)): at the default readouts of eight seeds, least-squares latents never cancel 9 channels exactly (smallest maximum error 0.027 rad); 8 channels cancel exactly (1e-14) only when they split 4 / 4 between the coherence classes.

E8.4 (same-direction moves). For a relation n_x a(x) + n_y a(y) = 0 (+1/-1: n = (1, 1); +1/+2: n = (-2, 1)): signed has one linear zero set; |.| with n_x n_y < 0 (no inverse pair) has two collective zero sets, {n_x z_x + n_y z_y = 0} and its mirror under z_y -> -z_y, both with non-blind points; with n_x n_y > 0 (an inverse pair) neither is non-blind. In the quadratic surrogate restricted to a sign region in which both coherence classes contain channels whose rows span R^r, the only latent stationary point is the blind origin, and the latent flow contracts toward it until some step reaches zero. *Proof.* As E8.2-E8.3; on the region the gradients are n_x (H_+ u_+ + H_- u_-) and n_y (H_+ u_+ - H_- u_-) with

H_+- positive definite, vanishing iff z_x = z_y = 0. [] For +1/-1 every channel's residual is the clock k_c >= 0, so the |.| latent gradient that removes the clock is a contraction toward the wall, while the signed one projects onto u = 0 and leaves the map alone (`prop_checks.py` (2): signed ends at |u| / |z| < 1e-3 with the map part unchanged; |.| ends at the blind origin).

E8.5 (rank; the clock shares every channel at rank D). One head, latent dimension r, move latents in R^r; for a D-dimensional map write the map frame U = [d_1 ... d_D] in R^{r x D} (half-differences of inverse pairs), rank U = D, and the common clock c (the latent step shared by every move, plus the mean observation step). Channel i reads a phase per unit displacement k_i = U^T f_i in R^D (f_i its readout row in diag(omega) W_out) and a drift tau_i = per move. (a) r = D, U invertible: c = U beta for a unique phantom displacement beta (cells per move), and tau_i = in every channel. A channel is drift-free iff its wavevector is orthogonal to beta, i.e. iff it does not resolve the drift direction. So while c != 0 no channel is both drift-free and a full D-dimensional map, and the only way to a clean code is c = 0 (the collective route). (b) r >= D + 1: split c = c_par + c_perp with c_perp orthogonal to range(U). If c_perp != 0, the readout change delta = -tau_i c_perp / |c_perp|^2 cancels channel i's drift with U^T delta = 0 exactly: the map is untouched to all orders (*parking*). Independent clock directions can be parked only up to r - D of them: on the torus a single common clock needs r >= 3, two independent pair clocks r >= 4. (c) At r = D the minimum-norm cancelling readout change sets k_i <- k_i - (k_i . bhat) bhat: it removes the channel's resolution along the drift direction (*hiding*). (d) At D = 1 hiding sets k_i = 0 (the channel dies), and the drift a channel sees in cells, tau_i / k_i = beta, is the same for every channel and every readout: no readout change alters the drift-to-map ratio; only shrinking c or growing U does.

Proof. (a) c in R^D = range(U) when U is invertible; = . (b) U^T c_perp = 0 by definition; for q independent clock directions solve = -tau_j, a linear system with independent coefficient vectors. (c) = ; Cauchy-Schwarz; the minimiser has U^T delta parallel to beta. (d) D = 1: k_i = U f_i, tau_i = c f_i, ratio c / U. [] Per head, because each head's channels read only that head's r-dimensional image: a block-diagonal rank-4 latent gives each head rank 2 and no c_perp. E8.5 and E8.2 use different clocks (the common step c, and the per-pair sums u); both vanish on a map code.

E8.6 (route inventory, stated once). For an inverse-pair map task, a class reaches the zero-clock set from a generic small start through the collective route iff its clock is a linear function of one latent vector whose zero set contains non-blind points, and through the per-channel route iff each channel has a clock-free, full-map readout reachable without crossing a fold. Then: signed, r >= D + 1: both (E8.2, E8.5(b)); signed, r = D: collective only (E8.5(a)); signed with frozen move latents: per-channel only (each channel can still cross zero through W_out); |.|: neither from a small start (E8.3: the latent route ends at a clock and no latent choice wraps more than 2r channels; E8.1 + E4.1: the per-channel zero-clock points lie at |k| = 2 pi n, across |k| = pi). A hybrid with readout |w_c| and signed latent, a_c(x) = omega_c |w_c| . z_x, has the signed class's collective route (E8.2 with W replaced by |W|) and its near-side cancelling point k = 0.

E9. Hiding at rank D (Proposition 5; Conjecture 1)

Assumptions. (A1) Small drift: for the gaps that carry gradient, g |tau_i| << 1, so the drift loss is L_d = (1/2) sum_i a_i tau_i^2 with a_i > 0. (A2) The map loss depends on the

frame only through the spectrum of the head's positional precision P = sum_i A_i k_i k_i^T, L_m = sum_j f(lambda_j(P)) with f decreasing and convex, plus a norm cost on readout rows. (A3) U near-isometric on the readout's span, so rotating k_i preserves |f_i|. Write r_beta = (bhat^T P bhat) / (tr P / D), the head's resolution along its own drift direction relative to its mean.

E9.1 (two routes). A change of parameters lowers tau_i only by changing c (shared by every channel of the head; the collective route) or by changing f_i with = -tau_i (the local route); E8.5 says what the local route costs.

E9.2 (Proposition 5; proved under A1-A3). (i) D >= 2, P isotropic: rotating any channel's k_i by a small angle eps from bhat toward a direction v orthogonal to bhat changes L_m by 0 at first order (a rotation leaves tr P unchanged) and changes L_d by -eps a_i |k_i|^2 |beta|^2 sin(2 phi_i) / 2 < 0 for every channel not already aligned with or orthogonal to bhat. So gradient flow on the readout moves resolution off the drift axis at once, for any beta != 0. (ii) D = 1: the only readout move that lowers L_d is shrinking |k_i|, whose first-order map cost is -f'(lambda) A_i > 0; it does not change beta (E8.5(d)). (iii) r >= D + 1 with c_perp != 0: parking lowers L_d at rate a_i |c_perp|^2 per channel with zero change in L_m at all orders. *Derivation.* k_i = |k_i| (cos phi bhat + sin phi v), tau_i = |k_i| |beta| cos phi, and d/deps (a_i / 2) tau_i^2 = -a_i |k_i|^2 |beta|^2 cos phi sin phi. []

E9.3 (Conjecture 1). Under A1 the gradient of L_d with respect to beta is P_a beta (P_a = sum_i a_i k_i k_i^T), so the collective route removes the drift along bhat at a rate proportional to bhat^T P_a bhat, the quantity hiding lowers. At rank D >= 2 this is a positive feedback (drift, then hiding, then slower removal, then persisting drift) that leaves the drift in place or produces a long plateau; the clean state (beta = 0, P isotropic) remains a stable fixed point, which is the projection that the rank-2 existence test installs and training holds. Two refinements: long gaps carry the loss but not the gradient (once g |tau_i| > 1 the cosine averages out), so drift removal is driven by short gaps while the failure shows at long gaps and wrap-only revisits; and the CLOCK and COLLAPSE end states of rank-D heads (partial and complete hiding) are points on one curve. *Status.* The first-order statement is proved; the feedback is a conjecture. A CPU pilot at T = 128 (n = 3) saw a failing rank-2 seed whose weak axis was present at initialisation, and at initialisation on the real task the drift is too large for A1 (App. G6).

E10. Exists, kept, not found (Remark 2; definitional)

Let Phi_B be the stochastic training map for budget B, S_eps = {theta : L(theta) <= eps} the solution set of a class and P_0 the initialisation distribution. Define the basin Bas = {theta : P(Phi_B(theta) in S_eps) >= 1/2}. *Exists:* S_eps is non-empty in the class (E2, constructed). *Kept:* the constructed points lie in Bas (ring 8/8, torus re-found 8/8, rank-2 projection 7/8). *Not found:* P_0(Bas) is small (SOLVED fractions 0-2/8 estimate it). Together they place the obstruction in the position of supp(P_0) relative to Bas, not in the emptiness or instability of S_eps. Two measurements of that position are in App. H7: the loss along straight paths from the start to a constructed solution, and the basin's extent along such a segment.

E11. Readers and codes (Proposition 6; proved)

E11.1 (one layer is a position-only reader). In a one-layer MapWM the pre-softmax score between the query at i and the key at j, head h, is s_h(i, j) = sum_c |q_c(x_i)| | k_c(x_j)| cos(theta_c(i) - theta_c(j) + psi_c(x_i, x_j)) / sqrt(d_head), where q_c, k_c are 2D blocks of W_q LN(e(x_i)), W_k LN(e(x_j)) and psi_c the angle between them. Both depend only on the token identities; position enters only through theta(i) - theta(j); the values depend on identity alone. So for scored revisit queries a one-layer model is a channel-weighted cosine kernel with token-type-dependent weights and offsets, a soft vote over the observations it lands on, and an MLP on the vote.

E11.2 (extra passes widen the reader, not the code). `MapFormerWM.forward` computes delta = action_to_lie(token_emb(x)) and cos, sin = path_integrator(delta) once and hands them to every layer; `MapFormerWM_Looped.forward` does the same and applies one block n_loops times. From the second pass on, q and k depend on the residual stream, so the weights |q_c||k_c| and offsets psi_c become context-dependent and values can carry retrieved content; theta is unchanged. A loop with the same parameters and initialisation is therefore the same code family with a larger reader function class. []

E11.3 (a score rule acts on the readout term). Any score of the form sum_c A_c cos(dphi_c + beta_c), with A_c >= 0, sees position only through dphi. Write the target coherence (the A-weighted cos(dphi + beta) on the target key) as C = C0 + (C - C0), with C0 the A-weighted cos(dphi) on the target, the code term. A score rule changes A and beta, so it changes C - C0 and can reweight channels within C0, but cannot change dphi. []

E12. Peak pinning (Proposition 7; proved given delta = 0)

RoPE / MapWM score: s_ij = sum_c Re(q_c conj(k_c) e^{i dphi_c}), so A_c = |q_c||k_c| and beta_c = arg q_c - arg k_c, both content-dependent. PoPE / MapPoPE (`model_pope.py`, cos(dphi - delta)): A_c = softplus(q_c) softplus(k_c), beta_c = -delta_c, content-free. If beta_c = 0 for all c, then for any content pair s_ij <= sum_c A_c, with equality iff dphi_c = 0 mod 2 pi on every channel with A_c > 0; as a function of displacement the kernel is a non-negative combination of cosines, positive-definite by Bochner, so content can reweight the frequencies but cannot move or invert the peak. *Corollary (index phase).* With PoPE's frequencies theta_c = 10000^(-c/64) <= 1 rad per token, each cos(theta_c D) decreases on D in [0, pi], so with beta = 0 every content pair's kernel strictly decreases over offsets 0-3: a one-layer pinned-peak index head cannot prefer the observation 3 tokens back to the one 1 token back, while the one-layer torus solution needs peaks at the retrace offsets 4j - 1. *Assumption check:* delta is in the clamp's dead zone on 98-100% of entries in every PoPE run (App. L).

E13. The used-channel ratio (Remark 3; light theory)

With token frequencies f_w, split every step into a common part and a content part: Delta_c(w) = mu_c + d_c(w), mu_c = sum_w f_w Delta_c(w), so sum_w f_w d_c(w) = 0. Over an interval of L tokens, Phi_c(t) - Phi_c(t') = mu_c L + (sum of d_c over the interval). The first term is a clock (Case I if carried by every token, Case II if only by selected tokens); the second can be a map (Case III) only if it cancels over closed loops. Define rho_c = |mu_c| L / range(content_c, L) at the training length L. E7.2 constrains mu_c only

in channels the readout uses. Prediction: in used channels rho_c << 1 when targets are keyed to a latent state reached at variable lags, and rho_c >> 1 when they are keyed to a token or event count. The pooled growth exponent of the summed step weights channels by their raw range, so a large tick in unused channels dominates it; the used-channel median of rho does not have this problem.

E14. Status

#	statement	status	check
E1	rotary position factor = Howard's propagator with s = i omega	proved (identity: Mamba-3 Prop. 3)	`a0_propagator.py`
E2 (a), (f), (g)	exact non-negative (	.	, softplus, exp) representation mod 2 pi with <= r stepping tokens
E3	relative-sign gauge; excluded codes are opposite-sign rows	proved	`t1_gauge.py`
E4	cut circle; sticky wall along s	proved	`p3_p5_geometry.py`
E5	CARoPE: every channel a clock or blind	proved	`p6_carope.py`
E6.1	prefix sum in the phase; one layer at any gap	proved	same-cell attention (App. F2)
E6.2	two-layer global count (normaliser, precision, width)	cited (Liu et al. Lemma 6; Yao et al.)	probes (App. F3)
E7	phase decomposition; position-factor bound; tick rule	proved	bound consistent with the strata, not a test of them; text-world drift (App. J)
E8.1-E8.4		.	clock floor; signed collective route; no
E8.5	rank D: no clock-free full-map channel; rank > D: parking	proved	end states (App. G)
E8.6	route inventory	proved (inventory)	Table 4; SIGNROUTE
E9.2	hiding free to first order at D >= 2	proved under A1-A3	drift on the least-resolved axis (App. G)
E9.3	hiding starves the collective route	conjecture	not tested
E10	basins	definitional	App. H7
E11	readers and codes	proved (architecture, algebra)	one-pass reader (App. M)
E12	peak pinning	proved given delta = 0 (measured)	DELTA_FREE (App. L)
E13	used-channel ratio	light theory	App. J4

Statements of the theory appendix v3 not used here: P1(c)-(e), P2 (rescaling sensitivity; its drift account is withdrawn), P5 (the pi-mirror) and P7 (the CoPE lattice); they are in `docs/paper/APPENDIX_theory_v3.md` with their check scripts.

F. Exchange-rate details

Dyck-2 (`DYCK_MDEPTH_RESULTS.md`). Task and metric as MapFormer v4 App. B.4, trained and scored at length L = 32 and nesting depth D = 12; primary metric A2f, Hewitt's distance-averaged closing accuracy over prefixes where a closer can be emitted (CE-optimal predictor 1.000, chance 0.500, best stack-free floor 0.594). 2 heads, d_model 128, 1-4 layers; path-integrated arms MapWM and MapPoPE at rank 2; index arms RoPE and PoPE at base 10000 (and base 32 in the 4-layer 3x cell); n = 8 paired seeds. Position main effect = ((MapWM - RoPE) + (MapPoPE - PoPE)) / 2.

condition	layers	RoPE	PoPE	MapWM	MapPoPE	position main effect
1x budget	1	0.648	0.640	0.996	0.998	+0.353
1x budget	2	0.889	0.850	1.000	1.000	+0.130
1x budget	3	0.964	0.946	1.000	1.000	+0.045
1x budget	4	0.982	0.971	1.000	1.000	+0.024
3x budget	4	0.998	0.997	1.000	1.000	+0.002
mixture of depths 4-12	4	0.898	0.881	0.999	0.999	+0.110

Every effect is positive on 8/8 seeds (sign-flip p 0.0078). Index base 32 minus base 10000 at four layers, 3x budget: +0.001 (RoPE), -0.001 (PoPE), within the MDE. Three times the budget raises the index arms by +0.021 at four layers (8/8).

Ring (C3). The index-one-layer accuracies are not monotone in p_plus (0.710, 0.677, 0.824, 1.000 on fresh seeds; every pair differs, p 0.0022-0.0108), so the exchange rate, not the one-layer accuracy, is the quantity that tracks whether steps are undone.

Torus (`FACTORIAL_RESULTS.md`, Fig. F1). Strata (short gap / long gap / wrap-only; mean and run range, n = 4): path one layer 1.000 / 1.000 / 1.000; index one layer 0.769 (0.735-0.787) / 0.480 (0.456-0.508) / 0.482 (0.470-0.495); four layers 0.943 (0.900-0.958) / 0.458 (0.367-0.506) / 0.457 (0.381-0.494); looped x4 0.938 (0.937-0.940) / 0.507 (0.496-0.531) / 0.509 (0.496-0.521). Floors on this stream (analysis): constant 0.508 / 0.515 / 0.517, retrace 0.912 on short gaps. Memorisation check: four index layers 0.897 on their own training map against 0.860 held out; looped 0.862 against 0.864; path 1.000 on both.

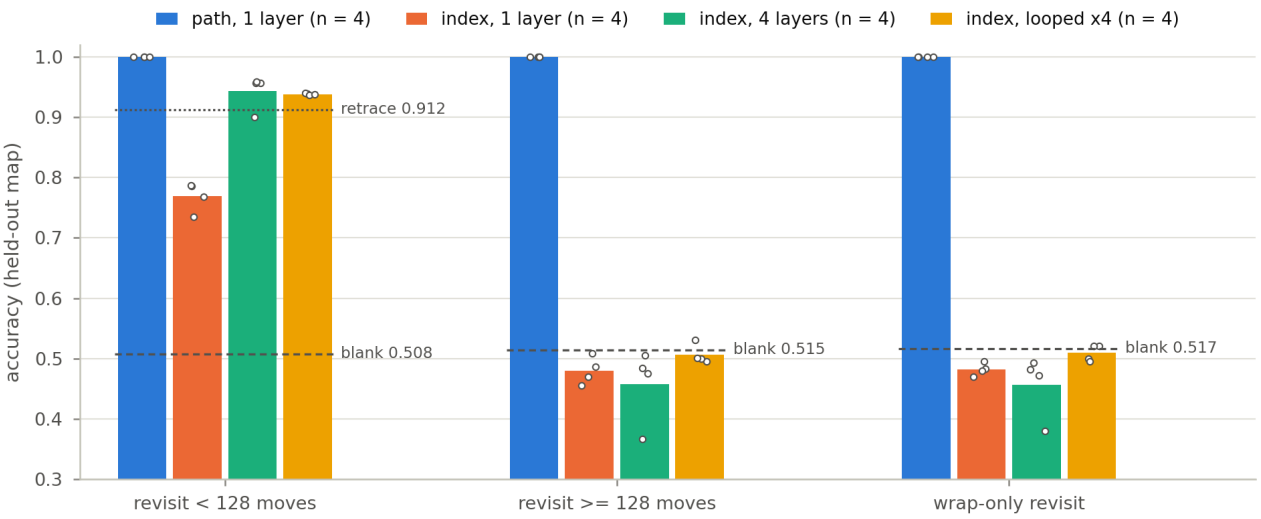


Figure F1 (`fig_factorial_strata`). Revisit accuracy by stratum on the torus at T = 1024 (trained and tested at T = 1024; bars: means of n = 4 runs, dots: runs). Dashed: always-blank within each stratum, 0.508 / 0.515 / 0.517; dotted: retrace within the short-gap stratum, 0.912 (elsewhere it equals the constant); floors on this batch's own evaluation walks. Path integration with one layer is at 1.000 in every stratum. Four index layers or a loop of four lift short-gap revisits to 0.94, just above retrace; at long gaps and wrap-only revisits every index model stays at or below the constant (0.46-0.51).

F1. Which revisits depth buys on the torus (analysis; `scripts/depth/torus_depth.py`, `summarize_torus.py` -> `out/torus_depth_out.txt`, `summarize_torus.py --segments` -> `out/torus_segments_out.txt`)

The FACTORIAL evaluation stream was regenerated (env seed 10000, np seed 1234 + s, 100 walks); overall accuracy per run equals the stored strata files' "all" to 4 decimals for all 16 runs, and the re-implemented forward equals the model's logits (max |diff| 0.0). Per revisit: s = the last visit to the cell; *retrace* = the current run reverses the previous one and has not overshot it; *runs* = maximal single-action runs among moves s+1..t. Means of per-seed accuracies (n = 4; range in brackets).

revisit class	share	index 1L	index 4L	loop x4	path 1L
short gap, retrace	0.680	0.833 (0.79-0.86)	0.996 (0.99-1.00)	1.000	1.000
short gap, not retrace	0.148	0.475	0.699 (0.51-0.77)	0.655	1.000
... 3-4 runs	0.031	0.477	0.926 (0.74-0.99)	0.977	1.000
... 5-8 runs	0.056	0.476	0.735 (0.52-0.81)	0.645	1.000
... 9-16 runs	0.052	0.474	0.561 (0.38-0.64)	0.499	1.000
... 17+ runs	0.009	0.472	0.493	0.502	1.000
long gap (>= 128)	0.090	0.480	0.458	0.507	1.000
wrap-only	0.083	0.482	0.457	0.509	1.000

Gap in moves (plain revisits): index 4L / loop 1.000 / 1.000 at 2-7, 0.978 / 0.986 at 8-31, 0.614 / 0.551 at 32-127, 0.458 / 0.506 at 128-511; one layer 0.935 at 2-7, 0.681 at 8-31, floor beyond. Runs at a fixed gap of 32-127 moves (plain revisits, retraces included; mean of per-seed accuracies, n pooled over the 4 seeds; the 2-run cells are all retraces of merged runs; pooled over seeds instead, the small cells read 4L 0.896 at 2 runs and 0.785 at 3-4):

runs spanned	n	index 1L	index 4L	loop x4
2	240	0.412	0.874	0.996
3-4	503	0.483	0.832	0.911
5-8	4565	0.464	0.688	0.575
9-16	6569	0.474	0.560	0.499
17+	1202	0.472	0.493	0.502

At gaps of 8-31 moves: 4L 0.993 / 0.940 / 0.818, loop 1.000 / 0.987 / 0.768 at 2 / 3-4 / 5-8 runs. Long-gap revisits span 17 or more runs in 98.5% of cases (median 37, minimum 11), wrap-only revisits in 99% (median 65). The deep models score below the constant floor there with high loss (2.1-4.8 nats): they predict a specific wrong object rather than

abstaining to blank. *Decomposition* (an identity, since short = retrace + not retrace): the short-gap retrace floor is (0.680 x 1 + 0.148 x ~0.5) / 0.828 = 0.911 (measured 0.912); four index layers give (0.680 x 0.996 + 0.148 x 0.699) / 0.828 = 0.943. The class contrasts at n = 4 (e.g. 4L against the loop at 3-4 runs; 4L range 0.74-0.99) carry no test.

F2. Where index attention looks (analysis; same script)

Attention at the revisit query (token a_t), final pass, per head; *same* = mass on earlier observation tokens of the same cell (uniform chance 0.001-0.003); *last* = mass on the last visit's observation and arrival tokens.

model, final pass	retrace: same / last	short, not retrace: same	long gap: same	wrap-only: same
index 1L	0.05 / 0.10 (h0), 0.06 / 0.12 (h1)	0.00	0.00	0.00
index 4L, layer 4	0.55 / 0.56, 0.58 / 0.58	0.22 / 0.22	0.01 / 0.01	0.00 / 0.00
loop x4, pass 4	0.50 / 0.52, 0.49 / 0.50	0.20 / 0.20	0.00 / 0.00	0.00 / 0.00
path 1L	0.83 / 0.70, 0.76 / 0.65	0.82 / 0.76	0.81 / 0.74	0.66 / 0.58

In the deep index models retrieval happens in the last pass only, and on retraces all of it is on the last visit: they find the mirror point, not the cell.

F3. No global count on the torus (analysis; `scripts/depth/torus_probe.py` -> `out/torus_probe_out.txt`)

Ridge probes from the residual stream at move tokens (train walks 0-29, test 30-39, t >= 16; out-of-sample R^2, mean of 4 seeds, x and y averaged; best pass):

target	index 1L	index 4L	loop x4	ring index 3L, p_plus 0.5 (t >= 8; see F5)
unwrapped displacement P_t	-0.06	0.10	-0.01	0.60
normalised count P_t / (t+1)	-0.03	0.21	0.06	0.69
1 / (t+1)	-0.01	0.78	0.74	0.89
displacement over the last 4 moves	0.90	0.96	0.95	0.97
... last 16	0.47	0.93	0.86	0.86
... last 64	0.09	0.77	0.56	0.70
... last 256	-0.01	0.37	0.14	--

The deep torus models know elapsed time (1/t) but carry no normalised count; they carry a recent-displacement code whose error grows with the window (0.5 cells at 4 moves, 1.8 at 16, 7.1 at 64, 25 at 256 for 4L; the d_w targets at t < w are the displacement since the start, so the rmse at w = 256 is approximate). Linear probes bound only what is linearly readable; the attention readout of F2 does not rely on them. The ring column's count-like R^2 is a window code (F5).

F4. Run length moved at evaluation (analysis, distribution-shifted; `scripts/depth/torus_runlen.py`)

The stored FACTORIAL models re-evaluated on walks whose run length is fixed at 2, 5 or 10 moves. Two-run returns are gap-invariant for deep index models (retraces at gaps of 16-63 moves with runs of 10: 0.978 / 0.995 for 4L / loop). Down a column of fixed gap (16-63 moves) 4L falls from 0.98 (2 runs) to 0.87-0.93 (3-4), 0.72-0.80 (5-8), 0.64-0.70 (9-16) and 0.58 (17+); within a run bin across gaps it moves by 0.02-0.2. Runs set the level and the gap trims it. Path 1.000 in every cell of every condition.

F5. The ring: rules, runs and a count (analysis; `scripts/depth/ring_depth.py`, `summarize_ring.py`)

CANCEL evaluation stream (overall accuracy equals each run's stored value for all 128 runs); fresh seeds 2-7. Classes: *retrace*; *lap* = the displacement over the last 32 moves is 0 mod 32 (a fixed one-lap offset lands on the cell); *neither*, split by runs. p_plus 0.5, mean (range):

class	share	index 1L	index 2L	index 3L	path 1L
retrace	0.716	0.797	0.987	1.000	1.000
lap, not retrace	0.035	0.497	0.977	1.000	1.000
neither, 3-4 runs (wrap-only included)	0.127	0.487	0.891 (0.72-0.94)	0.997	1.000
neither, 5-8	0.097	0.489	0.802 (0.55-0.87)	0.993	1.000
neither, 9-16	0.025	0.485	0.678 (0.48-0.75)	0.972	1.000
wrap-only	0.118	0.503	0.819	0.989	1.000

The "neither" rows include wrap-only revisits (29% / 47% / 56% of the three bins); on plain revisits only, 2L scores 0.910 / 0.828 / 0.742 and 3L 0.999 / 0.996 / 0.988 (`ring_plain_bins.py`; the conclusions, 2L decays and 3L is flat, are the same). p_plus 0.75 / 0.9 (index 1L): retrace 0.692 / 0.677, lap 0.987 / 0.994, neither 0.487 / 0.592, against best-constant floors of 0.493 / 0.491 / 0.502 for the "neither" class at p_plus 0.5 / 0.75 / 0.9 (`ring_class_floors.py`): at 0.9 one layer sits an unexplained +0.09 above it. At p_plus 1.0 every revisit is a lap revisit, 1.000 at every depth. *No global count on the ring either* (`ring_probe_window.py` -> `out/ring_probe_window_out.txt`). Probes read at t >= 8 on 128-move walks (P_t R^2 0.60, P_t / t 0.69, 1/t 0.89 for three layers; one layer 0.09 / 0.18 / 0.14; two 0.40 / 0.60 / 0.67) cannot separate a prefix count from a recent-window code, since half of P_t / t's variance sits at t <= 32 and for t < 64 the 64-move window equals P_t. Read late, three layers give P_t R^2 0.59 at t >= 64 (two layers 0.32), and at t >= 96 the displacement before the last 64 moves has R^2 0.01 (before the last 32: 0.31), while the windows are well coded (last 64 moves 0.72); the true 64-move displacement alone predicts P_t there with R^2 0.51, against the model's 0.53. Three ring layers carry a displacement code reaching 32-64 moves back, which on 128-move walks covers most revisits; their last-layer attention reaches the cell on wrap-only revisits (same-cell mass 0.36 / 0.38). *One-layer accuracy across p_plus (descriptive):* a p-invariant mixture does not reproduce it: per-class accuracies pooled over p predict 0.678 / 0.683 / 0.749 against the observed 0.710 / 0.677 / 0.824 at p_plus 0.5 / 0.75 / 0.9, because the class accuracies themselves move with p (retrace 0.797 -> 0.692 from 0.5 to 0.75, as large as the net dip of 0.033). No mechanism is claimed for the dip at 0.75.

F6. Dyck-2: blocks to skip (analysis; `scripts/depth/dyck_distance.py`, `dyck_crosstab.py`)

Closing accuracy by the number of complete balanced blocks between the top opener and the prefix (mean of 8 seeds):

blocks to skip (n prefixes)	RoPE 1L	2L	3L	4L	MapWM 1L
0 (3682)	0.998	0.999	0.999	0.999	0.999
1 (5839)	0.640	0.925	0.984	0.991	0.997
2 (572)	0.641	0.778	0.885	0.934	0.993
3-4 (161)	0.613	0.760	0.860	0.925	0.993

Dyck has a BOS token, so the count route's normaliser is available, and both blocks and distance matter for the index models; the path model is flat in both (0.987-0.998). 93% of scored prefixes need at most one block, which is why three index layers already reach 0.964 in aggregate.

G. Rank details

G1. Arms, curves and dimension

Arms (A1). Shared r2 (`Vanilla`), per-head r2 (`Vanilla_r2ph`), block-diagonal r4 (`Vanilla_r4mibd`: the shared rank-4 latent with W_out's off-diagonal blocks held at zero), per-head r3 (`Vanilla_r3ph`), shared r4 (`Vanilla_r4mi`), per-head r4 (`Vanilla_r4ph`), all built from the shared rank-2 model's initial weights at each seed. Initial angle-increment scale (std): 0.208 for block-diagonal r4 against 0.33-0.34 for the other arms, which fail (rank 2) or solve (rank 4) from the same scale. Per-seed results: C4.

Training curves (Fig. G1; `fig_loss_curves_rank_out.txt`). Shared r2 0/8 SOLVED (4 STALLED, 4 DESCENDING; median final loss 0.35) against shared r4 8/8 (2.8e-3); per-head r2 2/8 (median 0.22), r3 6/8 (3.5e-3), r4 8/8 (3.1e-3). The rank-3 and rank-4 medians cross the threshold at epochs 478-532; the rank-2 medians never do.

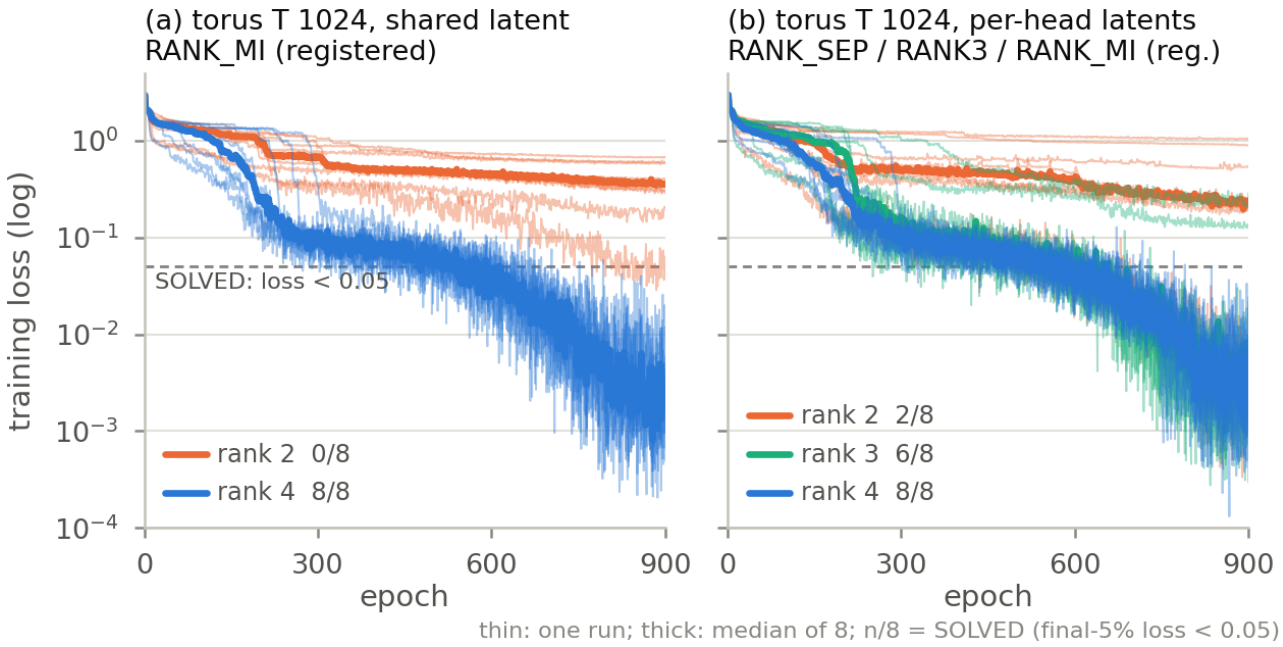


Figure G1 (`fig_loss_curves_rank`). Training loss per epoch on the torus (T = 1024) by the per-head rank of the content-to-angle map; thin lines are runs, thick lines medians of 8; dashed: the SOLVED threshold. (a) One latent shared by the heads: rank 2 solves 0/8, rank 4 8/8. (b) Per-head latents: rank 2 2/8, rank 3 6/8, rank 4 8/8.

Restart curves (`RANK_PROJ_ANALYSIS.txt`). Projected rank 2: peak training loss 0.37-0.46 after the restart, back below 0.05 at epochs 553-654 on 7/8 seeds. Rank-4 control: peak 0.26-0.30, back at 497-669 on 7/8. Rank 2 continued: peak 0.55-0.82, back on 1/8.

Rank against dimension (`RANK_ND_RESULTS.md`, `RANK_WRAP_RESULTS.md`, `RANK_TESTS_RESULTS.md`). Per-head rank, every arm from the same base, trained and tested at T = 1024, 900 epochs:

torus	D	per-head rank	wrap-only share of revisits	SOLVED	held-out accuracy
32 per side	2	2	0.37	1/8	0.769
32 per side	2	3	0.37	8/8	0.999
10 per side	3	3	0.71	1/8	0.800
10 per side	3	4	0.71	4/8	0.847
18 per side	3	4	0.35	8/8	1.000

2D rank 3 vs rank 2: 8/8 vs 1/8 (Fisher p 0.0014), +0.230 (permutation p 0.0014). 3D, 18 vs 10 per side at rank 4: +0.153 (p 0.0085). On a 100-cell 2D torus (10 per side) rank 3 memorises its training map (0.986 on it, 0.273 on an unseen map), so that cell does not measure map learning. 1D ring (32 cells, T = 128, 300 epochs): rank 1 7/8 (0.993), rank 2 8/8 (1.000); with the component common to the move steps removed by construction, rank 1 solves 8/8 in a median of 24 epochs (58 without).

Escape test (`RANK_TESTS_RESULTS.md`). Lifting the eight stalled shared-rank-2 runs to rank 4 (function-preserving) and training 600 epochs solves 3/8, against 1/8 for rank 2 continued

and 2/8 for a fresh rank-4 step map on the same networks; at this budget the contrasts are not resolved.

G2. PoPE's score does not rescue rank 2 (`SCORE_RANK_RESULTS.md`, registered NO RESCUE)

Torus at T = 1024, 900 epochs, 32 angles per head in every arm; rank-2 arms on seeds 10-21, rank-4 arms 10-17:

arm	accuracy	SOLVED
MapWM r2	0.848 +/- 0.178	3/12
PoPE score r2	0.946 +/- 0.040	3/12
MapWM r4	0.9998	8/8
PoPE score r4	0.9995	8/8

Primary: 3/12 vs 3/12 (Fisher p 1.00); the rank-2 deficit replicates on these fresh seeds under both scores (8/8 vs 3/12, p 0.0014 each). The stratum readouts are secondaries (short gaps +0.115, p 0.036; gap >= 128 +0.043; wrap-only -0.015).

G3. The clock at the end of training, per head (analysis; `scripts/sign/clock_table.py` -> `clock_table_out.txt`; latent column `scripts/sign/rank_clock.py` -> `rank_clock_out.txt`)

Clock / map ratio of Sec. 4 (E7.1') per run (usage pooled over both heads) and per head (usage normalised within the head); full models (observations step). The latent clock (|z_x + z_y| over the map frame, raw latent coordinates, mean over unsolved runs' heads) is a different quantity and not invariant to the GL(r) gauge of the latent (in the W_out image the unsolved values become 1.4-2.6).

arm	SOLVED	solved runs: run mean (heads median, range)	unsolved runs: run mean, median (heads median, range)	latent clock, unsolved
shared r2	0/8	--	0.595, 0.579 (0.619, 0.07-1.26)	1.72
per-head r2	2/8	0.236 (0.045, 0.03-0.54)	0.694, 0.792 (0.756, 0.02-1.98)	3.19
block-diagonal r4 (rank 2 per head)	2/8	0.335 (0.067, 0.04-0.68)	0.541, 0.575 (0.084, 0.01-1.41)	1.63
per-head r3	6/8	0.043 (0.038, 0.02-0.08)	0.936 (0.678, 0.12-1.25)	3.68
shared r4	8/8	0.049 (0.036, 0.01-0.10)	--	--
per-head r4	8/8	0.051 (0.030, 0.02-0.14)	--	--

The solved rank-2 run means hide a split: in each of the four solved rank-2 runs at least one head is clean (per-head r2 s1 h1 0.03, s2 0.05 / 0.05; block-diagonal s2 0.04 / 0.06, s6 h0 0.07), and two runs keep a drifting second head (per-head s1 h0 0.54, block-diagonal s6 h1 0.68). Not every failure is clock-coupled: 5 of the 20 unsolved rank-2 runs have both heads below 0.15 (shared s1; per-head s5, s6; block-diagonal s1, s7), and 4 of them (all but shared s1) are in the collapsed class of `rank/sweep_final_out.txt`: visible drift <=

0.01 per move with the two map axes near-collinear (frame conditioning 0.00-0.03), a map that has lost a dimension; shared s1 ends just above the SOLVED threshold (final loss 0.051). Heads below 0.15 among unsolved runs: shared 4/16, per-head 4/12, block-diagonal 7/12. The rank-3 unsolved runs are clock-coupled. The observation clock (mean observation step over the map; `rank_clock.py`) is 0.015-0.018 in solved rank-3/4 heads and 0.18-0.52 in unsolved heads. At r = 2 = D, rank_clock's "in-span" share of the clock is 1 by construction before wrapping, so it is not reported as a measurement.

G4. The leftover latent clock and hiding (analysis; `scripts/rank/`; a latent quantity, not the ratio of G3)

Here the clock is c = a + o, the mean move latent a plus the mean observation latent o (the drift a walk accumulates per move and observation), measured in the native latent of each head's read subspace against the map frame U (lam = |c| / |U|); "off-plane share" = |c_perp| / |c|; "readgain" = how strongly the readout sees the leftover clock relative to the map (1 = read like the map, 0 = invisible). This latent metric and G3's channel ratio agree that what is still read in solved heads is small; they are not numerically comparable. The native latent metric is not invariant to the GL(r) gauge of the latent; the independent audit found the balanced gauge gives the same picture (solved-set medians 0.94-1.00).

What the leftover clock is made of (`clock_decomp.py` -> `clock_decomp_out.txt`; medians over solved heads): a and o are each 0.017-0.037 of a map step and anti-parallel (cosine -0.79 to -0.999), so they cancel; at T = 1024 they lie mostly in the map plane (off-plane share 0.05-0.35 for a, 0.02-0.08 for o). Only their small residual c is off-plane. SEARCH's per-pair clocks z(-e_j) + z(+e_j), which leave the observation step out, are 0.034-0.057 of a map step and in-plane (off-plane 0.14-0.21 on the 2D sets, below the random-direction median); with the two observation steps of one out-and-back added, 0.73-0.99 off-plane. The per-pair clock therefore overstates the drift a walk accumulates; the two definitions agree on what is read.

set (solved heads)	|a| / |U|	|o| / |U|	cos(a, o)	off-plane share of a / o / c
torus G64 per-head r3	0.021	0.020	-0.988	0.15 / 0.07 / 0.98
torus G64 shared r4 (RANK_MI)	0.017	0.017	-0.980	0.19 / 0.08 / 1.00
3D G18 r4	0.021	0.022	-0.999	0.05 / 0.06 / 0.81
torus T = 128 r4	0.034	0.029	-0.897	0.48 / 0.40 / 0.99
ring rank 2	0.037	0.032	-0.793	0.47 / 0.17 / 0.99

Where the residual lies in solved heads of rank above D (`sweep_final.py`, `summarise_final.py`; consistent with parking, but removal through the latent, which takes the read components first, leaves the same end state, so this does not discriminate the two):

set (solved heads)	r - D	lam	off-plane share (random direction, median)	readgain
torus G64 per-head r3	1	0.004	0.978 (0.50)	0.15
torus G64 shared r4 (RANK_MI)	2	0.004	0.996 (0.71)	0.11
torus G64 per-head r4	2	0.003	0.988 (0.71)	0.16
	2	0.003	0.998 (0.71)	0.09

set (solved heads)	r - D	lam	off-plane share (random direction, median)	readgain
torus G64 shared r4 (SCORE_RANK)				
2D G32 r3	1	0.003	0.987 (0.50)	0.09
3D G10 r4	1	0.008	0.964 (0.40)	0.23
3D G18 r4	1	0.001	0.810 (0.40)	0.57
ring rank 2 (1D), 300 epochs	1	0.025	0.989 (0.71)	0.21
torus T = 128 r4, 300 epochs	2	0.021	0.988 (0.71)	0.27
escape, fresh r4 step map, 600 epochs	2	0.003-0.004	0.98 (0.71)	0.18

In the clean head of solved rank-D runs (ring rank 1, torus rank 2) the clock is removed (lam <= 0.0005 on the T = 1024 2D sets, <= 0.0013 on the ring), since it cannot lie off the plane; the other head of such runs often drifts (lam 0.15-1.75). The leftover is about 6x larger after 300 epochs than after 600-900, of the size decoupled weight decay alone gives an unread direction over a cosine schedule (consistent, not a test). The escape arms (below) show the latent clock shrinking, not held: in the fresh rank-4 step maps its absolute size falls 14-85x (relative to the map step 270-1300x), the off-plane part 15-72x in 7/8 runs (1.6x in s1, which started at 0.005), and in the three solved lifted runs 23-490x (`clock_decomp_out.txt`). Unsolved heads of rank above D read their clock in all three sets and hold it in the map plane in two (torus per-head r3: off-plane 0.27 against a random-direction median of 0.50, readgain 0.61; escape lifted runs: 0.08 against 0.71, readgain 0.76; 3D G10 r4: off-plane 0.53 against 0.40, readgain 0.78).

Drifting rank-D heads drift along their least-resolved axis (`c1_runlevel.py`): over 116 drifting heads in nine sets the median r_beta (resolution along the drift direction relative to the head's mean) is 0.087, and 85% are below 0.5 against 29% for a random direction given each head's own frame; at the run level 57 of 62 runs against 28.7 expected (P < 5e-5, 20,000 simulations); escape C continues rank_mi's own networks, so counting each network once gives 50 of 55 against 25.1 (`c1_runlevel_once.py`). Clean heads of rank above D sit at r_beta 0.79-1.10 (medians). This geometry is what E9 leads to and also what pruning of a visible drift after any collapse would leave; it fixes the geometry, not the order of events.

The escape test, re-read (`escape_start.py`; RANK_TESTS escape arms, 600 epochs on the same eight trained networks):

arm	runs with a drift-free head at the end	heads drift-free	off-plane share at end
fresh rank-4 step map (start drift 0.28-5.86 cells per move)	8/8	16/16	0.91-1.00
stalled rank 2 lifted to rank 4	3/8	6/16	1.00 in the 3 solved
rank 2 continued	1/8	2/16	--

Within the lifted arm the clock's off-plane share at the start (the lift adds two latent rows at default scale beside large trained rows) is 0.09 / 0.14 / 0.11 in the three runs that solve

and 0.01-0.06 in the five that do not (exact one-sided rank-sum p 0.018; post hoc). Two caveats make this suggestive only. The statistic was chosen after the fact: with the off-plane clock relative to the map step instead (|c_perp| / |U|), the solved runs rank 1st, 2nd and 8th (p 0.29). And s1's source also solves under plain continuation (arm C s1 SOLVED) and starts nearly clock-free (|c| / |U| 0.05), so its large share is that of a near-zero clock; without s1, p 0.048 (`escape_start_out.txt`). A seeded escape test (RANK_THEORY's G2) would test it. The fresh step maps remove the drift everywhere, and the residual geometry does not track SOLVED within that arm (the 12 unsolved heads show the same off-plane share, 0.985, and readgain, 0.19, as the 4 solved ones; audit); learning the rest of the map within 600 epochs (2/8 SOLVED) is a separate matter.

G5. The 1D exemption (registered + analysis)

On the ring (`RANK_TESTS_RESULTS.md`), rank 1 solves 7/8 (0.993) and rank 2 8/8 (1.000); with the drift forbidden by construction (opposite move latents tied, observation steps zero) rank 1 solves 8/8 in a median of 24 epochs (58 without). Every channel of a rank-1 head sees the same phantom drift in cells (E8.5(d)); over the 8 rank-1 runs the 11 clean heads (visible drift <= 0.01 per move) end at lam <= 0.0013, and the 5 drifting heads (visible drift 0.04-0.50 per move; two of them in the unsolved run s22) have the largest map steps (|U| 2.8-5.0 against 1.6-2.6), the only other global way to shrink the drift in cells.

G6. Dynamics (analysis, CPU pilot; `scripts/rank/train_snap.py`, `analyze_snap.py`, `pilot_t128/`)

Paper torus T = 128 (grid 64), batch 32, 40 epochs cosine, lr 1e-3, seeds 3000-3002 (outside every batch); arms shared r2, shared r4 (from r2's base) and R2Clock (shared r2 plus a fixed extra latent coordinate that every token steps by 1, giving each channel a learned clock rate; identical to r2 at initialisation). Snapshots every 49 steps.

arm	seed	final loss	accuracy T = 128 / 512	latent clock
r2	3000	0.005	1.000 / 0.937	removed in lockstep with the visible drift (lam 0.63 -> 0.004)
r2	3001	0.230	0.922 / 0.836	kept, on the unresolved axis (r_beta 0.00); frame near-collinear from step 0
r2	3002	0.003	1.000 / 0.981	removed in lockstep
r4	3000	0.005	1.000 / 0.954	off-plane 0.62 -> 0.96, then removed (lam falls 66x)
r4	3001	0.028	0.990 / 0.959	off-plane 0.55 -> 0.98, then removed (180x)
r4	3002	0.005	1.000 / 0.946	off-plane 0.81 -> 1.00, then removed (5.5x)
R2Clock	3000-3002	0.003-0.006	1.000 / 0.941-0.959	the slot holds a fixed unit clock by construction (off-plane ~1 from step 0)

At rank 4 the visible drift fell tenfold one or two snapshots before the latent clock did in 4/6 heads, and the latent clock was then removed too (5-180x within 40 epochs); in the rank-2 control the same order appears in 2/6 heads (n = 3). R2Clock is not counted: its slot clock cannot shrink tenfold by construction. At the same initial weights as the failing rank-2 seed 3001, R2Clock solved (loss 0.006 against 0.230) and repaired its frame (one matched pair). The failing seed's weak axis was present at initialisation; the readout

gradient at initialisation on the real T = 1024 task gives no consistent hiding direction (drift 0.4-2.7 cells per move, outside assumption A1; one stochastic sample of two batches, dropout on; `grad_hiding.py`). Initial frame conditioning does not separate solved from unsolved rank-D runs at T = 1024 (one-sided p 0.19; `init_frame.py`); initial drift is lower in the runs that solve (as a test only the independent SCORE_RANK seeds count: one-sided p 0.073; pooled over sets whose seeds share a base, 0.012, descriptive). MapFormer's 5D setting (T = 16, grid 5) solved at every rank under our recipe (rank 5, 6 and 5 + clock slot 3/3 each; index RoPE stalled), so it does not test rank here (`pilot_5d/`).

H. Sign details

H1. Learning rates (`LRSWEEP_RESULTS.md`; ring, observation-step-free, seeds 1500-1507)

learning rate	signed SOLVED, accuracy	|.| SOLVED, accuracy
1e-3	8/8, 1.000	1/8, 0.917
3e-3	8/8, 1.000	2/8, 0.963
1e-2	4/8, 0.972	1/8, 0.872
3e-2	1/8, 0.559	0/8, 0.584

Registered B1 (each arm at its best rate, selection redone in the null over all 12,870 relabellings): |.| 3e-3 2/8 vs signed 1e-3 8/8, -0.037, decision p 0.0070 (SOLVED) / 0.0057 (accuracy). The |.| optimum is interior to the grid. Three |.| runs at 3e-3 were still descending at 300 epochs. At 3e-2 both arms sit near the 0.50 blank floor. Per seed:

```
seed | 1e-3:SO | 1e-3:AO | 3e-3:SO | 3e-3:AO | 1e-2:SO | 1e-2:AO | 3e-2:SO | 3e-2:AO
1500 | 1.0000 SOLV | 0.9362 STAL | 1.0000 SOLV | 0.9111 STAL | 0.9460 DESC | 0.9936 DESC |
0.4960 STAL | 0.4958 STAL
1501 | 1.0000 SOLV | 0.8759 STAL | 1.0000 SOLV | 0.9662 DESC | 0.9460 STAL | 0.4960 STAL |
0.4960 STAL | 0.4960 STAL
1502 | 1.0000 SOLV | 0.8995 STAL | 1.0000 SOLV | 0.9428 STAL | 0.9407 STAL | 0.9703 DESC |
0.4960 STAL | 0.4960 STAL
1503 | 1.0000 SOLV | 0.9116 STAL | 1.0000 SOLV | 1.0000 SOLV | 1.0000 SOLV | 0.6000 STAL |
0.4960 STAL | 0.4960 STAL
1504 | 1.0000 SOLV | 0.9415 STAL | 1.0000 SOLV | 0.9686 STAL | 0.9460 STAL | 1.0000 DESC |
0.4960 STAL | 0.8159 DESC
1505 | 1.0000 SOLV | 1.0000 SOLV | 1.0000 SOLV | 0.9993 SOLV | 1.0000 SOLV | 0.9950 DESC |
0.9998 SOLV | 0.4961 STAL
1506 | 1.0000 SOLV | 0.8736 STAL | 1.0000 SOLV | 0.9595 DESC | 1.0000 SOLV | 0.9996 SOLV |
0.4960 STAL | 0.4960 STAL
1507 | 1.0000 SOLV | 0.8971 STAL | 1.0000 SOLV | 0.9587 DESC | 1.0000 SOLV | 0.9184 DESC |
0.4959 DESC | 0.8799 DESC
```

H2. Recipe cells (`ROBUST_RESULTS.md`, seeds 1400-1407)

cell	recipe	|.| accuracy, SOLVED	signed accuracy, SOLVED	d [95% CI]	perm p	Fisher p
ring	lr 1e-3, weight decay 0.05	0.911, 1/8	1.000, 8/8	-0.089 [-0.125, -0.055]	0.0002	0.0014
ring		0.873, 0/8	1.000, 8/8		0.0002	0.0002

cell	recipe	|.| accuracy, SOLVED	signed accuracy, SOLVED	d [95% CI]	perm p	Fisher p
	lr 1e-3, weight decay 0			-0.127 [-0.200, -0.075]		
ring	lr 3e-3, weight decay 0.05	0.987, 4/8	1.000, 8/8	-0.013 [-0.025, -0.005]	0.026	0.077
ring	lr 3.3e-4, weight decay 0.05	0.841, 0/8	0.950, 6/8	-0.109 [-0.215, +0.010]	0.089	0.0070
torus	2 layers, d_model 256	0.967, 2/8	0.999, 8/8	-0.032 [-0.055, -0.015]	0.0005	0.0070

At lr 3e-3 the registered contrast was below its decision threshold in this batch (4/8 vs 8/8); on fresh seeds in the sweep the same rate gives 2/8 vs 8/8 (H1). Declared secondaries: cancelling share (unweighted) |.| 0.04-0.13 in every cell, signed 0.43-1.00; torus strata (short / long / wrap-only) signed 1.000 / 0.999 / 0.992, |.| 0.989 / 0.911 / 0.799; on the torus 6/8 |.| runs were still descending at 900 epochs (final losses 0.018-0.144).

H3. Starts (`MECHANISM_RESULTS.md`, ring p_plus 0.5, seeds 800-807; `TORUS_START_RESULTS.md`, torus, seeds 1200-1207)

Observation-step-free models differing only in the initial move angles (every other weight identical at a seed):

initial move angles	signed (ring)	|.| (ring)	signed (torus)	|.| (torus)
default (small, signs independent)	8/8	0/8	8/8	1/8
the |.| start: both moves rotating the same way	7/8	0/8	--	--
large, opposite-move sums below pi, U(0.9, 1.5)	--	2/8	--	0/8
large, sums above pi, U(1.7, 4.5)	8/8	8/8	2/8	0/8

Ring, registered: same start, sign matters (0/8 vs 7/8, Fisher p 0.0014; -0.092 [-0.135, -0.045]); a start past pi rescues |.| (8/8 vs 0/8, +0.109 [+0.085, +0.135], p 0.0002); a large start with sums below pi does not (2/8; vs past pi Fisher p 0.0070); from past pi the sign makes no difference (8/8 vs 8/8). The large start also moves the angles 13-22x faster per Adam step (the sub-pi control about 5x), so sum and size are only partly separated. Analysis: signed models from the small starts end with 98-99% of channels opposite-signed and cancel through zero; from past pi both signs cancel by wrapping (signed 88%, |.| 100% of cancelling channels); the rescued |.| models cancel in 23% of channels. Torus, registered: the past-pi start does not rescue |.| (0/8 vs 1/8, +0.005 [-0.060, +0.070]), and it slows the signed model (2/8; 5/8 runs per large-start arm still descending at 900 epochs); the default-start sign cost replicates (1/8 vs 8/8, Fisher p 0.0014). The start lever is a ring result.

H4. Moves +1/-1 vs +1/+2 (`GAUGE_RESULTS.md`, ring, seeds 1300-1307)

By Lemma 1 (ii) a |.| step represents, without wrapping or crossing zero, a map whose two moves step the same way. With moves +1/-1 (an inverse pair) the cost replicates: |.| 0.931, 1/8 vs signed 1.000, 8/8 (-0.069 [-0.095, -0.045], Fisher p 0.0014). Moves +1/+2 need integration (index RoPE 0.649, 0/8) and every revisit is a lap of at least 16 moves;

there |.| scores 0.883, 2/8 against signed 0.947, 5/8 (-0.064 [-0.200, +0.075], Fisher p 0.31), a registered comparison that is not resolved because the task is harder for both arms. The step code is learned in both classes (analysis; `scripts/sign/latent_geometry.py`, `lattice_p.py`): in the channels attention uses the relation a(+2) = 2 a(+1) holds on 0.98-1.00 of usage in 8/8 |.| runs (signed 0.99-1.00 in 7/8; angle tolerance 0.05 rad, a post hoc choice: at 0.3 rad the lattice shares change a lot), and the latent relation in 5/8 |.| runs, one of them through the mirror copy z_2 = -2 z_1 of E8.4 (cosine -0.98), against 7/8 signed. The unsolved |.| runs and 2 of the 3 unsolved signed runs have the relation in place (signed s1300: 0.68); what they miss lies outside the step code. There is no evidence of a sign cost without an inverse pair. Signed models on +1/+2 end with 0.92 of channels having both move angles of one sign (0.01 on +1/-1), the sign-coherent code Lemma 1 allows.

H5. Three times the ring budget (`RING_BUDGET_RESULTS.md`, seeds 40-47)

The |.| step at 900 epochs solves 3/8, as at 300 epochs on the same seeds (3/8); the solved sets differ.

H6. Cancelling share and ring strata

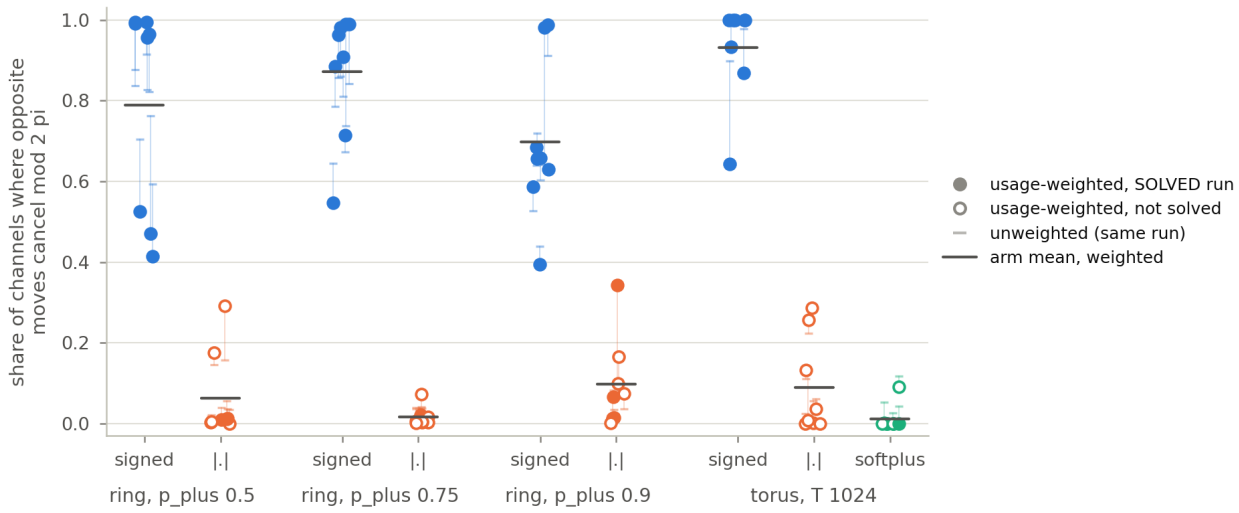


Figure H1 (`fig_wrap_share`; analysis). Share of rotary channels in which opposite moves cancel mod 2 pi, per run, weighted by how much attention uses each channel (dots; filled = SOLVED) and unweighted (faint dashes, joined to the same run). A channel cancels when |wrap(a(move) + a(opposite move))| < 0.1 rad; channels whose move angle has collapsed to within 0.05 rad of 0 mod 2 pi are excluded from the denominator. Usage weight: mean query norm of the channel at move tokens times mean key norm at observation tokens. Ring of 32 cells at three move biases (n = 8 each) and the torus at T = 1024 (n = 8). Bars: arm means of the weighted share. Signed steps 0.70-0.93; |.| steps 0.02-0.10; softplus 0.01. Seven of the eight non-negative runs that solve stay at or below 0.07.

cell	arm	SOLVED	weighted mean (range)	unweighted mean (range)
ring p_plus 0.5	signed	8/8	0.79 (0.41-0.99)	0.79 (0.59-0.92)
ring p_plus 0.5	|.|	2/8	0.06 (0.00-0.29)	0.06 (0.02-0.16)

cell	arm	SOLVED	weighted mean (range)	unweighted mean (range)
ring p_plus 0.75	signed	8/8	0.87 (0.55-0.99)	0.78 (0.64-0.86)
ring p_plus 0.75	|.|	1/8	0.02 (0.00-0.07)	0.03 (0.02-0.04)
ring p_plus 0.9	signed	8/8	0.70 (0.39-0.99)	0.64 (0.44-0.91)
ring p_plus 0.9	|.|	4/8	0.10 (0.00-0.34)	0.06 (0.02-0.10)
torus T 1024	signed	8/8	0.93 (0.64-1.00)	0.98 (0.90-1.00)
torus T 1024	|.|	0/8	0.09 (0.00-0.29)	0.11 (0.00-0.27)
torus T 1024	softplus	1/8	0.01 (0.00-0.09)	0.03 (0.00-0.12)

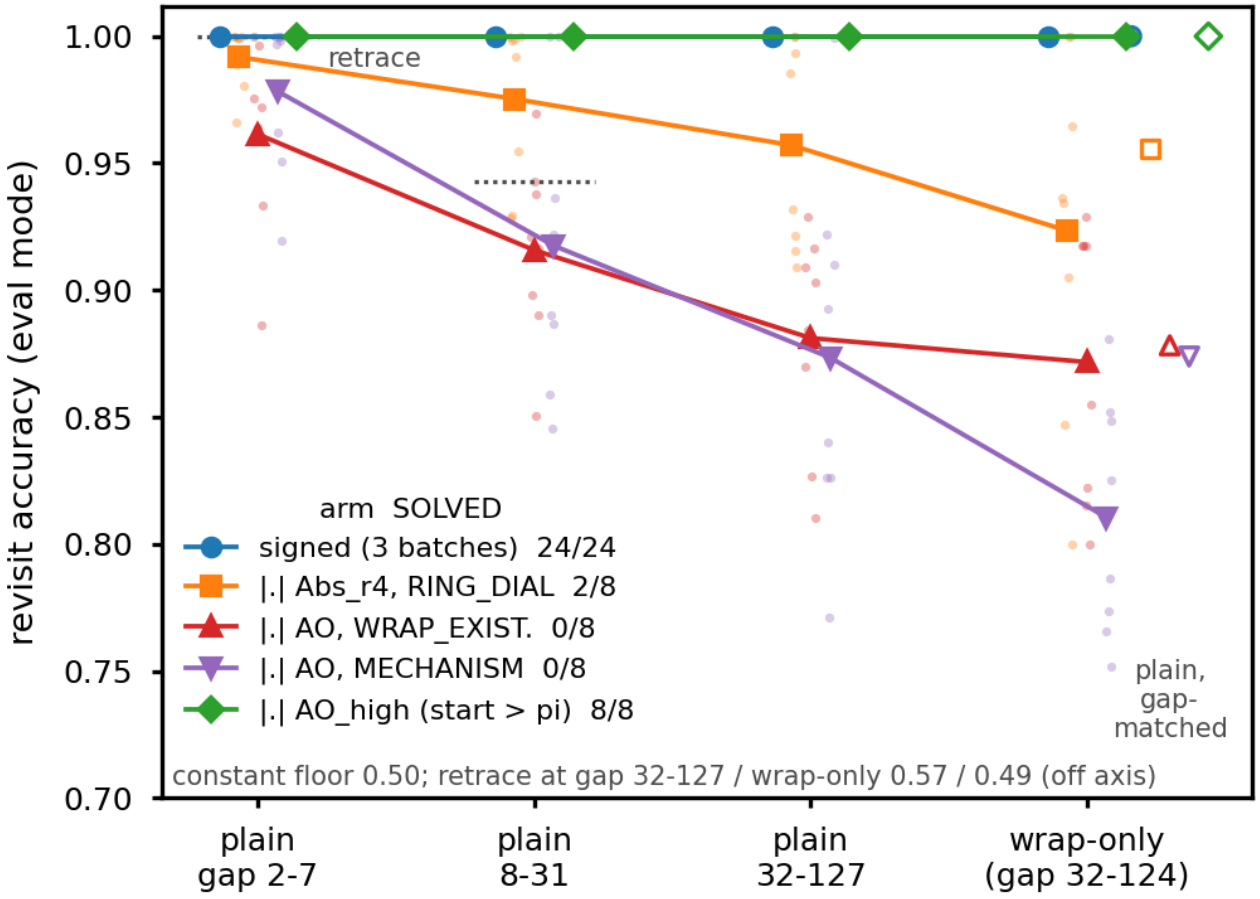


Figure H2 (`fig_ring_strata`; analysis). Revisit accuracy on the 1D ring by stratum (eval mode; 400 trajectories, T = 128; maps redrawn). "Plain" revisits have an earlier visit at the same unwrapped position; "wrap-only" revisits were seen only a lap away. Lines: arm means; dots: runs; hollow markers: plain revisits re-weighted to the wrap-only gap distribution. Signed steps (24 runs over three batches) and |.| steps started past pi are at 1.000 everywhere. |.| steps from the default start lose accuracy with the gap (0.96-0.99 at gaps 2-7, 0.87-0.96 at gaps 32-127). Dotted: retrace within each stratum, 1.000 at gaps 2-7 and 0.943 at gaps 8-31 (0.57 and 0.49 in the long-gap and wrap-only strata, off axis); constant floor 0.50.

H7. Routes, measured (analysis; `docs/paper/expansion/scripts/sign/`)

End-state clocks (`clock_table.py` -> `clock_table_out.txt`: the clock / map ratio of E7.1', one value per run; the usage share on cancelling channels from `sign_clock.py`; `sign_clock.py`'s

one-sided map wrap(a_y), the old column, gave 1.683 / 1.632 / 1.549 / 1.394 / 1.520 / 1.219 for the |.| and frozen rows, and is reproduced by `clock_table.py` as a check):

batch, arm	SOLVED	clock / map, arm mean	usage on cancelling channels
ring MECHANISM, signed (default start)	8/8	0.018	0.96
ring MECHANISM, signed from the |.| start	7/8	0.015	0.96
ring GAUGE, signed (+1/-1)	8/8	0.016	0.95
ring MECHANISM, |.| (default start)	0/8	1.471	0.08
ring MECHANISM, |.| (|.| start)	0/8	1.571	0.02
ring GAUGE, |.| (+1/-1)	1/8	1.559	0.06
ring ACCOUNTS T2, signed, move latents frozen, |.| start	1/8	1.181 (the solved run 0.058)	0.15
ring ACCOUNTS T2, signed, move latents frozen, default start	7/8	0.167	0.87
ring MECHANISM, |.| from a large sub-pi start	2/8	1.423 (solved runs 0.10, 0.59)	0.20
ring MECHANISM, |.| from a start past pi	8/8	0.317 (wrapped; runs 0.11-0.76)	0.82
ring MECHANISM, signed from a start past pi	8/8	0.166 (runs 0.004-0.48)	--
torus TORUS_START, signed	8/8	0.001	0.77
torus TORUS_START, |.|	1/8	1.248 (the solved run 0.024)	0.16
torus TORUS_START, signed from a start past pi	2/8	0.737 (solved runs 1.39, 1.78)	--

"Clock-free at the end" holds for runs that succeed from a small start (arm means 0.001-0.018 here, 0.043-0.051 for the rank arms of G3); runs that succeed from a large start can end with a large ratio. In the two solved signed torus runs from a start past pi, 0.43 and 0.99 of the usage sits on channels whose per-pair clock is near pi (|wrap(k)| > 2.6 rad; `clock_table_out.txt`), where E7.1's bound says nothing; the independent audit found the second of them (s1206) at revisit accuracy 1.000 in all four parity strata of (n_-x, n_-y) on 16 fresh walks although the correct key's score flips sign with the parity (mechanism not identified). For unwrapped |.| channels the ratio is at least 1 by construction (E8.1).

The |.| cone inequality |k| >= |m| holds with zero violations on every sub-pi channel (an identity for |.|, listed as a check of the code). Trained signed ring models end on the collective zero set: |z_x + z_y| / (|z_x| + |z_y|) 0.044-0.053, cosine -0.994 to -0.995 (`latent_geometry.py`), MapFormer's antipodal end state (their Figs. 3d and 9b).

The cone boundary. Sub-pi |.| ring channels within 10% of the cone boundary (|delta| >= 0.9 |k| / 2, one move's step near zero): 0.30 (MECHANISM default start), 0.36 (|.| start), 0.25 (GAUGE), against 0.056 at initialisation and 0.00-0.02 for the large starts. |.| models end with 18.8-22.2 one-sided and 13.2-16.0 blind channels of 64 (signed 2.1-4.1 and 8.1-8.9; initialisation 15.6 and 5.0); these counts are unweighted over all 64 channels,

unused ones included. The MECHANISM initialisation values (0.056; 15.6 / 5.0) were recomputed with the script's functions by the independent audit; the script itself prints GAUGE's (0.078; 19.8 / 3.1). At fixed clock the largest map a |.| channel can carry sits on that boundary, so collapse onto the wall is the boundary optimum of a clock-map trade-off as well as the sticky kink of E4.2.

Latent-reachable share at initialisation (`latent_reach.py`, MECHANISM seeds 800-807). Share of the needed change of the angle sums that a latent move can produce: signed (to zero) 1.000 on every seed (E8.2(a)); |.| (to the nearest wrap) 0.339 unweighted and 0.561 usage-weighted (random-subspace baseline 0.125). The rest must come from per-channel readout changes. In the held constructed |.| models the wrap targets are carried by the readout (`aobal_geometry.py`): they cancel in 9-13 channels carrying 92-98% of usage, and a readout drawn at initialisation contains 0.3-11% of that target (random-subspace baseline 6%).

Straight paths with the content of a solved model fixed (`path_profiles.py`; ring, MECHANISM signed seeds 800-807; balanced gauge, so a straight line in W_out is a straight line in angle space; 64 fixed walks):

path	**loss at tau = 0, 0.2, ..., 1 (mean of 8)**	**accuracy at tau = 0 -> 1**	**largest rise above the running minimum, per seed**
signed, |default| start -> signed solution	9.46 9.18 8.65 7.50 6.41 0.00	0.30 -> 1.00	0.27 0.04 0.30 0.40 0.48 0.38 0.05 0.36
|.|, same start -> wrapped solution	9.46 8.23 7.87 7.96 7.44 0.00	0.30 -> 1.00	1.92 0.87 1.41 0.60 1.09 1.01 1.14 0.74
signed, collective latent route only	9.46 9.39 9.49 9.25 8.35 2.98	0.30 -> 0.72	0.03 0.00 0.00 1.36 2.15 1.44 0.21 0.02
|.|, the same latent route	9.46 9.41 9.66 9.90 9.68 9.21	0.30 -> 0.30	0.17 1.11 0.18 2.40 2.23 1.68 1.33 0.51

Every |.| seed's largest rise exceeds every signed seed's (complete separation, exact two-sided permutation p 1.6e-4).

The step map's basin with the content frozen (exploratory; `step_basin.py`, `basin_summary.py`; ring, seeds 800-807; content of the solved signed model frozen, only W_in, W_out and omega trained for 2000 steps of batch 32, about 2% of a full run; fresh-stream accuracy). These CPU runs are not deterministic and the step-map search is bistable: re-runs of single cells in the independent audit gave S_abs s801 0.944 -> 0.509 and 0.939, A_abs s803 0.548 -> 0.450 and A_high s804 0.921 -> 0.839, so the per-seed values and the "runs >= 0.9" column do not reproduce; only contrasts much larger than these swings are read:

start	**class**	**accuracy start -> end (mean)**	**runs >= 0.9**
default angles	signed	0.33 -> 0.83	3/8
|default| (the |.| start)	signed	0.29 -> 0.47	1/8
|default|, move latents frozen	signed	0.29 -> 0.40	0/8
|default|	|.|	0.29 -> 0.36	0/8
past pi, U(1.7, 4.5)	|.|	0.36 -> 0.64	1/8

start	class	accuracy start -> end (mean)	runs >= 0.9
halfway along the straight segment to the wrapped solution	|.|	0.32 -> 0.61	0/8

With the content fixed the start decides the step map's search (default vs |.| start, signed: +0.365, p 0.005; past pi vs small start, |.|: +0.280, p 0.0007; both survive Holm over the eight exploratory tests, two others at p 0.012-0.013 do not). The signed escape from the same-direction start that full training shows (7/8) is not reproduced by the step map alone (signed vs |.| from that start +0.104, p 0.32): in full training the content co-adapts. No run from halfway along the straight segment to the wrapped solution reached 0.9 (end accuracies 0.42-0.82), so within this budget the wrapped solution's basin did not extend that far; a re-run could change single runs.

H8. The sign route (`SIGNROUTE_RESULTS.md`, registered SUPPORTED; per seed in C10)

Six observation-step-free rank-4 arms on the ring at lr 1e-3 and 3e-3 (LRSWEEP's recipe), fresh seeds 1800-1807. The *hybrid* confines the readout to the non-negative orthant, a_c(x) = omega_c |w_c| . z_x, with the latent and hence the increment signed. *abs_tied* is |.| with the move latents tied antipodal, z_y := -z_x, so both moves step omega |w . z| and its code is a pure clock.

lr	signed	|.|	hybrid	abs_tied	primary (hybrid vs |.|, gated on signed vs |.|)
1e-3	8/8, 1.000	0/8, 0.918	8/8, 1.000	0/8, 0.739	Fisher p 0.0002; d +0.082
3e-3	8/8, 1.000	2/8, 0.958	8/8, 1.000	0/8, 0.748	Fisher p 0.007; d +0.042

Hybrid vs signed: d 0.000 at both rates. Descriptive end states (medians): hybrid u-ratio \| z_x + z_y\| / (\|z_x\| + \|z_y\|) 0.043 / 0.002 and cosine(z_x, z_y) -0.996 / -1.000, as signed (0.055 / 0.001; -0.994 / -1.000); |.| 0.661 / 0.664 and -0.133 / -0.165; the hybrid's effective readout is 0% negative and fully row-sign-coherent. Both the absence of the collective route (E8.3) and the far-side cancelling point (E8.1, E4.1) predict this outcome, since the hybrid restores both; the batch localises the cost to the increment and shows that non-negativity in the readout costs nothing.

H9. Exp and softplus increments (`SIGNROUTE_RESULTS.md`, registered breadth cells at fixed rates)

lr	arm	SOLVED	accuracy	d vs signed	label (Holm per test, 4 cells)
1e-3	exp	1/8	0.935	-0.065	signed learns better (Fisher 0.0056, perm 0.0006)
1e-3	softplus	1/8	0.879	-0.122	signed learns better (Fisher 0.0056, perm 0.0042)
3e-3	exp	1/8	0.958	-0.042	signed learns better (Fisher 0.0056; SOLVED test only; 5/7 unsolved runs still descending)
3e-3	softplus	3/8	0.966	-0.034	signed learns better (Fisher 0.0256; SOLVED test only; 2/5 unsolved still descending)

Both start every token at a positive step (mean move step at initialisation exp +1.051, softplus +0.706, signed +0.003), i.e. as index RoPE with a rescaled omega; no best-rate sweep was run for them.

I. CARoPE, CoPE and Selective RoPE

CARoPE (`model_mono_baselines.py`, `CARoPEF`). Per head h and token t a scalar rate f_h(x_t) = 1 / (softplus(x_t W)_h + 1) in (0, 1) replaces RoPE's base frequency, so channel i advances by f_h(x_t)^i per token; initialised as index RoPE (W = 0, a per-head bias with 1 / (softplus(b) + 1) = 10000^(-2 / d_head)). Every token, observations included, advances the phase; the layer is MapWM's, so only the phase differs. Proposition 8 (E5): no channel can read every return. Registered (GENERALITY): torus 0.667 (0/8), below index RoPE (0.722, -0.055, p 0.040); ring 0.741 against index 0.730 (+0.011, not resolved).

CoPE (`CoPEF`). Golovneva et al.'s Listing 1 in torch: per layer one module shared by the heads; gates g_ij = sigmoid(q_i . k_j / sqrt(d_h)); count p_ij = sum_{k=j}^{i} g_ik, clamped to 2047; position logit by linear interpolation of q_i . e[floor p] and q_i . e[ceil p], table e at zeros initially; added to the attention logits; no rotary position. *Representability (constructed):* gates 17/16 and 15/16 per (move, observation) pair make p_ij mod 2 a function of net displacement, read by a period-2 table (1.000 revisit accuracy). The exact solutions form a lattice (P7 of v3: period 2 or 3).

batch	task	CoPE	SOLVED	comparison
GENERALITY (700-707)	ring	0.887	0/8	signed 1.000, 8/8 (-0.113, p 0.0002); index 0.730 (+0.157, p 0.0002)
WRAP_COPE (900-907)	ring, table at zeros / sinusoid past pi / below pi / slope-matched	0.840 / 0.913 / 0.867 / 0.943	1/8, 2/8, 1/8, 0/8	contrasts not resolved (bimodal seeds)
ACCOUNTS T4 (1100-1115)	256-cell line (never wraps) / ring	0.914 / 0.873	1/16 / 0/16	line: signed 1.000, 8/8 (Fisher p < 1e-4); index 0.772, 0/8

On both line and ring several CoPE seeds reach 0.99-1.00 held-out accuracy without meeting the loss criterion (at a held-out accuracy cut of 0.999 the line reads 5/16 and the ring 2/16; analysis). MapFormer v4 Table 2 reports CoPE failing 1D and 2D navigation (one layer 0.63 +/- 0.21 in 1D and 0.65 in 2D; four layers 0.88 and 0.76).

Selective RoPE (`SROPE_TORUS_RESULTS.md`, registered). Faithful Selective RoPE (Movahedi et al. 2026), the module reproduced from the authors' code to 6e-6, in the authors' best language-model configuration (SRoPE), the same without the SiLU (SRoPE_NoSiLU), the authors' state-tracking configuration (SRoPE_ST: low rank 32, no SiLU, no gate, a learnable bias of +1 rad per token per channel at initialisation) and our earlier generator swap (SRoPEGen), against MapWM (signed, rank 4) and index RoPE in one batch; torus, T = 1024, 900 epochs, seeds 1700-1707; own-stream floors blank 0.509, n-gram 0.578, retrace 0.845.

arm	accuracy T = 1024	SOLVED	classes	short / long / wrap-only	T = 2048
MapWM	1.000	8/8	all SOLVED	1.000 / 1.000 / 0.999	0.992
SRoPEGen	0.985	5/8	5 SOLVED, 2 DESC, 1 STAL	0.997 / 0.975 / 0.865	0.951
SRoPE_NoSiLU	0.969	2/8	2 SOLVED, 6 DESC	0.987 / 0.952 / 0.805	0.861
SRoPE (best LM config)	0.961	0/8	all DESCENDING	0.992 / 0.937 / 0.659	0.807
SRoPE_ST	0.769	0/8	7 STAL, 1 DESC	0.788 / 0.705 / 0.628	0.724
index RoPE	0.725	0/8	all STALLED	0.775 / 0.470 / 0.465	0.561

Registered: SRoPE vs MapWM fails to learn the map (d -0.039 [-0.050, -0.030], perm p 0.0002; 0/8 vs 8/8, Fisher p 0.0002); SRoPE beats index RoPE (+0.236, p 0.0002); it uses its phase (angle set to zero: 0.961 -> 0.507, p 0.0078). Declared secondaries: the SiLU makes no detected difference (p 0.57); SRoPEGen vs MapWM not resolved (5/8 vs 8/8, p 0.20); SRoPE_ST fails against MapWM and is not distinguished from index RoPE (+0.044, p 0.23); at a loss threshold of 0.1 SRoPE vs MapWM is still 0/8 vs 8/8. Descriptive readouts: an observation's step is 0.86-0.96 of a move's in SRoPE / NoSiLU (0.45 SRoPEGen, 0.027 MapWM); opposite moves cancel (wrap-aware) in 0.75-0.80 of channel pairs (MapWM 1.00); the angles vary with context through the causal convolution (context CV 0.11-0.13, MapWM 0.00); SRoPE_ST's learned per-token angles have negative-mass share 0.00 for every token class and cancel in 0.00 of channels, so its bias keeps the step effectively non-negative.

J. Text world

Task (`TEXTWORLD_RESULTS.md`, `train_textworld.py`). The 64 x 64 torus walk rendered as English: 58 words, three synonyms per direction, movement-free verbs, fillers and asides; T = 1024 words (about 132 steps); held-out map; predict the observation word at revisits. A direction word appears only inside a movement clause (context-free steps). Floors: best constant 0.505, reversal-copy 0.597, retrace 0.934 (analysis; J6). Recipe: the torus recipe (A2), path MapWM r4 at one layer against index RoPE at one and two layers; 200 evaluation walks.

Step tables per seed (declared secondaries, `fig_textworld_out.txt`; "minus common" = opposition after removing the component shared by the four direction steps; ablation columns: held-out accuracy, eval mode without the dropout scale):

seed	type	raw opposition	minus common	common / north	accuracy	non-direction steps zeroed	direction steps zeroed
0	clock	1.730	0.0071	0.711	0.881	0.191	0.388
1	map	0.083	0.0029	0.041	0.998	0.893	0.505
2	clock	1.434	0.0064	1.144	0.999	0.665	0.482

seed	type	raw opposition	minus common	common / north	accuracy	non-direction steps zeroed	direction steps zeroed
3	clock	1.707	0.0022	0.480	0.995	0.673	0.463
4	clock	1.768	0.0052	0.501	0.999	0.539	0.445
5	map	0.120	0.0050	0.060	1.000	0.898	0.505
6	map	0.080	0.0117	0.040	1.000	0.950	0.505
7	map	0.119	0.0057	0.059	0.882	0.586	0.498

On the clock seeds the common component is parallel to the verbs' step (cosine +0.999 to +1.000), and 31-38 of 64 phase channels drift between visits to the same cell (2-4 on the map seeds). Synonym cosine 0.9996-1.0000 on every seed.

The paragraph and table above use the registered batch. The analyses below (`docs/paper/expansion/scripts/language/`) read the 8 registered text-world path runs (seeds 0-7) and the 8 MapWM, 8 NormStep and 8 DirOnly runs of TW_NORMSTEP (seeds 10-17) on a held-out map (np seed 424242, 40 walks of 1024 words unless stated). The manual forward used for the manipulations reproduces every model's logits exactly, and the role-annotated generator reproduces the task generator token for token. "Re-scored" = with the x1.111 attention-dropout scale (C9).

J1. The step table by word class (analysis; `tw_step_account.py`)

Steps omega-scaled, relative to the clause-level direction differential (one move):

word class	map-type models (14 learned-step)	clock-type models (10)
direction words (differential)	1.00 by normalisation; opposites cancel 0.001-0.013	same
adverbs, fillers	0.004-0.03	0.004-0.025
`and` / `a` / `she` (shared by fillers, seeing phrases, asides)	0.02-0.07	0.02-0.08
verbs, seeing words, object words, `.` (once per move)	0.02-0.14	0.24-1.49
aside-only words (`thought`, `about`, `remembered`, `story`)	0.07-0.16 per word	0.41-1.33 per word
aside sentence, net (usage-weighted; equal by construction to the aside noun's offset from its place)	0.05-0.12 of a move	0.04-0.16 of a move
DirOnly oracle: every non-direction word	0 by construction	--

The clause tick (what every move adds whatever its direction) is 0.03-0.23 of a move on map-type models and 3.3-6.8 moves (raw norm, all channels) on clock-type models, parallel to the verbs' step (cosine +0.999 to +1.000 on all 10).

J2. The tick predicts the drift; the clock counts moves (analysis; `tw_step_account.py`, `tw_clock_unit.py`)

Predicted drift per channel = mean |wrap(lag x tau_c)| over consecutive-visit pairs (median lag 8 moves, 1,221 pairs); observed = mean |wrap(dtheta_c)| at those pairs.

Classified at 1 rad they agree on 0.98-1.00 of channels on each of the 32 models. On the 22 map and DirOnly models both are near zero, so the agreement there is close to an identity; the informative evidence is the 10 clock models, where the predicted and observed drifting-channel counts match exactly. Clock models drift in 26-38 channels between consecutive visits (registered text world s0 / s2 / s3 / s4: 26 / 29 / 35 / 36; MapWM s11 / s13 / s14 / s15: 38 / 27 / 28 / 30; NormStep s11 / s16: 27 / 28); map models in 0-3. (The registered 31-38 of the CORRECTED block in `TEXTWORLD_RESULTS.md` counts drift from the first visit.) Regressing each move's net phase in the drifting channels on direction, aside present, filler count and adverb present: an aside sentence adds 0.009-0.062 of a tick, a filler word 0.001-0.007, an adverb word 0.001-0.007 (10 clock models). A word counter would give fillers about 1/8 of a tick each and a sentence counter would give asides about one tick: the clock counts moves. On all 10 clock models 2-6 channels tick by exactly pi per move (e.g. -3.141, -3.140 rad); on the even 64-torus a pi tick encodes the parity of the cell and is invisible to revisits (E7.2), and these channels carry up to 0.24 of retrieval usage. The registered finding that zeroing every non-direction step costs the clock seeds 0.33-0.69 and the map seeds 0.05-0.30 tracks the direction words' common component in the used channels (0.06-0.34 of a move on clock seeds, 0.01-0.03 on map seeds), which the other words cancel within the move (Spearman +0.905 across the 8 registered seeds, exact permutation p 0.0046; `tw_common_used.py`, p from `scripts/whatwhere/whatwhere_stats.py`; the used channels are those whose tick is 0 mod pi, a post hoc rule).

J3. The clock is idle (analysis; `usage_tick.py`, `tw_attention_asides.py`)

79-99% of retrieval usage sits in channels whose tick is 0 mod pi. Removing the drifting channels' q and k components from the score (re-scored accuracy):

model	intact	clock channels removed	as many map channels removed (5 draws)	only clock channels kept
text world s0 (eval mode 0.875, below ceiling)	0.973	0.717	0.43 [0.24, 0.58]	0.07
text world s2	0.999	0.990	0.25 [0.03, 0.63]	0.03
text world s3	0.998	0.993	0.06 (all 29 map channels)	0.06
text world s4	0.999	0.999	0.03 (all 28)	0.03
MapWM s11	0.989	0.985	0.48 (all 26)	0.48
MapWM s13	0.997	0.996	0.65 [0.50, 0.84]	0.48
MapWM s14 (eval mode 0.960)	0.995	0.902	0.25 [0.03, 0.72]	0.03
MapWM s15	0.995	0.994	0.07 [0.02, 0.23]	0.02
NormStep s11	0.998	0.996	0.46 [0.20, 0.60]	0.44
NormStep s16 (eval mode 0.835)	0.982	0.809	0.22 [0.04, 0.48]	0.04

On the 7 clock models at ceiling in eval mode the clock costs at most 0.009 to remove; the 3 below ceiling still lean on it (0.09-0.26).

J4. The used-channel ratio across data (analysis; `crossover_usage.py`, `usage_tick.py`, `tw_alpha_drift.py`)

Usage-weighted median of rho_c (E13) at the training length, all MapWM with a signed step:

data	used-channel rho	usage in clock-dominated channels	seeds
2D torus, 1 layer, matched length (SIGN_MATCHED signed)	0.064-0.079	0.000	8
text world, 1 layer (23 of 24 learned-step models; the unconverged tw s0: 2.5)	0.06-0.23	0.01-0.30	24
code, bytes, 9 layers (trained at 2048)	4.5-6.6	0.50-0.55	3
TinyStories, words, 6 layers (trained at 512)	9.9-12.3	0.42-0.43	3
k-back recency with fillers, 1 layer (signed rank 4)	21.8-22.1	0.93-1.00	8

Clock-dominated: rho_c > 1 and more than half a turn per training window. The pooled growth exponent alpha does not separate map from clock data: it reads 0.61-0.74 on text-world map seeds (the true walk: 0.668), 0.98-1.02 on text-world clock seeds, which carry a perfect map, 1.00-1.01 on TinyStories and 0.99-1.01 on code. The used-channel rho separates the clock seeds (0.07-0.23) from TinyStories (9.9-12.3).

J5. Asides: the offset is what beats the oracle (analysis; `tw_attention_asides.py`, `tw_attention_asides_dironly.py`)

`TW_NORMSTEP_RESULTS.md` (registered) found that DirOnly, whose steps are restricted to the twelve direction words, stops at 0.970-0.974 on 8/8 seeds because an aside's noun sits at the place's phase, and that learned-step models make none of these errors; re-scored, every learned-step arm beats DirOnly (-0.024, p 0.0002). Two manipulations at evaluation, each keeping the phase after the aside unchanged:

- *Collapse* (learned models): every aside word is put at its clause's object-slot phase. The 19 learned-step models at ceiling in eval mode fall from 0.989-1.000 to 0.952-0.975, and 71-100% of their errors then name an object told in an aside at that cell (before: 0-13 errors, one of that kind, in MapWM s13). The 5 models below ceiling respond partly (0.966-0.994). The 19 / 5 split at eval-mode accuracy 0.98 is a post hoc cut (it falls in a gap, 0.960 against 0.989).
- *Lift* (DirOnly): intact, DirOnly scores 0.962-0.981 on this stream and 100% of its errors are aside errors on 8/8 seeds. Offsetting its aside words by half a move in-plane gives 0.999-1.000 on 8/8; by one move off the map plane 1.000 on 7/8 (one seed 0.972); by a quarter move off-plane 0.963-0.996 (a variant added after the main run). The lift sizes are free choices, not fitted.

At each revisit query the 19 learned-step models at ceiling put 0.55-0.76 of their attention on the object slot of the most recent visit to the cell and 0.004-0.035 on aside nouns told there; DirOnly puts 0.041-0.077 on those aside nouns.

J6. Retrace on the text world (analysis; `tw_retrace_floor.py`, `tw_strata_200.py` -> `tw_strata_200walks_out.txt`)

The retrace rule of B, ported to the text world, scores 0.9337 on the registered evaluation stream (0.9365 on TW_NORMSTEP's), where the constant and reversal-copy rules reproduce the registered 0.5054 / 0.5965. On all 200 walks of that stream (6,381 revisits) the rule applies to 5,525 (86.6%) and is never wrong when it applies; on the other 856 (13.4%), places reached by another route than undoing the last run, it predicts 'nothing', which is right on the 50.6% of them whose cell is blank. Accuracy by stratum (mean over 8 seeds; fresh seeds 2-7 in brackets):

model (registered batch, n = 8)	all	retrace applies (5,525)	retrace cannot reach (856)	... cell holds an object (423)	... blank cell (433)
path, 1 layer, re-scored	0.993	0.999	0.956 (0.978; min 0.786 at s0)	0.925 (0.967; s0 0.617, s7 0.872)	0.985
path, 1 layer, eval mode	0.969	0.975	0.934 (0.958)	0.884 (0.927)	0.984
index RoPE, 2 layers	0.772	0.828	0.411 (0.34-0.55)	0.129 (0.09-0.15)	0.687
index RoPE, 1 layer	0.505	0.513	0.454	0.004	0.894
best constant	0.505 ('nothing')	--	0.506 ('nothing')	0.088 ('cat')	1.000 ('nothing')

The object-cell column is the stratum "retrace wrong" of the first version of this analysis (`tw_strata.py`, first 100 walks: 5.9% of 3,055 revisits, path 0.920 / 0.961 fresh, RoPE 2L 0.138, constant 0.112). Its best constant excludes 'nothing' by construction, and one index layer scores near zero there because it predicts 'nothing'. On the whole non-retrace stratum, which needs no outcome-defined split, two index layers score below the constant. Two-layer index RoPE learns part of retrace (its best seeds reach 0.927 and 0.926 overall, against retrace's 0.934) and nothing that needs the map. Seeds at ceiling score 0.936-1.000 on the object-cell stratum.

K. What and where; the leak

K1. What and where at T = 1024 (analysis; `docs/audits/2026-10-20/WHAT_WHERE_T1024.md`, `causal_whatwhere_T1024.py`)

The causal what/where test re-run in the regime where a map is needed: the SIGN_MATCHED checkpoints (torus, trained and tested at T = 1024, one layer, rank 4, n = 8 per arm; floors on these walks blank 0.506, n-gram 0.575, retrace 0.843). The rebuilt forward equals the model logits exactly on all 32 checkpoints; intact accuracy matches the registered evaluation to <= 0.0007. Held-out accuracy when the score is rebuilt:

condition	signed (learns the map)	|.|	softplus	index RoPE
intact	0.998	0.821	0.780	0.731

condition	signed (learns the map)	|.|	softplus	index RoPE
token identity within type removed from the score (token type kept)	0.966 (-0.032; 8/8 lower, sign-flip p 0.008, uncorrected)	0.544 (-0.277)	0.476 (-0.305)	0.479 (-0.252)
one shared position kernel x per-pair content gain plus offset	0.998 (-0.000)	0.584 (-0.237)	0.675 (-0.106)	0.439 (-0.293)
additive position + content	0.895 (-0.104)	0.350	0.512	0.492
token type removed too (one global mean)	0.113	0.295	0.302	0.399
content only (no rotation)	0.505	0.399	0.385	0.490
SS(P) / SS(S): within-row score sum of squares carried by the content-averaged score	0.925	0.644	0.577	0.618
content x position interaction share	0.061	0.187	0.174	0.353

The model that learns the map factorises what and where: one position kernel per head, scaled by a per-pair content gain plus offset (2 x 441 parameters per head, fitted on the calibration walks, so this condition nests the additive one), reproduces it exactly; removing token identity within type (which also removes action identity within the action class and blank against object within the observation class) costs 0.032 (8/8 seeds lower; sign-flip p 0.008, the design's minimum; paired t p 0.012; Holm over the 24 contrasts 0.08; the effect sits at the MDE of 0.031); token type is essential. The failing models do not factorise: removing identity costs 0.25-0.31 on average (|.|: 7 of 8 runs lose at least 0.15; Abs_r4 s0 loses 0.020), and a shared kernel times a gain does not restore them. Within the non-negative arms, across 16 runs, accuracy correlates with the share of it that survives removing token identity (Spearman 0.84, p 5e-5; a post hoc statistic, `scripts/whatwhere/whatwhere_stats.py`); the two non-negative runs that reach > 0.95 keep 0.97 and 0.99 of their accuracy, the four lowest softplus runs fall to 0.03-0.07 (5-12% of their accuracy), and the |.| runs keep 0.22-0.77 of theirs except Abs_r4 s0 (0.98 of 0.931). (Intact softplus reads 0.780 on this rebuilt forward against the registered 0.781. The `.txt` of the source was written by `--report` after the run; the JSON is the run's.)

K2. What and where at T = 128 (analysis; `docs/WHAT_WHERE_CHECKS.md`; the 48 PAPER2X2 checkpoints)

score used	RoPE	PoPE	MapWM r2	MapWM r4	MapPoPE r2	MapPoPE r4
intact	0.805	0.679	0.971	1.000	0.999	1.000
position only (object identity removed, token type kept)	0.463	0.476	0.837	0.989	0.974	0.986
shared position kernel x content gain	0.430	0.435	0.791	1.000	0.988	1.000

On short walks, where retrace already scores 0.934, converged path models keep 0.974-0.989 with object identity removed and 0.988-1.000 with a shared kernel times a content gain; index models fall below the n-gram floor (0.598). Separation as such is MapFormer's (their Fig. 9).

K3. The what-to-where leak (`LEAK_RESULTS.md`, `GAIN_PHASE_RESULTS.md`, registered)

On a task with a fresh map and 16 fresh objects per sequence (32 x 32 torus, T = 1024, rank 4), object tokens leak a small step into the phase. Removing the step from object tokens (ActOnly) or normalising the step input (NormStep: the step reads LayerNorm(embedding)) lets the model converge: 16/16 SOLVED (unseen-object accuracy 0.9997 / 0.9998) against 0/8 for MapWM (0.9890, still descending). In distribution the remedies gain +0.0107 in unseen-object accuracy (p 0.0002, 8/8), equal to MapWM's leak; the accuracy and loss are strongly coupled there (r(loss, accuracy) -0.94). On fresh seeds the convergence result replicated (NormStep 8/8 against MapWM 0/8) and the accuracy gain did not (+0.0045, p 0.46; 0.9940 against 0.9895). A score change alone leaves the leak in place (GAIN_PHASE).

L. PoPE's score

L1. Registered

batch	result	file
PAPER2X2 (T = 128, r2, n = 8)	RoPE 0.805, PoPE 0.679, MapWM 0.971, MapPoPE 0.999; interaction +0.154 (8/8); at r4 +0.125 with both path arms at 1.000	`PAPER2X2_RESULTS.md`
MAPPOPE_PAIR (T = 128, r2, n = 16)	the score rule, not the doubled angle count: +0.0243 (p 0.012; SOLVED 16/16 vs 10/16), plausibly convergence speed, +0.009 on fresh seeds; the angle count +0.0002 at a ceiling (0.9995-0.9997), uninformative	`MAPPOPE_PAIR_RESULTS.md`, `GAIN_GRAIN_RESULTS.md`
SCORE_RANK (T = 1024)	no rank-2 rescue: 3/12 vs 3/12 (App. G2)	`SCORE_RANK_RESULTS.md`
DELTA_FREE (T = 128, r2, n = 8, fresh seeds 1600-1607)	PEAK-PINNING SUPPORTED: freeing delta raises the index model by +0.122 [90% CI +0.076, +0.167] (8/8, sign-flip p 0.0078), concentrated in multi-step reversals (R3+ +0.180, Holm p 0.031), and leaves the path model at ceiling (-0.018, p 0.31; SOLVED 6/8 vs 8/8, Fisher p 0.47)	`DELTA_FREE_RESULTS.md`

DELTA_FREE arms (accuracy at T = 128 / 512 / 1024, SOLVED): PoPE-Flat (index, delta pinned: zero initialisation held at 0 by the clamp's dead zone) 0.649 / 0.615 / 0.599, 0/8; PoPE-FreeDelta (delta initialised U(-2 pi, 0), clamp removed) 0.771 / 0.536 / 0.442, 0/8; MapPoPE-Flat (path, pinned) 0.999 / 0.980 / 0.933, 8/8; MapPoPE-FreeDelta 0.981 / 0.901 / 0.831, 6/8; MapPoPE-FrozenDelta (path, delta frozen at the free arms' draw) 0.999 / 0.929 / 0.832, 8/8. The delta x phase interaction is -0.140 [-0.186, -0.092] (0/8 seeds positive). Declared secondaries: with delta pinned 0% of index content pairs prefer offset 3 over offset 1 (kernel argmax at offset 1 in 1074/1088 pairs), freed 50-100% per seed with argmax spread over offsets 3, 4, 7, 8; attention on the newest observation (never the target) 0.287 -> 0.003; index gain by stratum R1 +0.039 (n.s.), R2 +0.117, R3+ +0.180, unvisited-run +0.014; path target alignment C 0.987 pinned vs 0.616 free and 0.614 frozen, with the code term C0 0.993 / 0.798 / 0.884; pinned arms 96-100% in the dead zone; free arms moved mostly by weight decay. Beyond the training length the path robustness contrasts (free or frozen minus pinned at T = 1024: -0.101, -0.102) are

below the registered threshold, and the free index arm's retrace kernel does not extrapolate (-0.158 at T = 1024).

L2. The mechanism, probed (analysis; `docs/paper/expansion/scripts/mappope/kernel_probe.py, probe_stats.py`)

Layer 0 rebuilt by hand; every run's logits equal the model's own (difference 0.0); per-seed accuracies reproduce the stored evaluation. T = 128, PAPER2X2 checkpoints, 100 walks per seed:

readout	RoPE	PoPE	MapWM r2	MapPoPE r2
target coherence C (path arms)	--	--	0.703	0.995 (p 0.0002, 8/8)
code term C0 (path arms)	--	--	0.905	0.996 (p 0.0002)
same-cell attention mass (path arms; retrieval head)	--	--	0.747	0.925 (p 0.0008)
attention on action keys (path arms; retrieval head; over both heads 0.223 vs 0.147, p 0.15)	--	--	0.179	0.041 (p 0.0002)
content pairs with K(3) > K(1) (index arms)	53-100% per seed	0% on every seed	--	--
attention on the newest observation (index arms; the more recency-weighted head)	0.002	0.307 (p 0.0002)	--	--
R3+ accuracy (index arms)	0.779	0.566 (p 0.0002)	--	--

The argmax of the content-pair kernel over the odd offsets 1-15, where an observation key can sit, is at offset 1 for 1068/1088 PoPE pairs; for RoPE it is at offsets 3 (569), 7 (342), 11 (156) and 15 (21): every RoPE pair peaks on a retrace offset 4j - 1, a comb (`kernel_argmax_odd.py`; the argmax over all offsets 1-16 also lands on even offsets, where no observation key sits). The PoPE - RoPE accuracy gap decomposes over retrace strata as R1 -0.006, R2 -0.005, R3+ -0.110 and non-retrace -0.004. MapWM's content phase costs C0 - C = +0.20 (7/8 seeds; a statement about target coherence on the retrieval head, not an accuracy prediction). About 80% of the +0.154 interaction is the index side, (MapPoPE - MapWM) - (PoPE - RoPE) = (+0.028) - (-0.126).

T = 1024, SCORE_RANK checkpoints (r2 n = 12, r4 n = 8; 40 of the 100 registered walks per seed; each stratum read on the head with the most same-cell attention in that stratum, `HEAD_RULE=stratum` -> `kernel_probe_score_rank_hstratum_out.txt`, `probe_stats_hstratum_out.txt`; the first version read every stratum on the head chosen over all revisits):

stratum	MapWM r2: C / C0	PoPE score r2: C / C0	MapWM r4: C / C0	PoPE score r4: C / C0
plain gap < 128	0.617 / 0.900	0.960 / 0.960	0.926 / 1.000	0.997 / 1.000
plain gap >= 128	0.538 / 0.795	0.908 / 0.905	0.923 / 0.993	0.991 / 0.993
wrap-only	0.144 / 0.399	0.485 / 0.482	0.904 / 0.967	0.969 / 0.970

On short gaps the score raises C by +0.34 (p 0.0002) and C0 by +0.06 (0.900 vs 0.960, p 0.16): the repair is mostly in the readout term. On gap >= 128 revisits the score raises C (0.538 to 0.908, p 0.005) and leaves C0 low (0.80 / 0.91, p 0.32, against 0.99 at rank 4); on wrap-only revisits C0 is 0.40 / 0.48 under the two scores (p 0.64), and rank 4 raises it to 0.97 under either (MapWM +0.568, p 0.0009; PoPE score +0.488, p 0.012). In both rank-4 models the wrap-only revisits are carried by the head that is not the short-gap head (C0 0.97 there). At T = 128 MapPoPE's C0 is also higher than MapWM's (0.996 vs 0.905), so on short walks a better readout and a better code co-occur. About 60 contrasts are computed in this appendix without a multiplicity correction; the p 0.0002 contrasts survive Holm over the family, those at p 0.03-0.04 (e.g. MapPoPE - MapWM short-gap accuracy +0.114, p 0.036, consistent with the registered +0.115 on the same checkpoints) do not.

M. Loops and depth on a rank-2 code

M1. Registered (`LOOP_RANK_RESULTS.md`; torus, T = 1024, 900 epochs, n = 8, one batch)

arm	parameters	passes	SOLVED	accuracy	vs A: Fisher / permutation
A: r2, 1 layer	204,373	1	0/8	0.894	--
L: r2, looped x4	204,373	4	2/8	0.973	0.47 / 0.0034 (+0.079, CI [+0.026, +0.131])
L4: r2, 4 real layers	799,189	4	5/8	0.990	0.0256 / 0.0012 (+0.096)
C: r4, 1 layer	204,757	1	8/8	0.998	reference

L and A are identical in parameters and initialisation at every seed (maximum weight difference 0.0). Holm-adjusted p 0.0034 / 0.0028 / 0.0028; L4 - L +0.017 (p 0.0009). The registered verdict is UNMEASURED (it required L to solve >= 4/8). Final losses form three regimes: A 0.051-0.675, L 0.045-0.127, L4 0.011-0.070, C 0.002-0.010. Strata (short / long / wrap-only, registered secondaries): A 0.950 / 0.671 / 0.557; L 1.000 / 0.927 / 0.746; L4 1.000 / 0.990 / 0.887; C 1.000 / 0.995 / 0.977.

M2. Per pass (analysis; `scripts/recurrence/pass_strata.py`, `attn_offsets.py`)

The re-implemented forward matches each model's own forward and its stored evaluation (max |diff| <= 3.1e-4). L at `n_loops = j` is exact; L4's intermediate readouts are a logit lens.

arm	stratum	pass 1	2	3	4	5	6	8
L	all	0.381	0.887	0.964	0.973	0.971	0.954	0.847
L	long	0.363	0.612	0.875	0.927	0.921	0.871	0.731
L	wrap-only	0.393	0.519	0.700	0.747	0.730	0.699	0.623
L4	all	0.507	0.575	0.935	0.990			
L4	wrap-only	0.504	0.507	0.601	0.887			

One pass of L's trained weights scores below the always-blank floor (0.381 vs 0.51; mean, per seed 0.09-0.67, 7 of 8 below); the loop is a fixed-depth computation, and past four passes accuracy falls. Same-cell attention mass (best head per seed, `pass_strata_besthead.py` -> `out/PASS_STRATA_BESTHEAD.txt`; head labels are a symmetry, and at pass 2 the retrieval head is head 1 on three seeds) rises over L's passes 0.03 -> 0.50 -> 0.64 -> 0.63, and the share of queries whose top-attended key is a same-cell observation 0.35 -> 0.86 -> 0.88 -> 0.92: retrieval forms at passes 2-3 and pass 4 sharpens it. Attention entropy falls 6.1 -> 2.4 nats (head mean, 10 walks per seed; `attn_offsets.py`). On wrap-only revisits L4's last layer puts its top attention on a same-cell observation 81% of the time (best head), from a code whose own one-pass reader finds the cell 12% of the time (the reader's top-1 cell, a different quantity). Mass on same-cell action tokens stays at 0.01-0.15 (best-head means over passes 1-4). With an index code every pass works on short gaps only: the index loop's long-gap and wrap-only accuracy sits at 0.50-0.52 after every pass while short gaps climb 0.41 -> 0.94 (`index_loop_passes.py`, FACTORIAL checkpoints).

M3. Code quality, read without the block (analysis; `phase_reader_soft.py`, `phase_code.py`)

The one-pass reader of E11.1 (token-type phase offsets on query and key, a key bias, a soft vote over observations), fitted to each model's own theta on 24 separate walks and scored on the 50 evaluation walks:

arm	model (all / short / long / wrap)	one-pass reader	model - reader
A	0.894 / 0.950 / 0.671 / 0.558	0.866 / 0.940 / 0.558 / 0.448	+0.028 / +0.010 / +0.113 / +0.110
L	0.973 / 1.000 / 0.927 / 0.747	0.893 / 0.971 / 0.608 / 0.406	+0.080 / +0.029 / +0.319 / +0.341
L4	0.990 / 1.000 / 0.990 / 0.887	0.897 / 0.967 / 0.653 / 0.453	+0.094 / +0.033 / +0.337 / +0.435
C	0.998 / 1.000 / 0.995 / 0.977	1.000 / 1.000 / 1.000 / 0.997	-0.002 / 0.000 / -0.005 / -0.020

Code quality L - A +0.027 (permutation p 0.28), L4 - A +0.031 (p 0.27), both below the exact-t MDE (0.074 / 0.080; long-gap +0.050 / +0.095 against 0.27-0.28; `reader_mde.py` -> `out/READER_MDE.txt`): whether the loop and depth learn a code one pass reads better is unmeasured; C - L +0.107, C - L4 +0.103 (p 0.0002 each). The model is scored on 100 walks and the reader on the first 50 of them; on matched walks the margins change by <= 0.004. The reader under-fits on some runs: with five times the fitting steps on seeds 0, 1 and 3 the independent audit found L's long-gap reader accuracy up by 0.03-0.06, so L's long-gap margin (+0.32) is probably 0.03-0.05 too large; no verdict changes. Model-minus-reader margin against A's: L overall p 0.0036, long 0.0025, wrap-only 0.0002; L4 0.0031, 0.0045, 0.0002. A simpler top-1 kernel reader gives the same verdicts.

M4. Step-map geometry (analysis; `geometry.py`)

Per run, the omega-weighted x- and y-step phase vectors; *2D map*: common mode below 0.1 of the map step and the second singular value above 0.3 of the first; *1D map*: common

mode below 0.1, x and y steps collinear (sigma_2 / sigma_1 <= 0.002 in every such run); *drifting*: common mode >= 0.1.

arm	2D map	1D map	drifting
A	1/8	0/8	7/8
L	0/8	4/8 (code x + kappa y, |kappa| 1.38-1.42)	4/8
L4	1/8	4/8 (|kappa| 0.62-1.72)	3/8
C	8/8	0/8	0/8

2D map, aids vs rank 4: 1/16 vs 8/8 (Fisher p 1.2e-5); the aids do shift the class, 1D cancelling 8/16 against 0/8 for plain rank 2 (Fisher p 0.022, post hoc), so they change which code is learned within the family but rarely reach rank 4's. The class thresholds are post hoc and stable (any common-mode cut in (0.08, 0.34) and any second-singular-value cut in (0.002, 0.57) give the same classes; closest call C s0 at 0.08). In the reader's terms the 1D code is the collapsed end state of a rank-D head with the common mode removed. At initialisation the loop multiplies the step map's gradient norm by 2.9 and leaves its batch-to-batch coherence unchanged (0.46 -> 0.41; `grad_at_init.py`): the loop does not deliver a stronger or more consistent signal to the code, and Adam removes the scale.

N. EM's recency deficit: a contrast case

Single-origin MapEM (one p_0, an ablation of the paper-faithful model) on k-back recency (T = 1024, 300 epochs):

leg	result	file
exists	the rewind installed and frozen scores 1.000 on 8/8 at T = 1024 and 2048	`WARM_RESULTS.md` (W1)
kept	installed at the 1/64 scale and trained: 0.642; at 8x the scale 0.941 (4/8 >= 0.95); fully kept (1.000, 8/8) with `w_in`'s content columns held at zero	`WARM_RESULTS.md`, `NOLEAK_RESULTS.md`
from scratch	0.600, against 0.975 for MapWM and 0.837 for paper-faithful MapEM	`RECENCY_EM_RESULTS.md`
found per token	with one offset k every query shares, EM finds a 63-symbol rewind on 7/8 seeds (0.985), faster than MapWM	`SEARCH_RESULTS.md`
budget	four times the budget moves the 64-offset task from 0.578 to 0.928 (6/8 near 1.000)	`SPREAD_RESULTS.md` (an arm of a registered batch; the budget contrast itself was not registered)

EM's deficit is a search problem of a different kind: exposure per query token, which budget closes. Rank 2, by contrast, solves 0/8 with twice the budget on the torus (5/8 runs still descending at 1800 epochs), and |.| on the ring 3/8 at three times the budget (`RING_BUDGET_RESULTS.md`).

O. Analysis scripts

Every number marked (analysis) comes from one of these scripts (CPU, eval-only, on stored checkpoints of registered batches unless stated). "Verification" = the check the script runs before any new readout. Independent audit: every script below was audited results-blind and re-run on CPU (`docs/paper/review_draft5/ANALYSIS_SCRIPT_AUDIT.md`); every committed output re-derived, except `sign/step_basin.py`, which is not deterministic (its numbers are labelled exploratory). The audit's corrections are applied in this version; the scripts marked "new" were added to apply them, so that every quoted number has a committed generator.

script (under `docs/paper/expansion/scripts/`)	what it computes	used in	verification
`depth/torus_depth.py`, `summarize_torus.py` (`--segments`: new)	torus revisit classes, runs spanned, attention masses; fixed-gap segments and runs spanned by long-gap revisits	Table 3; Sec. 6; F1-F2	stored accuracy to 4 decimals (16 runs); forward = model logits
`depth/torus_probe.py`	ridge probes of the residual stream	F3	same stream
`depth/torus_runlen.py`	run-length-shifted evaluation	F4	-- (distribution-shifted)
`depth/ring_depth.py`, `summarize_ring.py`	ring classes, probes, one-layer mixture test	F5	stored accuracy for all 128 runs (< 2e-3)
`depth/ring_probe_window.py`, `ring_class_floors.py`, `ring_plain_bins.py` (new)	late-position probes and old-displacement targets; per-class floors; plain-only run bins	Sec. 6; F5	same stream as `ring_depth.py`
`depth/dyck_distance.py`, `dyck_crosstab.py`	Dyck closing accuracy by blocks and distance	F6	arm means equal the registered ones
`sign/clock_table.py` (new)	the clock / map ratio of E7.1' for every sign and rank arm, per run and per head	Table 4; Sec. 7.1; G3; H7	reproduces `sign_clock.py`'s one-sided ratios as a check
`sign/sign_clock.py`, `rank_clock.py`	cancelling usage share; observation and latent clocks	G3; H7	cone inequality as an identity (0 violations)
`sign/latent_geometry.py`, `lattice_p.py`	latent end states; +1/+2 relations	H4; H7	initialisation rebuilt bit-exact
`sign/latent_reach.py`, `aobal_geometry.py`	latent-reachable share; readout targets	H7	--
`sign/prop_checks.py`	numerical checks of E8.3-E8.4	E8	--
`sign/path_profiles.py`	loss along straight paths	H7	--
`sign/step_basin.py`, `basin_summary.py`	step-map-only training from fixed content (CPU; not deterministic: exploratory)	Sec. 7.3; H7	exact solutions held (pilot)

script (under docs/paper/expansion/scripts/)	what it computes	used in	verification
rank/rank_geom.py, sweep_final.py, summarise_final.py, c1_runlevel.py	off-plane share, readgain, r_beta	G4	drift and frame conditioning equal the registered head statistics to 4 decimals
rank/escape_start.py, init_frame.py, grad_hiding.py	escape re-read; initial conditions	G4; G6	--
rank/clock_decomp.py, c1_runlevel_once.py (new)	move / observation parts of the leftover clock; escape-arm clock sizes; C1 counting each network once	Sec. 7.2; G4	--
rank/train_snap.py, analyze_snap.py, train_nd_cpu.py	CPU pilots on seeds 3000-3002, 4000-4002	G6	outside every registered batch
recurrence/pass_strata.py, attn_offsets.py, index_loop_passes.py; pass_strata_besthead.py (new)	per-pass accuracy and attention; best head per seed	M2	forward and stored evaluation to <= 3.1e-4 (1e-6 on FACTORIAL)
recurrence/reader_mde.py (new)	MDE of the reader's code-quality contrasts	Sec. 7.4; M3	--
recurrence/phase_reader_soft.py, phase_code.py	one-pass reader fits	Sec. 7.4; M3	fitted on separate walks
recurrence/geometry.py, training_curves.py, grad_at_init.py	step-map classes; curves; gradients at initialisation	M4	--
mappope/kernel_probe.py, probe_stats.py (HEAD_RULE=stratum / SUFFIX=_hstratum: new); kernel_argmax_odd.py (new)	kernel, coherence C and C0, attention by offset; per-stratum head; kernel argmax over odd offsets	Sec. 7.4; L2	rebuilt logits equal the model's (0.0); per-seed accuracies reproduce the stored evaluation
language/tw_lib.py and the tw_*.py scripts; tw_strata_200.py (new: all 200 walks, the non-retrace stratum and its floors)	step table, drift, clock unit, asides, attention, retrace, strata	Table 7; Sec. 8; J1-J6	manual forward logit-identical; generator token-identical; N_WALKS=100 reproduces tw_strata.py
language/usage_tick.py, crossover_usage.py, tw_alpha_drift.py	used-channel rho; growth exponent	Sec. 9; J4	--
docs/audits/2026-10-20/causal_whatwhere_T1024.py; whatwhere/whatwhere_stats.py (new)	what/where at T = 1024; its sign-flip p, SS(P)/SS(S) and Spearman, and the text-world Spearman p	Sec. 8; Sec. 9; J2; K1	rebuilt forward exact on 32 checkpoints; intact accuracy within 0.0007

Post hoc choices (rule 7; from the audit's Sec. 5): the best-head rule of the kernel probes and the per-stratum head rule that replaces it at T = 1024; 40 of the 100 registered walks at T = 1024 (L2); the r_beta < 0.5 cut, the 10x criteria for t_kappa / t_lam and the off-plane share as the escape statistic (G4, G6); geometry.py's class thresholds (M4, stable over wide ranges); sign_clock.py's 0.1 / 0.05 rad cuts and the near-pi cut 2.6 rad of clock_table.py (H7); the lattice_p.py tolerance (H4); the path-profile rise statistic (H7); the depth run-length bins (F1, F4); the text-world 0.98 ceiling cut, the blank exclusion of the object-cell stratum and the lift sizes (J5, J6); the what/where Spearman (K1); the 0 mod pi

used-channel rule (J2-J4). The audit found that none of its corrections depends on a threshold choice.